A Physical Introduction to Higher Mathematics

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Contents

Part I

Symmetry

1 Spectral Theory

Experimental observations.

- Small transverse disturbances on a taut string superpose: overlapping pulses add, and scaling an initial deflection scales its later motion. A finite string fixed at both ends has discrete resonant modes, while a pulse on an effectively infinite string requires a continuous range of frequencies.

1.1 Linear structures, duality, and determinants

Definition 1.1 (Vector spaces and linear maps)**.** Let k be a field. A *k-vector space* is an abelian group V with an action $k \times V \to V$ satisfying distributivity, associativity, and $1v = v$. A map $T : V \to W$ is *linear* if $T(av + bw) = aT(v) + bT(w)$.

Exercise 1.1 (String profiles and superposition)**.** Let V be the real-valued functions on $[0, L]$. Prove that the fixed-end profiles u satisfying $u(0) = u(L) = 0$ form a vector subspace. Express the opening string observation as closure under addition and scalar multiplication. Show that sampling a profile at finitely many points defines a linear map $V \to \mathbb{R}^n$.

Definition 1.2 (Subspaces, kernels, images, and quotients)**.** A *subspace* of V is a subset closed under linear combinations. For a linear map $T : V \to W$, define

$$\ker T = \{v : T(v) = 0\}, \qquad \operatorname{im} T = \{T(v) : v \in V\}.$$

For a subspace $U \subseteq V$, the *quotient* V/U is the vector space of cosets $v + U$. The *direct sum* $V \oplus W$ is $V \times W$ with componentwise operations.

Exercise 1.2 (Kernels, images, and quotients)**.** Prove that $\ker T$ and $\operatorname{im} T$ are subspaces. Prove that a linear map $f : V \to X$ factors uniquely through V/U exactly when $U \subseteq \ker f$. Deduce $V/\ker T \cong \operatorname{im} T$.

Exercise 1.3 (Direct sums and isomorphism theorems)**.** Prove the product and coproduct universal properties of $V \oplus W$. For subspaces $A, B \subseteq V$, prove $A/(A \cap B) \cong (A + B)/B$. Deduce the remaining vector-space isomorphism theorems by applying this identity to nested subspaces.

Definition 1.3 (Duals and pairings)**.** The *dual* of V is $V^* = \operatorname{Hom}_k(V, k)$. A *pairing* of V and W is a bilinear map $V \times W \to k$. It is *nondegenerate* if each nonzero argument pairs nontrivially with some argument in the other space.

Definition 1.4 (Tensor products)**.** A *tensor product* of V and W is a vector space $V \otimes W$ with a bilinear map $(v, w) \mapsto v \otimes w$ such that every bilinear map $b : V \times W \to X$ factors uniquely through a linear map $\tilde{b} : V \otimes W \to X$. Tensor powers and multilinear maps are defined by iteration. Such a space exists: let $k^{(V \times W)}$ be the free vector space on symbols $[v, w]$, and quotient by the span of

$$[v + v', w] - [v, w] - [v', w], \qquad [v, w + w'] - [v, w] - [v, w'],$$
$$[av, w] - a[v, w], \qquad [v, aw] - a[v, w]$$

for $a \in k$. The class of $[v, w]$ is $v \otimes w$, and sending $[v, w]$ to $b(v, w)$ proves the universal property. It also shows that any two tensor products are uniquely isomorphic in a way preserving pure tensors.

Definition 1.5 (Finite-dimensional duality, trace, and dimension)**.** The *evaluation* map is

$$\mathrm{ev}_V : V^* \otimes V \longrightarrow k, \qquad \varphi \otimes v \longmapsto \varphi(v).$$

The vector space V is *finite-dimensional* if there is a coevaluation map $\mathrm{coev}_V : k \to V \otimes V^*$ satisfying the snake identities

$$(\mathrm{id}_V \otimes \mathrm{ev}_V) \circ (\mathrm{coev}_V \otimes \mathrm{id}_V) = \mathrm{id}_V,$$

$$(\mathrm{ev}_V \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{V^*} \otimes \mathrm{coev}_V) = \mathrm{id}_{V^*}.$$

Let $s_{V,V^*} : V \otimes V^* \to V^* \otimes V$ exchange the tensor factors. For $T \in \mathrm{End}(V)$, define

$$\mathrm{tr}(T) = \mathrm{ev}_V \circ s_{V,V^*} \circ (T \otimes \mathrm{id}_{V^*}) \circ \mathrm{coev}_V \in k, \qquad \dim V = \mathrm{tr}(\mathrm{id}_V).$$

Exercise 1.4 (Duality, trace, and dimension)**.** Prove directly from the snake identities that the coevaluation is unique and that dualizable vector spaces are closed under direct sums, tensor products, and duals. For finite-dimensional V, W, prove $\mathrm{tr}(gf) = \mathrm{tr}(fg)$ for maps $f : V \to W$ and $g : W \to V$, and prove

$$\dim(V \oplus W) = \dim V + \dim W, \qquad \dim(V \otimes W) = \dim V \, \dim W.$$

Carry out these proofs using evaluation, coevaluation, symmetry, and their universal properties, without choosing coordinates.

Exercise 1.5 (Duality and multilinearity)**.** For finite-dimensional spaces, construct the natural maps

$$V^* \otimes W \longrightarrow \mathrm{Hom}(V, W), \qquad V^* \otimes W^* \longrightarrow (V \otimes W)^*.$$

Prove that they are isomorphisms, identify the evaluation pairing, and prove the tensor–hom adjunction $\mathrm{Hom}(U \otimes V, W) \cong \mathrm{Hom}(U, \mathrm{Hom}(V, W))$.

Definition 1.6 (Exterior powers, determinant, and characteristic polynomial)**.** The *tensor algebra* and *exterior algebra* of V are

$$T(V) = \bigoplus_{r \geq 0} V^{\otimes r}, \qquad \Lambda V = T(V)/\langle v \otimes v : v \in V \rangle;$$

write $\Lambda^r V$ for the degree-r component. For finite-dimensional V, define its *rank* to be the unique integer $n \geq 0$ such that $\Lambda^n V \neq 0$ and $\Lambda^{n+1} V = 0$; the exercise below proves that this integer exists and that $\Lambda^n V$ is one-dimensional. For $T \in \mathrm{End}(V)$, the induced map $\Lambda^n T$ is multiplication by a scalar, the *determinant* $\det(T)$. The *characteristic polynomial* is

$$\chi_T(t) = \det(t \, \mathrm{id}_V - T).$$

Its root multiplicity at λ is the *algebraic multiplicity* of λ; the dimension of $\ker(T - \lambda \, \mathrm{id}_V)$ is its *geometric multiplicity.*

Exercise 1.6 (Top exterior powers and determinant identities)**.** Using only the universal property of alternating maps and the duality maps of V, prove that the rank n exists, that $\Lambda^n V$ is one-dimensional, and that $\Lambda^r V = 0$ for $r > n$. Prove

$$\det(ST) = \det(S)\det(T), \qquad \det(T \oplus R) = \det(T)\det(R).$$

Relate the scalar dimension to rank by $\dim V = (\operatorname{rank} V)1_k$ and explain why the scalar dimension alone need not distinguish ranks in positive characteristic. For $V = \mathbb{R}^n$, identify $|\det T|$ with the volume-scaling factor and the sign with orientation. Deduce the change-of-variables rule for parallelepipeds and Cramer's rule.

In finite dimensions, an inner product turns linear superposition into a geometry of orthogonal modes and makes spectral decomposition possible.

1.2 Inner products and finite-dimensional spectral theory

Definition 1.7 (Inner products and orthogonality)**.** An *inner product* on a real vector space is a positive-definite symmetric bilinear form. On a complex vector space it is a positive-definite sesquilinear form, conjugate-linear in the first variable. Write $\|v\| = \sqrt{\langle v, v\rangle}$. Vectors are *orthogonal* when their inner product is zero, and $U^\perp = \{v : \langle u, v\rangle = 0 \text{ for every } u \in U\}$.

Definition 1.8 (Projections and adjoints)**.** A *projection* is an endomorphism P with $P^2 = P$. An *orthogonal projection* is a projection with $\ker P = (\operatorname{im} P)^\perp$. The *adjoint* of $T : V \to W$ is a map $T^* : W \to V$ satisfying $\langle Tv, w\rangle = \langle v, T^* w\rangle$.

Exercise 1.7 (Orthogonal decomposition and best approximation)**.** Let V be finite-dimensional and $U \subseteq V$. Prove $V = U \oplus U^\perp$, construct the orthogonal projection P_U, and prove that $P_U v$ is the unique element of U minimizing $\|v - u\|$. Prove that P is an orthogonal projection if and only if $P = P^* = P^2$.

Definition 1.9 (Isometries and spectral classes)**.** A linear map is *orthogonal* over $\mathbb{R}$, or *unitary* over $\mathbb{C}$, if it preserves the inner product. An endomorphism A is *self-adjoint* if $A = A^*$, *normal* if $AA^* = A^*A$, and *positive* if $A = A^*$ and $\langle v, Av\rangle \geq 0$ for every v. A scalar λ is an *eigenvalue* if $\ker(A - \lambda \operatorname{id}) \neq 0$; a subspace U is *invariant* if $A(U) \subseteq U$. In finite dimensions, $\operatorname{spec}(A)$ denotes the set of eigenvalues of A in the scalar field.

Definition 1.10 (Complete commuting spectral projections)**.** A family (P_λ) of commuting orthogonal projections is *complete and mutually orthogonal* if

$$P_\lambda P_\mu = \delta_{\lambda\mu} P_\lambda, \qquad \sum_\lambda P_\lambda = \operatorname{id}.$$

A *spectral decomposition* of A is such a family satisfying

$$A = \sum_{\lambda \in \operatorname{spec}(A)} \lambda P_\lambda.$$

Exercise 1.8 (Finite-dimensional spectral theorem)**.** Prove that adjoints exist uniquely. Prove that a complex endomorphism is normal if and only if it admits a complete family of mutually orthogonal commuting spectral projections. Deduce the reality of self-adjoint spectra, the unit modulus of unitary spectra, and nonnegative spectra for positive operators. Prove that every positive operator A has a unique positive square root $A^{1/2}$ by applying the square-root function to its spectral projections. For a map $T : V \to W$ between finite-dimensional inner-product spaces, let (Q_s) be the complete spectral projections of $|T| = (T^*T)^{1/2}$. For each $s > 0$, construct an isometric isomorphism $U_s : Q_s V \to T(Q_s V)$ satisfying $TQ_s = sU_s Q_s$, and obtain the coordinate-free singular-value decomposition

$$T = \sum_{s>0} sU_s Q_s.$$

Exercise 1.9 (Commuting normal operators)**.** Prove that every spectral projection of a normal operator is a polynomial in that operator and its adjoint. For a finite commuting family of normal operators, construct a single complete family of mutually orthogonal commuting projections (P_α) and scalars $\lambda_j(\alpha)$ such that

$$A_j = \sum_\alpha \lambda_j(\alpha) P_\alpha$$

for every member A_j of the family. Interpret $P_\alpha V$ as the joint spectral subspaces without choosing coordinates.

The finite spectral theorem turns coupled equations of motion into independent normal modes. A chain of masses provides a finite approximation to the same decomposition for a continuous string.

1.3 Oscillators, normal modes, and vibrating strings

Definition 1.11 (Linear oscillator)**.** A *linear oscillator* on a finite-dimensional real inner-product space V is an equation

$$M\ddot{q} + Kq = 0,$$

where M is positive definite and K is self-adjoint and positive semidefinite. A *normal mode* is a nonzero $v \in V$ for which $q(t) = a(t)v$ solves the equation for some nonconstant a.

Exercise 1.10 (The harmonic oscillator)**.** Solve $m\ddot{q} + kq = 0$ for $m, k > 0$. Prove conservation of $E = \frac{1}{2}m\dot{q}^2 + \frac{1}{2}kq^2$, determine the period, and deduce from the measured period of a small-amplitude mass–spring system the ratio k/m.

Exercise 1.11 (Critical damping and a repeated root)**.** For $m, b, \kappa > 0$, apply the exponential ansatz to $m\ddot{x} + b\dot{x} + \kappa x = 0$ and solve the equation in the underdamped, overdamped, and critically damped regimes. Obtain the second critical solution $te^{-bt/(2m)}$ directly from the repeated-root limit. Show that critical damping is exactly $b^2 = 4m\kappa$. Among $b^2 \geq 4m\kappa$, show that the critical value maximizes the magnitude of the slow decay rate, explaining the fastest asymptotic return without oscillatory eigenvalues.

Exercise 1.12 (Coupled oscillators and normal modes)**.** Set $x = M^{1/2}q$ and $L = M^{-1/2}KM^{-1/2}$. Use the spectral projections of L to derive the normal-mode decomposition. For a fixed-end chain of N equal masses joined by equal springs, compute the mode shapes and frequencies and relate them to the modes of a taut string.

Definition 1.12 (Vibrating string)**.** For length L, tension τ, and linear density ρ, a *vibrating string* is a function $u : [0, L] \times \mathbb{R} \to \mathbb{R}$ satisfying fixed-end conditions $u(0, t) = u(L, t) = 0$ and

$$\rho\, \partial_t^2 u = \tau\, \partial_x^2 u.$$

Exercise 1.13 (String modes and measured pitch)**.** Derive the vibrating-string equation from transverse force balance in the small-slope limit. Determine all separated normal modes and their frequencies. Prove that the fundamental frequency is $f_1 = (2L)^{-1}\sqrt{\tau/\rho}$ and compare the predicted ratios $f_n/f_1 = n$ with the harmonics of a stretched string.

A continuous string requires infinite-dimensional spaces of functions. Measure and integration produce the spaces L^p, and completeness turns L^2 into a Hilbert space.

1.4 Measure, integration, and Hilbert spaces

Definition 1.13 (Measure spaces and simple functions)**.** A *measure space* is a triple (X, Σ, μ) in which Σ is a σ-algebra and $\mu : \Sigma \to [0, \infty]$ is countably additive on disjoint families, with $\mu(\varnothing) = 0$. A function is *measurable* when inverse images of Borel sets are in Σ. A nonnegative *simple function* is a finite sum $\sum_j a_j 1_{E_j}$ with $a_j \geq 0$ and $E_j \in \Sigma$.

Definition 1.14 (L^p spaces)**.** For $1 \leq p < \infty$, the space $L^p(X, \mu)$ consists of measurable functions modulo equality almost everywhere with

$$\|f\|_p = \left(\int_X |f|^p \, d\mu \right)^{1/p} < \infty;$$

L^∞ uses the essential supremum. For Lebesgue measure on $\mathbb{R}^n$ and normalized arc measure on $\mathbb{T}$, write $L^p(\mathbb{R}^n)$ and $L^p(\mathbb{T})$. The inner product on $L^2(\mathbb{T})$ is $\langle f, g \rangle = (2\pi)^{-1} \int_0^{2\pi} \overline{f(x)} g(x) \, dx$. The exercises below establish the analytic properties used in this section.

Exercise 1.14 (Integration and monotone convergence)**.** Define the integral of a nonnegative simple function and then of an arbitrary nonnegative measurable function by monotone approximation. Prove monotone convergence and Fatou's lemma.

Exercise 1.15 (Dominated convergence and product measures)**.** Construct the integral of an integrable complex function from its positive and negative real and imaginary parts. Prove dominated convergence. Construct product measure for σ-finite spaces and prove Tonelli's and Fubini's theorems.

Exercise 1.16 (L^p inequalities)**.** Prove Hölder's and Minkowski's inequalities for $1 \leq p < \infty$.

Exercise 1.17 (L^p completeness and approximation)**.** Prove completeness of L^p for $1 \leq p < \infty$. For Lebesgue measure, prove density first of simple functions with finite-measure support and then of $C_c(\mathbb{R}^n)$. Deduce that the inner-product norm makes L^2 complete.

Definition 1.15 (Hilbert spaces and bounded operators)**.** From this point onward, Hilbert spaces are complex. A *Hilbert space* is a complete inner-product space. A linear map $T : H \to K$ is *bounded* if $\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty$. Write $B(H, K)$ for the space of bounded maps and $B(H) = B(H, H)$. For $T \in B(H)$, the *resolvent set* consists of $\lambda \in \mathbb{C}$ for which $T - \lambda \,\mathrm{id}$ has a bounded inverse; its complement is the *spectrum* $\operatorname{spec}(T)$.

Exercise 1.18 (Hilbert-space foundations)**.** Prove the projection theorem for closed subspaces and the Riesz representation theorem. Deduce $H = M \oplus M^\perp$ for closed M and prove existence, uniqueness, and $\|T^*\| = \|T\|$ for bounded adjoints.

Orthogonal bases turn infinite-dimensional functions into sequences of modes. On a finite string this gives Fourier series; on an unbounded domain, the mode parameter becomes continuous and the series becomes a transform.

Definition 1.16 (Orthogonal expansions)**.** Let H be an inner-product space. For an orthogonal family (e_n), the *orthogonal coefficients* of f are $\widehat{f}(n) = \langle e_n, f \rangle / \|e_n\|^2$. A *Fourier series* on $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ is the formal expansion $f \sim \sum_{n \in \mathbb{Z}} \widehat{f}(n) e^{inx}$, where $\widehat{f}(n) = (2\pi)^{-1} \int_0^{2\pi} f(x) e^{-inx} \, dx$.

Exercise 1.19 (Bessel and Parseval criteria)**.** Prove Bessel's inequality for every finite orthogonal family. Prove that an orthonormal family (e_n) is complete precisely when

$$\|f\|^2 = \sum_n |\langle e_n, f \rangle|^2$$

for every f.

Exercise 1.20 (Completeness of Fourier modes)**.** Construct the Fejér kernels, prove that they form an approximate identity, and use convolution with them to prove completeness of $(e^{inx})_{n\in\mathbb{Z}}$ in $L^2(\mathbb{T})$. Deduce Parseval's identity and L^2 convergence of Fourier series.

Exercise 1.21 (Spectral evolution of a string)**.** For smooth fixed-end initial displacement and velocity satisfying the endpoint compatibility conditions, expand the data in sine modes and derive the classical solution of the wave equation. Prove uniqueness and conservation of wave energy from the spectral coefficients.

Exercise 1.22 (Spectral evolution of heat)**.** For smooth fixed-end initial temperature, use the same sine modes to derive the heat-equation solution $\partial_t u = \kappa \partial_x^2 u$. Prove uniqueness and monotone decay of L^2 energy from the spectral coefficients.

1.5 Fourier analysis and spectral evolution

Definition 1.17 (Schwartz functions, Fourier transform, and convolution)**.** A *Schwartz function* on $\mathbb{R}^n$ is a smooth function f such that $\sup_x |x^\alpha \partial^\beta f(x)| < \infty$ for every pair of multi-indices α, β. The Schwartz space is denoted $\mathcal{S}(\mathbb{R}^n)$. For $f, g \in C_c(\mathbb{R}^n)$, define the *Fourier transform* and *convolution* by

$$\mathcal{F}f(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} f(x) e^{-ix\cdot\xi}\, dx, \qquad (f * g)(x) = \int_{\mathbb{R}^n} f(y) g(x-y)\, dy.$$

The estimates $\|\mathcal{F}f\|_\infty \le (2\pi)^{-n/2}\|f\|_1$ and $\|f * g\|_1 \le \|f\|_1 \|g\|_1$ extend these operations to the L^1 completion.

Exercise 1.23 (Fourier inversion and convolution)**.** For Schwartz functions, prove Fourier inversion and $\mathcal{F}(f * g) = (2\pi)^{n/2}(\mathcal{F}f)(\mathcal{F}g)$.

Exercise 1.24 (Plancherel theorem)**.** Prove on $\mathcal{S}(\mathbb{R}^n)$ that $\mathcal{F}$ preserves the L^2 inner product. Prove that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ and extend $\mathcal{F}$ uniquely to a unitary operator and deduce Plancherel's identity.

Exercise 1.25 (Wave and heat propagators)**.** Use the Fourier transform to solve the Cauchy problems for

$$\partial_t^2 u = c^2 \Delta u, \qquad \partial_t u = \kappa \Delta u$$

on $\mathbb{R}^n$. Identify their spectral-mode evolution, derive the Gaussian heat kernel and its convolution law, and prove finite propagation speed for the one-dimensional wave equation.

1.6 Time–bandwidth concentration and Slepian functions

Real measurements observe a pulse only during a finite time window and through a receiver with finite bandwidth. No nonzero signal can be both exactly time-limited and band-limited, so the useful question is quantitative: among signals obeying one cutoff, which place the largest fraction of their energy inside the other? The answer turns simultaneous localization into an eigenvalue problem.

Definition 1.18 (Time and band limiting)**.** On $L^2(\mathbb{R})$, define the time-limiting and band-limiting projections by

$$(Q_T f)(x) = \mathbf{1}_{[-T,T]}(x) f(x), \qquad B_W = \mathcal{F}^{-1}\mathbf{1}_{[-W,W]}\mathcal{F},$$

where $T, W > 0$. The *time–band concentration operator* is

$$K_{T,W} = Q_T B_W Q_T.$$

After identifying $Q_T L^2(\mathbb{R})$ with $L^2([-T,T])$, its normalized eigenfunctions are the time-limited representatives of the *prolate spheroidal wave functions*, also called *Slepian functions*. The dimensionless parameter $c = TW$ is the time–bandwidth product.

Exercise 1.26 (Optimal simultaneous concentration)**.** Use Plancherel's theorem to prove that Q_T and B_W are orthogonal projections. Starting from the Fourier-transform convention above, derive

$$(K_{T,W} f)(x) = \int_{-T}^{T} \frac{\sin W(x-y)}{\pi(x-y)} f(y)\, dy, \qquad |x| \leq T,$$

where the diagonal value of the kernel is W/π. Prove that $K_{T,W}$ is a positive compact self-adjoint contraction. If its positive eigenvalues are $\lambda_0 \geq \lambda_1 \geq \cdots > 0$, use the Rayleigh–Ritz principle to show

$$\lambda_0 = \sup_{\substack{f \neq 0 \\ \operatorname{supp} f \subseteq [-T,T]}} \frac{\displaystyle\int_{-W}^{W} |\widehat{f}(\xi)|^2\, d\xi}{\displaystyle\int_{\mathbb{R}} |f(x)|^2\, dx}.$$

Show likewise that $B_W Q_T B_W$ has the same nonzero eigenvalues and that, for a band-limited eigenfunction g_n, λ_n is the fraction of its energy lying in $[-T,T]$. Thus successive eigenfunctions solve the optimal concentration problem subject to orthogonality to the preceding solutions.

Exercise 1.27 (The prolate equation and the Shannon number)**.** Rescale $x = Tt$ to $[-1,1]$ and write the concentration kernel as a function of $c = TW$. Verify, by differentiating the kernel and integrating by parts, that its integral operator commutes with the Sturm–Liouville operator

$$\mathcal{L}_c = -\frac{d}{dt}\left((1-t^2)\frac{d}{dt}\right) + c^2 t^2.$$

Conclude that the Slepian functions may be chosen to satisfy the prolate spheroidal equation

$$-\frac{d}{dt}\Big((1-t^2)\psi_n'(t)\Big) + c^2 t^2 \psi_n(t) = \chi_n \psi_n(t), \qquad -1 < t < 1,$$

with the solutions bounded at the endpoints. Recover the Legendre equation when $c = 0$.

Use the diagonal of the sinc kernel to prove the exact trace identity

$$\sum_{n \geq 0} \lambda_n = \operatorname{Tr} K_{T,W} = \frac{2TW}{\pi}.$$

Deduce that at most $2TW/(\pi\alpha)$ mutually orthogonal modes can have concentration at least α. Interpret $2TW/\pi$ as the effective number of well-concentrated signal degrees of freedom available within the time and frequency windows; see Slepian–Pollak, *Prolate Spheroidal Wave Functions, Fourier Analysis and Uncertainty—I* and Landau–Pollak, *Prolate Spheroidal Wave Functions, Fourier Analysis and Uncertainty—II*, Bell System Technical Journal **40** (1961).

The concentration operator composes a localization projection with a spectral projection. The same mechanism will reappear in Index Theory, where a Dirac spectral cutoff compresses an algebra of observables to an operator system and turns finite resolution into geometric data.

Fourier series and transforms are concrete spectral decompositions. The spectral theorem isolates their common structure and replaces sums over modes by integration against projection-valued measures.

1.7 Spectral measures and unbounded operators

Definition 1.19 (Projection-valued measures)**.** A *projection-valued measure* on a measurable space X assigns an orthogonal projection $E(B)$ to each measurable set, with $E(X) = \mathrm{id}$ and countable additivity in the strong operator topology. For bounded measurable f, the operator $\int_X f\,dE$ is defined by uniform approximation from simple functions.

Exercise 1.28 (Bounded spectral theorem)**.** Prove that every bounded self-adjoint operator A has a unique projection-valued measure on $\mathrm{spec}(A) \subset \mathbb{R}$ such that $A = \int \lambda\,dE(\lambda)$. Extend the statement to bounded normal operators, derive continuous functional calculus, and characterize unitary and projection spectra.

The differential operators governing waves and translations are not bounded. To extend spectral theory to them, the domain must become part of the operator.

Definition 1.20 (Unbounded operators and domains)**.** A *densely defined operator* T has a dense linear domain $\mathrm{Dom}(T) \subset H$. It is *closed* when its graph is closed in $H \oplus H$; its graph norm is $\|x\|_T^2 = \|x\|^2 + \|Tx\|^2$. The domain of T^* consists of those y for which $x \mapsto \langle Tx, y\rangle$ is bounded in the Hilbert norm on $\mathrm{Dom}(T)$; for such y, the Riesz theorem gives the unique T^*y satisfying $\langle Tx, y\rangle = \langle x, T^*y\rangle$. The operator is *symmetric* when $\langle Tx, y\rangle = \langle x, Ty\rangle$ on its domain and *self-adjoint* when $T = T^*$, including equality of domains.

Definition 1.21 (Weak derivatives and first Sobolev spaces)**.** A function $g \in L^1_{\mathrm{loc}}(0, L)$ is the *weak derivative* of $f \in L^1_{\mathrm{loc}}(0, L)$ when

$$\int_0^L f\phi'\,dx = -\int_0^L g\phi\,dx \qquad (\phi \in C_c^\infty(0, L)).$$

The space $H^1(0, L)$ consists of the $f \in L^2(0, L)$ having weak derivative $f' \in L^2(0, L)$, with norm $\|f\|_{H^1}^2 = \|f\|_2^2 + \|f'\|_2^2$.

Exercise 1.29 (Completeness and representatives of H^1)**.** Prove that $H^1(0, L)$ is complete. Prove that every class has a unique absolutely continuous representative and that its classical derivative agrees almost everywhere with its weak derivative.

Exercise 1.30 (Traces and H_0^1)**.** Prove that endpoint evaluation on the absolutely continuous representative is continuous, establish integration by parts, and identify $H_0^1(0, L)$ as the closure of $C_c^\infty(0, L)$ and as the subspace with zero endpoint values.

Exercise 1.31 (Self-adjoint momentum and boundary holonomy)**.** On $L^2([0, L])$, let $P_\theta = -i\hbar\,d/dx$ on the absolutely continuous functions with derivative in L^2 and boundary condition $\psi(L) = e^{i\theta}\psi(0)$. Use integration by parts and the adjoint domain to prove that P_θ is self-adjoint. Determine its mutually orthogonal spectral projections $(E_n)_{n\in\mathbb{Z}}$, prove that they sum strongly to the identity, and show that their spectral values are

$$p_n = \frac{\hbar}{L}(2\pi n + \theta), \qquad n \in \mathbb{Z}.$$

Interpret θ as a boundary holonomy, later realized by magnetic flux through a ring, and explain why changing the domain changes the measured spectrum.

Exercise 1.32 (Cayley transform and spectral measure)**.** Use the Cayley transform and the bounded spectral theorem to prove that every self-adjoint H has a unique projection-valued measure with

$$H = \int_{\mathbb{R}} \lambda\,dE_H(\lambda), \qquad \mathrm{Dom}(H) = \left\{\psi : \int_{\mathbb{R}} \lambda^2\,d\langle\psi, E_H(\lambda)\psi\rangle < \infty\right\}.$$

Exercise 1.33 (Measurable functional calculus)**.** For a self-adjoint H with spectral measure E_H, define $f(H)$ first for bounded measurable f and then on the domain

$$\left\{\psi : \int_{\mathbb{R}} |f(\lambda)|^2 \, d\langle\psi, E_H(\lambda)\psi\rangle < \infty\right\}.$$

Prove the algebraic, adjoint, and composition rules wherever the natural domains make them meaningful.

Exercise 1.34 (Stone's theorem)**.** Prove that $U_t = e^{-itH/\hbar}$ is a strongly continuous unitary group for every self-adjoint H. Conversely, recover a unique self-adjoint generator H from every strongly continuous unitary group and identify its domain by the strong derivative at $t = 0$.

Exercise 1.35 (Fourier spectra and free quantum evolution)**.** Use the Fourier transform to realize $-i\hbar\partial_j$ and $-\Delta$ as multiplication by $\hbar\xi_j$ and $|\xi|^2$ on their natural domains. Deduce self-adjointness, identify their spectral projections, and derive the free Schrödinger evolution for $H = -(\hbar^2/2m)\Delta$. Prove preservation of the domain, norm, and expected energy, and use this concrete multiplication-operator model to verify the preceding spectral and Stone theorems, including their domain statements. Identify the continuous energy spectrum and relate it to the continuum of outgoing-electron energies above an ionization threshold.

Exercise 1.36 (Spectral lines from unitary evolution)**.** If $H|n\rangle = E_n|n\rangle$, use $U_t = e^{-itH/\hbar}$ to show that $\langle A(t)\rangle$ contains frequencies $(E_m - E_n)/\hbar$. Explain why spectroscopic peaks measure energy differences and why line strengths depend on matrix elements between the corresponding spectral subspaces. Convert the Balmer wavelengths 656.3, 486.1, 434.0, and 410.2 nm to energy differences using $E = hc/\lambda$. Identify the levels E_n, rather than their differences, with atoms of the spectral measure of H, and identify the transition frequencies with spectral values of the Liouvillian $A \mapsto [H, A]/\hbar$.

2 Representation Theory

Experimental observations.

- Symmetric molecules can have distinct vibrational modes with the same frequency. Replacing one atom can lower the symmetry and split the corresponding spectral lines.

Spectral theory decomposes a Hamiltonian into energy subspaces, but by itself does not explain why a degeneracy persists or why some transitions are absent. A symmetry commuting with the Hamiltonian preserves each energy subspace; its representation there organizes degeneracies, while the symmetry type of an observable constrains its matrix elements.

2.1 Groups, actions, and linear representations

Definition 2.1 (Groups, homomorphisms, and quotients)**.** A *group* is a set G with an associative multiplication, an identity, and inverses. A *homomorphism* $\phi : G \to H$ preserves multiplication. A subgroup $N \leq G$ is *normal* if $gNg^{-1} = N$ for every $g \in G$; the *quotient group* G/N consists of cosets with induced multiplication.

Exercise 2.1 (The group isomorphism theorems)**.** Prove that kernels are normal and that a normal subgroup is the kernel of the quotient homomorphism. Prove the three isomorphism theorems for groups and the universal property of G/N.

Definition 2.2 (Actions, orbits, and stabilizers)**.** A *left action* of G on X is a homomorphism $G \to \mathrm{Bij}(X)$. The *orbit* and *stabilizer* of $x \in X$ are $Gx = \{gx : g \in G\}$ and $G_x = \{g : gx = x\}$.

Exercise 2.2 (Orbit–stabilizer and class equation)**.** Construct a natural bijection $G/G_x \to Gx$. For finite G, deduce the orbit–stabilizer formula and the class equation from the conjugation action. Use these results to prove that every group of prime order is cyclic.

Definition 2.3 (Representations and intertwiners)**.** A *representation* of G on V is a homomorphism $\rho : G \to \mathrm{GL}(V)$. An *intertwiner* $T : (V, \rho) \to (W, \sigma)$ satisfies $T\rho(g) = \sigma(g)T$. A subspace is *invariant* if it is preserved by every $\rho(g)$; a nonzero representation is *irreducible* if it has no proper nonzero invariant subspace. Direct sums and tensor products carry the actions $\rho \oplus \sigma$ and $g \mapsto \rho(g) \otimes \sigma(g)$.

Exercise 2.3 (Maschke's theorem)**.** Let G be finite and let the characteristic of k not divide $|G|$. Average an arbitrary projection over G to construct an invariant complement to every invariant subspace. Deduce that every finite-dimensional representation is a direct sum of irreducible representations.

Exercise 2.4 (Schur's lemma)**.** Prove that a nonzero intertwiner between irreducible complex finite-dimensional representations is an isomorphism and that every endomorphism of an irreducible representation is scalar. Deduce uniqueness of irreducible multiplicities in a decomposition.

Definition 2.4 (Characters)**.** The *character* of a finite-dimensional representation V is the class function $\chi_V(g) = \mathrm{Tr}(\rho_V(g))$. For class functions on a finite group, define

$$\langle f, h \rangle_G = \frac{1}{|G|} \sum_{g \in G} \overline{f(g)} h(g).$$

Exercise 2.5 (Character orthogonality and decomposition)**.** Use averaging of linear maps and Schur's lemma to prove orthonormality of the irreducible characters. Prove that they form a complete orthonormal family in the class functions and that $\langle \chi_U, \chi_V \rangle_G = \dim \mathrm{Hom}_G(U, V)$. Deduce the multiplicity formula and the character table constraints.

Exercise 2.6 (Symmetry projections)**.** For an irreducible character χ of a finite group, prove that

$$P_\chi = \frac{\dim \chi}{|G|} \sum_{g \in G} \overline{\chi(g)} \rho(g)$$

is the projection onto the χ-isotypic subspace. Apply these projections to the displacement space of a molecule whose equilibrium configuration is preserved by G.

Exercise 2.7 (Molecular symmetry)**.** For the equilibrium geometry of NH_3, determine the C_{3v}-action on nuclear displacements, remove translations and rotations, and prove

$$V_{\mathrm{vib}} \cong 2A_1 \oplus 2E.$$

Deduce four fundamental frequencies, two of them doubly degenerate. For NH_2D, restrict to C_s and prove $E|_{C_s} \cong A' \oplus A''$. Deduce that symmetry no longer enforces either degeneracy and compare with the observed band splitting.

The molecular examples use finite groups. Rotations vary continuously, so their symmetry groups require compatible topological and smooth structures.

2.2 Topological groups and infinitesimal symmetry

Definition 2.5 (Topological spaces)**.** A *topology* on X is a family of subsets containing $\varnothing, X$ and closed under arbitrary unions and finite intersections. A map is *continuous* when inverse images of open sets are open. A space is *Hausdorff* when distinct points have disjoint neighborhoods, *second countable* when its topology is generated by a countable family of open sets, and *compact* when every open cover has a finite subcover.

Definition 2.6 (Topological and smooth manifolds)**.** A *topological manifold* of dimension n is a Hausdorff, second-countable space locally homeomorphic to $\mathbb{R}^n$. A *smooth manifold* is a topological manifold with a maximal atlas whose transition maps are smooth. A map of smooth manifolds is *smooth* when its chart representatives are smooth.

Definition 2.7 (Tangent and cotangent spaces)**.** The *tangent space* T_pM is the vector space of derivations $v : C^\infty(M) \to \mathbb{R}$ at p, satisfying $v(fg) = f(p)v(g) + g(p)v(f)$. The *cotangent space* is $T_p^*M = (T_pM)^*$. The differential of $F : M \to N$ is $dF_p(v)(f) = v(f \circ F)$. The disjoint union $TM = \coprod_{p \in M} T_pM$ has a canonical smooth vector-bundle structure: if $x = (x^1, \ldots, x^n)$ is a coordinate chart on U, its bundle chart is

$$TM|_U \longrightarrow x(U) \times \mathbb{R}^n, \qquad v \in T_pM \longmapsto (x(p), v(x^1), \ldots, v(x^n)).$$

The chain rule makes the transition maps smooth and fiberwise linear. The dual bundle charts similarly define the cotangent bundle T^*M.

Exercise 2.8 (Intrinsic tangent vectors)**.** Prove that derivations at p, equivalence classes of smooth curves through p, and the usual coordinate tangent vectors are naturally isomorphic. Prove the chain rule $d(G \circ F)_p = dG_{F(p)} \circ dF_p$ and compute tangent spaces of level sets $F^{-1}(q)$ at points where dF is surjective.

Definition 2.8 (Vector fields and flows)**.** A *vector field* is a smooth section $X : M \to TM$. A *flow* of X is a smooth local action Φ of $\mathbb{R}$ on M satisfying $\frac{d}{dt}\big|_0 \Phi_t(p) = X_p$. The *Lie bracket* is the vector field $[X, Y](f) = X(Yf) - Y(Xf)$.

Exercise 2.9 (Flows and brackets)**.** Prove local existence and uniqueness of flows. Prove that $[X, Y] = 0$ if and only if their local flows commute, and that the Lie derivative $\mathcal{L}_X Y = [X, Y]$ is natural under diffeomorphisms.

Definition 2.9 (Lie groups and Lie algebras)**.** A *Lie group* is a smooth manifold G whose multiplication and inversion are smooth. Its *Lie algebra* is $\mathfrak{g} = T_eG$ with bracket induced by left-invariant vector fields. A *one-parameter subgroup* is a smooth homomorphism $\mathbb{R} \to G$. The *exponential map* $\exp : \mathfrak{g} \to G$ sends X to the value at 1 of the one-parameter subgroup with initial velocity X.

Exercise 2.10 (One-parameter subgroups and the exponential map)**.** Prove existence and uniqueness of the one-parameter subgroup generated by $X \in \mathfrak{g}$. For a closed matrix group, prove that it is $t \mapsto e^{tX}$, identify the commutator bracket, and compute the first nontrivial terms of the Baker–Campbell–Hausdorff formula.

Definition 2.10 (Infinitesimal representations)**.** For a finite-dimensional smooth representation $\rho : G \to \mathrm{GL}(V)$, its *infinitesimal representation* is

$$d\rho : \mathfrak{g} \to \mathrm{End}(V), \qquad d\rho(X) = \frac{d}{dt}\bigg|_0 \rho(\exp tX).$$

Exercise 2.11 (Differentiating symmetry)**.** Prove that $d\rho$ preserves brackets and that $\rho(\exp X) = \exp(d\rho(X))$. Prove that a connected Lie-group representation is determined by its infinitesimal representation whenever the latter integrates uniquely.

2.3 Rotations, spin, and quantum selection rules

Definition 2.11 (Rotations and spin)**.** Define

$$SO(3) = \{R \in \mathrm{End}(\mathbb{R}^3) : R^*R = \mathrm{id},\ \det R = 1\},$$
$$SU(2) = \{U \in \mathrm{End}(\mathbb{C}^2) : U^*U = \mathrm{id},\ \det U = 1\}.$$

An *angular-momentum representation* is a unitary representation of $SU(2)$; with Planck constant $\hbar$, its infinitesimal generators J_1, J_2, J_3 are normalized by $[J_i, J_j] = i\hbar\varepsilon_{ijk}J_k$.

Exercise 2.12 ($SU(2)$ double-covers $SO(3)$)**.** Let $SU(2)$ act by conjugation on the real space of traceless Hermitian 2×2 operators. Prove that the resulting homomorphism $SU(2) \to SO(3)$ is surjective with kernel $\{\pm\,\mathrm{id}\}$. Deduce the effect of a 2π rotation in integer- and half-integer-spin representations and the 4π restoration observed by neutron interferometry.

Exercise 2.13 (Irreducible representations of $SU(2)$)**.** Using raising and lowering operators, prove that every finite-dimensional irreducible smooth complex representation is indexed by $j \in \{0, \frac{1}{2}, 1, \ldots\}$, has dimension $2j+1$, and has J_3-eigenvalues $m\hbar$ for $m = -j, -j+1, \ldots, j$. Compute the spectrum of $J^2 = J_1^2 + J_2^2 + J_3^2$. For $j = \frac{1}{2}$, relate the two J_3 outcomes to the two Stern–Gerlach beams.

Exercise 2.14 (Clebsch–Gordan decomposition)**.** Prove

$$V_{j_1} \otimes V_{j_2} \cong \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} V_j.$$

Construct coupled angular-momentum eigenstates from highest-weight vectors and compute the singlet and triplet decomposition of $V_{1/2} \otimes V_{1/2}$.

Definition 2.12 (Quantum symmetries)**.** Let a compact group G act continuously and unitarily by $U : G \to U(H)$ on a Hilbert space with self-adjoint Hamiltonian $\mathcal{H}$. It is a *symmetry group* when $U_g\,\mathrm{Dom}(\mathcal{H}) = \mathrm{Dom}(\mathcal{H})$ and $\mathcal{H}U_g\psi = U_g\mathcal{H}\psi$ for every $g \in G$. A *quantum number* is an eigenvalue of a self-adjoint operator whose spectral projections commute with those of $\mathcal{H}$. A bounded operator multiplet A has *symmetry type* W when its components define an intertwiner from W to $B(H)$ under conjugation by G.

Exercise 2.15 (Symmetry, degeneracy, and selection rules)**.** Prove that every finite-dimensional eigenspace of $\mathcal{H}$ is symmetry-invariant and decomposes into irreducible G-representations. Deduce symmetry-forced degeneracies. Prove that a transition matrix element $\langle f, Ai\rangle$ can be nonzero only if the trivial representation occurs in $V_f^* \otimes W \otimes V_i$, and derive the angular-momentum rules for an electric-dipole operator.

Exercise 2.16 (Atomic spectra)**.** For the Coulomb Hamiltonian, use rotational symmetry and separation into irreducible $SO(3)$-types to obtain the quantum numbers ℓ, m and their degeneracies. With the radial spectrum $E_n = -m_e e^4/(2(4\pi\varepsilon_0)^2\hbar^2 n^2)$ as derived input, compute the Balmer wavelengths and compare them with the observed hydrogen lines. Apply the electric-dipole rule to identify forbidden transitions.

3 Symplectic Geometry

Representation theory studies how transformations act on states and observables. Classical mechanics singles out a more specific question: which transformations of phase space preserve the structure that determines Hamilton's equations and Poisson brackets? In the Kleinian, or Erlangen, viewpoint, that preserved structure is an alternating form and its linear transformation group is the symplectic group.

3.1 Symplectic linear algebra and differential forms

Definition 3.1 (Symplectic vector spaces and groups)**.** A *symplectic vector space* is a finite-dimensional real vector space V equipped with a nondegenerate alternating bilinear form ω. Its *symplectic group* is

$$\mathrm{Sp}(V,\omega) = \{A \in \mathrm{GL}(V) : \omega(Au, Av) = \omega(u, v)\}.$$

For $V = \mathbb{R}^n_q \oplus \mathbb{R}^n_p$, the standard form is $\omega_0((q,p),(q',p')) = q \cdot p' - p \cdot q'$.

Exercise 3.1 (Symplectic bases and transformations)**.** Prove that every symplectic vector space has even dimension and admits a basis in which the form has matrix $\left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$. Show that A is symplectic exactly when $A^*JA = J$, and compute the Lie algebra of $\mathrm{Sp}(2n, \mathbb{R})$. Identify translations and linear symplectic maps as the transformations preserving the affine symplectic geometry of $\mathbb{R}^{2n}$.

Definition 3.2 (Differential forms and pullback)**.** A *differential k-form* on M is a smooth section of $\Lambda^k T^*M$; write $\Omega^k(M) = \Gamma(\Lambda^k T^*M)$. The *pullback* of $\omega \in \Omega^k(N)$ by $F : M \to N$ is

$$(F^*\omega)_p(v_1, \ldots, v_k) = \omega_{F(p)}(dF_p v_1, \ldots, dF_p v_k).$$

Definition 3.3 (Exterior derivative)**.** The *exterior derivative* is the unique family of maps $d : \Omega^k(M) \to \Omega^{k+1}(M)$ satisfying $df(X) = Xf$, the graded Leibniz rule, $d^2 = 0$, and naturality $F^*d = dF^*$.

Exercise 3.2 (Cartan calculus)**.** Construct d locally and prove its uniqueness. For contraction ι_X, prove Cartan's identity $\mathcal{L}_X = d\iota_X + \iota_X d$ and the naturality of d, wedge products, and pullbacks.

3.2 Hamiltonian dynamics and phase space

Definition 3.4 (Symplectic manifolds and dynamics)**.** A *symplectic form* on a manifold M is a closed, nondegenerate two-form ω. The *Hamiltonian vector field* of $H \in C^\infty(M)$ is defined by $\iota_{X_H}\omega = dH$. Its flow is the *Hamiltonian flow*, and the *Poisson bracket* is $\{f, g\} = \omega(X_f, X_g)$.

Theorem 3.5 (Darboux; used without proof)**.** *Every point of a symplectic manifold has local coordinates in which ω is the standard symplectic form ω_0.*

Definition 3.6 (Configuration and phase spaces)**.** A *configuration space* is a smooth manifold Q. Its *phase space* is the cotangent bundle T^*Q. The tautological one-form on T^*Q is $\theta_\alpha(v) = \alpha(d\pi(v))$, and its canonical symplectic form is $\omega = -d\theta$.

Definition 3.7 (Classical states and observables)**.** On a phase space M, a *pure classical state* is a point $x \in M$, and a *classical observable* is a smooth real-valued function $f \in C^\infty(M, \mathbb{R})$. A *statistical state* is a Borel probability measure μ on M, with expectation

$$\mathbb{E}_\mu[f] = \int_M f \, d\mu$$

whenever the integral exists. If ϕ_t is a Hamiltonian flow, Schrödinger evolution pushes states forward by $\mu_t = (\phi_t)_*\mu_0$, while Heisenberg evolution pulls observables back by $f_t = f \circ \phi_t$; hence $\int_M f\, d\mu_t = \int_M f_t\, d\mu_0$.

Exercise 3.3 (Hamilton's equations and Poisson algebra)**.** Prove that the canonical form on T^*Q is symplectic. In canonical coordinates, derive Hamilton's equations. Prove antisymmetry, the Leibniz rule, and the Jacobi identity for the Poisson bracket, and prove $\dot{f} = \{f, H\} + \partial_t f$ along Hamiltonian flow.

3.3 Momentum maps, reduction, and conservation laws

Definition 3.8 (Symplectic actions and momentum maps)**.** An action of a Lie group G on (M, ω) is *symplectic* if it preserves ω. A *momentum map* is an equivariant map $J : M \to \mathfrak{g}^*$ such that $d\langle J, \xi\rangle = \iota_{\xi_M}\omega$ for every $\xi \in \mathfrak{g}$.

Definition 3.9 (Symplectic reduction)**.** Let $J : M \to \mathfrak{g}^*$ be a momentum map, let ξ be a regular value, and let $G_\xi = \{g \in G : \mathrm{Ad}_g^*\, \xi = \xi\}$. If G_ξ acts freely and properly on $J^{-1}(\xi)$, the *reduced phase space* is

$$M//_\xi G = J^{-1}(\xi)/G_\xi.$$

Writing $i : J^{-1}(\xi) \hookrightarrow M$ and $q : J^{-1}(\xi) \to M//_\xi G$, its *reduced symplectic form* is the form ω_ξ characterized by $q^*\omega_\xi = i^*\omega$.

Exercise 3.4 (Noether's theorem)**.** Prove that if a Hamiltonian is invariant under a symplectic G-action with momentum map J, then every component of J is conserved. Compute the momentum maps for translations and rotations of a particle and recover linear and angular momentum conservation.

Exercise 3.5 (Reduction and central-force dynamics)**.** Under the hypotheses of symplectic reduction, prove that $i^*\omega$ is basic for the G_ξ-action and that ω_ξ exists uniquely and is symplectic. For a G-invariant Hamiltonian H, prove that X_H is tangent to $J^{-1}(\xi)$, that H descends to a Hamiltonian H_ξ, and that the flow of H_ξ is the quotient of the flow of H.

Apply this to the cotangent-lifted rotation action on $T^*(\mathbb{R}^3 \setminus \{0\})$, whose momentum map is $J(q, p) = q \times p$. For a nonzero angular momentum ℓ, identify $J^{-1}(\ell)/SO(3)_\ell$ with the radial phase space having coordinates (r, p_r), prove that its symplectic form is $dr \wedge dp_r$, and reduce

$$H(q, p) = \frac{\|p\|^2}{2m} + V(\|q\|)$$

to

$$H_\ell(r, p_r) = \frac{p_r^2}{2m} + V(r) + \frac{\|\ell\|^2}{2mr^2}.$$

For $V(r) = -\kappa/r$, derive the elliptical bound orbits and the equal-area law.

Exercise 3.6 (Gauss constraints and transverse modes)**.** Let $d : V \to W$ be a linear map between finite-dimensional real inner-product spaces. On $M = W \oplus W$ set

$$\omega((a, e), (a', e')) = \langle a, e'\rangle - \langle a', e\rangle.$$

Let the additive group $(\ker d)^\perp$ act by $\chi \cdot (A, E) = (A + d\chi, E)$. Prove that this action is free and Hamiltonian with momentum map

$$\langle J(A, E), \chi\rangle = \langle d\chi, E\rangle = \langle \chi, d^*E\rangle.$$

Show that reduction at zero imposes $d^*E = 0$ and identifies

$$M//_0(\ker d)^\perp \cong (W/\operatorname{im} d) \oplus \ker d^* \cong \ker d^* \oplus \ker d^*.$$

Interpret d as a finite-dimensional approximation to the exterior derivative: $d^*E = 0$ is the vacuum Gauss constraint, $A \sim A + d\chi$ is gauge equivalence, and the reduced variables are the transverse electromagnetic modes.

Exercise 3.7 (Liouville's theorem)**.** On a $2n$-dimensional symplectic manifold, prove that Hamiltonian flow preserves ω, the phase-space volume $\omega^n/n!$, and every density depending only on H. Deduce that Hamiltonian evolution cannot compress phase volume.

Exercise 3.8 (Betatron phase portraits)**.** Model one transverse degree of freedom in a storage ring by $H = (p^2 + \omega^2 q^2)/2$. Derive the closed elliptical phase portraits and show that their enclosed area is preserved. Interpret the motion as a betatron oscillation about a closed orbit.

4 Differential Geometry

Experimental observations.

- A changing magnetic flux through a conducting loop produces an electromotive force proportional to the rate of flux change.
- GW150914 produced a peak interferometer strain of order 10^{-21} near $150\,\mathrm{Hz}$ in two widely separated detectors.

To define geometry by its allowed transformations at each point, first assemble the local models and their changes of frame into bundles.

4.1 Bundles, sections, and geometric structures

Definition 4.1 (Smooth bundles and sections)**.** A *smooth fiber bundle* with fiber F is a smooth map $\pi : E \to M$ locally isomorphic over M to $U \times F \to U$. A *smooth vector bundle* has vector-space fibers and linear transition maps. A *smooth principal G-bundle* has a free right G-action, transitive on each fiber, and local equivariant trivializations. A *section* is a smooth map $s : M \to E$ with $\pi s = \mathrm{id}_M$. For a representation $G \to \mathrm{GL}(V)$, the *associated vector bundle* is $P \times_G V = (P \times V)/((pg, v) \sim (p, gv))$.

The transformation group specifies which frames, and therefore which measurements, are geometrically equivalent.

Definition 4.2 (Frame bundles and G-structures)**.** The *frame bundle* of an n-manifold is the principal $\mathrm{GL}(n,\mathbb{R})$-bundle

$$\mathrm{Fr}(M)_p = \mathrm{Iso}(\mathbb{R}^n, T_pM).$$

For a closed subgroup $G \le \mathrm{GL}(n,\mathbb{R})$, a *$G$-structure* is a principal G-subbundle of $\mathrm{Fr}(M)$. It is a choice of the frames related by the transformations in G. Riemannian, Lorentzian, and almost symplectic structures correspond respectively to reductions to $O(n)$, $O(1, n-1)$, and $\mathrm{Sp}(2n,\mathbb{R})$; an oriented, time-oriented spacetime has structure group $SO^+(1,3)$.

Exercise 4.1 (Geometry from transformation groups)**.** Prove the stated correspondences by identifying the stabilizers of a positive inner product, a form of signature $(1, n-1)$, and a nondegenerate alternating form. Explain why an $\mathrm{Sp}(2n, \mathbb{R})$-structure is only almost symplectic until its associated two-form is closed. Show that the proper orthochronous Poincaré group acts transitively on Minkowski spacetime and identify

$$\mathbb{R}^{1,3} \cong (\mathbb{R}^{1,3} \rtimes SO^+(1,3))/SO^+(1,3).$$

Explain why a curved spacetime retains local Lorentz frames but generally has no transitive Poincaré action.

Definition 4.3 (Tensor fields and metrics)**.** A (r,s)-*tensor field* is a smooth section of $TM^{\otimes r} \otimes T^*M^{\otimes s}$. Equivalently, an $O(p,q)$-structure is a smooth nondegenerate symmetric tensor g of signature (p,q). The cases $(n,0)$ and $(1,n-1)$ give Riemannian and Lorentzian metrics. The length of a Riemannian curve is $\int \sqrt{g(\dot\gamma,\dot\gamma)}\,dt$; the proper time of a timelike curve is $c^{-1}\int \sqrt{-g(\dot\gamma,\dot\gamma)}\,dt$. A metric and an orientation define a volume form vol_g.

Exercise 4.2 (Proper time)**.** Prove invariance of length and proper time under reparametrization. In Minkowski spacetime, prove that the inertial path maximizes proper time among piecewise smooth future-directed timelike paths with fixed timelike-separated endpoints, and derive special-relativistic time dilation. Using the muon rest lifetime $\tau_\mu \simeq 2.2\,\mu$s, derive the mean laboratory travel distance $\gamma v \tau_\mu$ and compare it with the distance $v\tau_\mu$ predicted without time dilation.

Integration and Stokes' theorem turn local differential equations into laws for flux through regions and their boundaries.

4.2 Differential forms, cohomology, and Hodge theory

Definition 4.4 (Integration of forms)**.** An *orientation* of an n-manifold is a reduction of its frame bundle to $\mathrm{GL}^+(n,\mathbb{R})$, equivalently a continuous choice of component of nonzero elements in $\Lambda^n T_p^* M$. The integral $\int_M \omega$ of a compactly supported top form is defined using orientation-preserving charts and a partition of unity.

Exercise 4.3 (Stokes' theorem)**.** Prove that the integral is independent of charts and partition of unity. For an oriented manifold with boundary, prove

$$\int_M d\omega = \int_{\partial M} \omega.$$

Deduce the fundamental theorem of calculus, Green's theorem, the divergence theorem, and the classical curl theorem. For time-dependent magnetic two-form B and electric one-form E, prove the equivalence of

$$\int_{\partial S} E = -\frac{d}{dt}\int_S B \qquad \text{and} \qquad dE = -\partial_t B,$$

recovering the observed flux–emf law.

Definition 4.5 (de Rham cohomology)**.** A differential form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η. Since $d^2 = 0$, the *de Rham cohomology* of M is

$$H^k_{\mathrm{dR}}(M) = \frac{\ker(d : \Omega^k(M) \to \Omega^{k+1}(M))}{\mathrm{im}(d : \Omega^{k-1}(M) \to \Omega^k(M))}.$$

Pullback induces maps on de Rham cohomology, and the wedge product makes $H^*_{\mathrm{dR}}(M)$ a graded-commutative algebra.

Exercise 4.4 (de Rham classes and periods)**.** On a star-shaped open subset of $\mathbb{R}^n$, construct a homotopy operator and prove the Poincaré lemma: every closed form of positive degree is exact. Use Stokes' theorem to prove that an exact k-form has zero integral over every closed oriented k-manifold mapped smoothly into M. Prove $H^1_{\mathrm{dR}}(S^1) \cong \mathbb{R}$, with the isomorphism given by integration.

On $\mathbb{R}^2 \setminus \{0\}$, show that

$$\alpha = \frac{-y\,dx + x\,dy}{x^2 + y^2}$$

is closed and has integral 2π around the unit circle, hence is not exact. Compute $H^1_{\mathrm{dR}}(\mathbb{R}^2 \setminus \{0\}) = \mathbb{R} \cdot [\alpha]$. Deduce that a closed electromagnetic field strength has local potentials, while it has a global potential $F = dA$ exactly when $[F] = 0$ in de Rham cohomology.

Definition 4.6 (Hodge star)**.** On an oriented pseudo-Riemannian n-manifold, the metric induces a nondegenerate pairing $\langle \cdot, \cdot \rangle_g$ on differential forms. The *Hodge star* is the bundle map

$$* : \Omega^p(M) \longrightarrow \Omega^{n-p}(M)$$

characterized by

$$\alpha \wedge *\beta = \langle \alpha, \beta \rangle_g \ \mathrm{vol}_g$$

for p-forms α, β.

Exercise 4.5 (Metric volume and Hodge star)**.** Construct vol_g and the Hodge star intrinsically and derive their coordinate expressions. For a metric with q negative directions, prove on p-forms that

$$*^2 = (-1)^{p(n-p)+q}\,\mathrm{id}\,.$$

Compute $*$ on the standard coordinate forms of Euclidean three-space and Minkowski four-space, stating the orientation and signature conventions.

A connection compares fields in different fibers. Its curvature measures the failure of parallel transport to be path independent.

4.3 Connections, curvature, and gauge fields

Definition 4.7 (Connections and parallel transport)**.** A *connection* on a vector bundle $E \to M$ is an $\mathbb{R}$-bilinear map $\nabla : \Gamma(TM) \times \Gamma(E) \to \Gamma(E)$, linear over smooth functions in the first variable and satisfying $\nabla_X(fs) = X(f)s + f\nabla_X s$. Along a curve γ it induces a covariant derivative D/dt; its solutions $Ds/dt = 0$ define *parallel transport* between endpoint fibers. On TM, the torsion is $T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$, and a *geodesic* satisfies $D\dot{\gamma}/dt = 0$.

Definition 4.8 (Curvature and holonomy)**.** The *curvature* of ∇ is the $\mathrm{End}(E)$-valued two-form

$$R^\nabla(X,Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X,Y]} s.$$

The *holonomy group* at p consists of the automorphisms of E_p obtained by parallel transport around loops based at p.

Exercise 4.6 (Parallel transport and curvature)**.** Prove existence, uniqueness, composition, and inverse properties of parallel transport. Prove that curvature is tensorial and gives the leading failure of transport around an infinitesimal parallelogram to return a vector to itself. Deduce that a flat connection has path-independent transport on every simply connected coordinate domain.

Definition 4.9 (Principal connections and gauge fields)**.** Let $P \to M$ be a principal G-bundle. A *principal connection* is an equivariant $\mathfrak{g}$-valued one-form A on P reproducing fundamental vertical fields. It induces connections on every associated bundle, and its *curvature* is $F_A = dA + \frac{1}{2}[A \wedge A]$. A *gauge transformation* is a G-equivariant bundle automorphism covering id_M. A reduction of the frame bundle describes spacetime geometry; a separate principal bundle describes an internal gauge symmetry.

Exercise 4.7 (Gauge covariance and Bianchi identity)**.** Choose local sections s_i and derive transition maps $g_{ij} : U_i \cap U_j \to G$ and the transformation laws for the local potentials $A_i = s_i^* A$ and curvatures $F_i = s_i^* F_A$. Prove $d_A F_A = 0$ and gauge invariance of holonomy up to conjugacy. For $G = U(1)$, recover $A \mapsto A + d\chi$ and $F = dA$. On $\mathbb{R}^2 \setminus \{0\}$, compute the curvature and holonomy of $A = (\Phi/2\pi) d\theta$ and derive the Aharonov–Bohm phase shift $q\Phi/\hbar$ despite $F = 0$ along the paths.

Definition 4.10 (Electromagnetic field)**.** An *electromagnetic gauge field* is a connection on an internal principal $U(1)$-bundle over an oriented Lorentzian four-manifold. Its field strength is the curvature two-form F; locally $F = dA$. Given a current one-form J, Maxwell's equations are

$$dF = 0, \qquad d{*}F = \mu_0 {*}J.$$

For the SI vector decomposition on Minkowski spacetime, fix

$$(x^0, x^1, x^2, x^3) = (ct, x, y, z), \qquad g = -(dx^0)^2 + dx^2 + dy^2 + dz^2,$$

with orientation $dx^0 \wedge dx \wedge dy \wedge dz$, and set

$$\begin{aligned} F &= \frac{1}{c}\, dx^0 \wedge (E_x\, dx + E_y\, dy + E_z\, dz) \\ &\quad - (B_x\, dy \wedge dz + B_y\, dz \wedge dx + B_z\, dx \wedge dy), \\ J &= c\rho\, dx^0 - j_x\, dx - j_y\, dy - j_z\, dz. \end{aligned}$$

With these sign conventions, the force law for a particle of charge q and four-velocity u is $m\nabla_u u = q(\iota_u F)^\sharp$.

Exercise 4.8 (Maxwell theory and electromagnetic waves)**.** Derive charge conservation from Maxwell's equations. Using the stated Minkowski conventions, recover the electric and magnetic vector equations and verify that the covariant force law reduces to $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$. Define the experimentally measured vacuum permittivity ε_0 by the electrostatic Gauss law $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$; comparison with the covariant equation should give $\mu_0 \varepsilon_0 c^2 = 1$. In vacuum, impose Lorenz gauge and derive wave equations with speed c. Prove transverse polarization and compare $c = (\mu_0\varepsilon_0)^{-1/2}$ with the measured speed of light.

Exercise 4.9 (Yang–Mills equations and instantons)**.** For $G = SU(r)$, use $\langle X, Y \rangle = -2\,\mathrm{Tr}(XY)$ and define

$$S_{\mathrm{YM}}(A) = \frac{1}{2g^2} \int_M \langle F_A \wedge {*}F_A \rangle$$

on a closed oriented Riemannian four-manifold. Prove gauge invariance and derive $d_A^* F_A = 0$, where $d_A^* = -*\, d_A *$. Decompose $F_A = F_A^+ + F_A^-$ and, for

$$k(A) = -\frac{1}{8\pi^2} \int_M \mathrm{Tr}(F_A \wedge F_A),$$

prove $S_{\mathrm{YM}}(A) \geq 8\pi^2 |k(A)|/g^2$, with equality precisely for $F_A = \pm * F_A$. Use the Bianchi identity to show that these instantons solve the Yang–Mills equation.

Gravity is carried by the Lorentz reduction of the frame bundle itself. Its connection compares local inertial frames, and its curvature measures the departure from the flat Minkowski model.

4.4 Riemannian geometry and gravitation

Definition 4.11 (Levi–Civita geometry)**.** An $O(p,q)$-structure determines a unique torsion-free connection preserving the metric, the *Levi–Civita connection.* On the orthonormal frame bundle it is a principal $O(p,q)$-connection; the solder form identifies its associated vector bundle with TM. Its curvature gives the *Riemann tensor* $\mathrm{Riem}(X,Y,Z,W) = g(R^{\nabla}(X,Y)Z,W)$, whose contractions are the *Ricci tensor* Ric and scalar curvature S.

Exercise 4.10 (Levi–Civita connection and holonomy)**.** Prove existence and uniqueness of the Levi–Civita connection and derive the Koszul formula. Prove that its geodesics are precisely the critical curves of the energy functional with fixed endpoints and that its parallel transport preserves the metric. Compute the rotation obtained by transporting a tangent vector around a latitude on the unit sphere and relate it to the enclosed curvature.

Exercise 4.11 (Curvature identities and geodesic deviation)**.** For the Levi–Civita connection, prove the algebraic symmetries of the Riemann tensor and the first and second Bianchi identities. For a variation through geodesics with separation field J, prove

$$\frac{D^2 J}{dt^2} + R^{\nabla}(J,\dot{\gamma})\dot{\gamma} = 0.$$

Deduce the leading relative acceleration of nearby freely falling bodies.

Definition 4.12 (Einstein equation)**.** The *Einstein tensor* of a Lorentzian metric is $G = \mathrm{Ric} - \frac{1}{2}Sg$. A *stress–energy tensor* is a symmetric two-tensor T whose contraction with an observer's velocity gives the observer's energy–momentum current. The *Einstein equation* with cosmological constant Λ is

$$G + \Lambda g = \frac{8\pi G_N}{c^4}T.$$

Exercise 4.12 (Conservation and the Newtonian limit)**.** Use the contracted Bianchi identity to derive $\nabla^a T_{ab} = 0$. Linearize $g = \eta + h$ for a static weak field and recover Poisson's equation and the Newtonian inverse-square law with the stated coupling constant.

Exercise 4.13 (Gravitational radiation)**.** Linearize the vacuum Einstein equation about Minkowski spacetime, impose transverse–traceless gauge, and obtain two tensor polarizations satisfying a wave equation. Apply geodesic deviation to a ring of freely falling test masses and derive the strain measured by an orthogonal-arm interferometer.

Exercise 4.14 (Gravitational-wave detection)**.** For a quasi-circular binary, derive the leading quadrupole waveform and the chirp relation between frequency, frequency derivative, and chirp mass. For GW150914, use the reported detector-frame component-mass medians $36M_{\odot}$ and $29M_{\odot}$ to infer the chirp mass. Compare with the model-independent reconstruction $\mathcal{M} \simeq 30M_{\odot}$ and with the reported peak strain $h \simeq 10^{-21}$ near $150\,\mathrm{Hz}$ in Abbott et al., *Physical Review D* **93** (2016) and the LIGO GW150914 fact sheet. Limit the conclusion to agreement of the observed waveform with general relativity in that regime.

5 Harmonic Analysis

Experimental observations.

- Monochromatic X-rays scattered by a crystal form sharp peaks only at selected directions determined by wavelength and lattice spacing.
- Electrical networks patterned on a hyperbolic tiling have been used as "hyperbolic drums": their measured modes and pulse propagation distinguish negative curvature from an ordinary planar circuit.

Wave fields decompose into modes adapted to spacetime symmetry. Invariant integration turns representations into a generalized Fourier analysis. For translation symmetry, its dual variables are the reciprocal coordinates measured by diffraction. Applied to phase-space translations, the same idea will turn the symplectic structure of classical mechanics into the operator relations of quantum mechanics.

5.1 Haar measure and abelian Fourier duality

Definition 5.1 (Haar measure and convolution)**.** Let G be a locally compact Hausdorff group. A *left Haar measure* is a nonzero regular Borel measure μ satisfying $\mu(gE) = \mu(E)$. On a compact group, a Haar measure may be normalized by $\mu(G) = 1$. For $f, g \in C_c(G)$, define

$$(f * g)(x) = \int_G f(y)g(y^{-1}x)\,d\mu(y).$$

The *left regular representation* on $L^2(G)$ is $(\lambda_g f)(x) = f(g^{-1}x)$.

Theorem 5.2 (Haar; used without proof)**.** *Every locally compact Hausdorff group has a left Haar measure, unique up to multiplication by a positive scalar.*

Exercise 5.1 (Haar measure and the regular representation)**.** Identify Haar measure on $\mathbb{R}^n$, $\mathbb{Z}^n$, and $\mathbb{T}^n$. Construct normalized Haar measure on a compact Lie group from a left-invariant metric. Prove associativity of convolution, the L^1 convolution inequality, and $\|f * g\|_2 \le \|f\|_1\|g\|_2$. Prove that $g \mapsto \lambda_g$ is a strongly continuous unitary representation. For a finite-dimensional representation of a compact group, average an inner product over G to obtain an invariant one.

Definition 5.3 (Pontryagin duality)**.** For a locally compact abelian group G, its *Pontryagin dual* $\widehat{G}$ is the group of continuous characters $\chi : G \to U(1)$, with pointwise multiplication and the compact-open topology. For $f \in L^1(G)$, define

$$\widehat{f}(\chi) = \int_G f(x)\overline{\chi(x)}\,d\mu(x).$$

With compatible Haar measures on G and $\widehat{G}$, Fourier inversion and Plancherel extend the earlier theorems. Pontryagin duality identifies G naturally with $\widehat{\widehat{G}}$.

Exercise 5.2 (Abelian Fourier duality)**.** Identify

$$\widehat{\mathbb{R}^n} \cong \mathbb{R}^n, \qquad \widehat{\mathbb{Z}^n} \cong \mathbb{T}^n, \qquad \widehat{\mathbb{T}^n} \cong \mathbb{Z}^n,$$

and verify the double-dual map in each case. Show that the group Fourier transform recovers Fourier transforms on $\mathbb{R}^n$ and Fourier series on $\mathbb{T}^n$. For a closed subgroup $H \le G$, define its annihilator $H^\perp \subseteq \widehat{G}$ and prove $\widehat{G/H} \cong H^\perp$.

Exercise 5.3 (Interference and Fourier optics)**.** For coherent scalar waves with complex amplitudes a_1, a_2, derive $|a_1 + a_2|^2$ and the fringe spacing in Young's double-slit geometry. In the Fraunhofer regime, derive that the far-field amplitude is the Fourier transform of the aperture. Compute the single-slit diffraction intensity, locate its zeros, and compare the predicted angular spacing with measured optical diffraction.

5.2 Lattices, Poisson summation, and diffraction

Definition 5.4 (Lattices and reciprocal lattices)**.** A *lattice* in $\mathbb{R}^n$ is a subgroup $\Gamma = A\mathbb{Z}^n$ for some invertible linear map $A : \mathbb{R}^n \to \mathbb{R}^n$. Its covolume is $\mathrm{vol}(\Gamma) = |\det A|$, and a fundamental cell is $F = A[0,1)^n$. Its *reciprocal lattice* is the annihilator

$$\Gamma^* = \{\xi \in \mathbb{R}^n : \xi \cdot \gamma \in 2\pi\mathbb{Z} \text{ for every } \gamma \in \Gamma\}.$$

Exercise 5.4 (Fourier modes of a lattice)**.** Prove $\Gamma^* = 2\pi(A^{-1})^*\mathbb{Z}^n$. Show that $e_\xi(x) = e^{i\xi\cdot x}$ is Γ-periodic exactly when $\xi \in \Gamma^*$. With inner product

$$\langle f, g\rangle_F = \frac{1}{\mathrm{vol}(\Gamma)} \int_F \overline{f(x)} g(x)\, dx,$$

prove that $(e_\xi)_{\xi\in\Gamma^*}$ is a complete orthonormal family in $L^2(\mathbb{R}^n/\Gamma)$. Interpret this as $\widehat{\mathbb{R}^n/\Gamma} \cong \Gamma^*$.

Definition 5.5 (Periodic summation and structure factors)**.** For $f \in \mathcal{S}(\mathbb{R}^n)$, its Γ-*periodization* is

$$P_\Gamma f(x) = \sum_{\gamma\in\Gamma} f(x+\gamma).$$

For a finite motif $B \subset \mathbb{R}^n$ with scattering weights c_b, its *structure factor* is

$$S_B(q) = \sum_{b\in B} c_b e^{-iq\cdot b}.$$

Exercise 5.5 (Poisson summation)**.** Prove that $P_\Gamma f$ converges with all derivatives and compute its Fourier coefficients. With the Fourier-transform normalization above, prove

$$P_\Gamma f(x) = \frac{(2\pi)^{n/2}}{\mathrm{vol}(\Gamma)} \sum_{\xi\in\Gamma^*} \mathcal{F}f(\xi) e^{i\xi\cdot x}.$$

Set $x = 0$ to obtain the Poisson summation formula. Derive the Jacobi theta transformation by applying it to a Gaussian.

Exercise 5.6 (Crystal diffraction and Bragg peaks)**.** Let Γ_N be a finite rectangular portion of Γ. In the single-scattering approximation, prove that the amplitude at momentum transfer q factors as

$$\mathcal{A}_N(q) = S_B(q) \sum_{\gamma\in\Gamma_N} e^{-iq\cdot\gamma}.$$

Use geometric sums and Poisson summation to show that the normalized intensity concentrates at $q \in \Gamma^*$ as the crystal grows. For elastic scattering with incident and outgoing wavevectors of magnitude $2\pi/\lambda$, derive the Laue condition and, for planes spaced by d, the Bragg relation

$$2d\sin\theta = m\lambda.$$

Show how zeros of S_B produce missing reflections. Compare the peak condition with W. H. Bragg and W. L. Bragg, *Proceedings of the Royal Society A* **88** (1913).

Exercise 5.7 (Electron diffraction)**.** For an electron accelerated through voltage V, use $eV = p^2/(2m_e)$ and $p = h/\lambda$ to derive

$$\lambda = \frac{h}{\sqrt{2m_e eV}}.$$

Combine this relation with the reciprocal-lattice and Bragg conditions to predict how the diffraction angles change with V. Compare the predicted sharply directed beams with Davisson–Germer, *Physical Review* **30** (1927).

5.3 Compact-group harmonic analysis

Definition 5.6 (Compact-group Fourier transform)**.** For a compact group G, let $\widehat{G}$ denote the equivalence classes of irreducible finite-dimensional unitary representations. For $f \in L^1(G)$ and $\pi \in \widehat{G}$, define

$$\widehat{f}(\pi) = \int_G f(g)\pi(g)^* \, dg \in \mathrm{End}(V_\pi).$$

Thus the Fourier transform is scalar-valued for abelian G and matrix-valued in general.

Exercise 5.8 (Peter–Weyl theorem)**.** Use Haar averaging and Schur's lemma to prove orthogonality of matrix coefficients of irreducible unitary representations of a compact group. Prove that their span is dense in $C(G)$ and $L^2(G)$ and obtain

$$L^2(G) \cong \widehat{\bigoplus_{\pi \in \widehat{G}}} V_\pi \otimes V_\pi^*.$$

Derive Fourier inversion, the Plancherel formula

$$\|f\|_2^2 = \sum_{\pi \in \widehat{G}} (\dim V_\pi) \|\widehat{f}(\pi)\|_{\mathrm{HS}}^2,$$

and $\widehat{f * g}(\pi) = \widehat{g}(\pi)\widehat{f}(\pi)$. Recover ordinary Fourier series for $G = \mathbb{T}$. For $G = SO(3)$, identify the matrix coefficients with the Wigner functions D^ℓ_{mn}.

Exercise 5.9 (Spherical harmonics and atomic orbitals)**.** Identify S^2 with $SO(3)/SO(2)$ and $L^2(S^2)$ with the right $SO(2)$-invariant subspace of $L^2(SO(3))$. Use Peter–Weyl to obtain

$$L^2(S^2) \cong \widehat{\bigoplus_{\ell=0}^{\infty}} V_\ell$$

and identify a basis of V_ℓ with $Y_{\ell m} \propto D^\ell_{m0}$. Derive $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1) Y_{\ell m}$ and completeness of the spherical harmonics. Separate the Coulomb Hamiltonian in spherical coordinates to obtain $\psi_{n\ell m}(r, \theta, \phi) = R_{n\ell}(r) Y_{\ell m}(\theta, \phi)$. Determine the angular nodes of the s, p, and d orbitals and relate them to symmetry-enforced zeros in photoelectron angular distributions in the plane-wave final-state approximation. Compare with Matsui et al., *Physical Review B* **72** (2005).

Exercise 5.10 (Molecular rotational spectra)**.** Model a rigid molecule with orientation $R \in SO(3)$ and principal moments I_1, I_2, I_3. On $L^2(SO(3))$, derive

$$H_{\mathrm{rot}} = \frac{J_1^2}{2I_1} + \frac{J_2^2}{2I_2} + \frac{J_3^2}{2I_3}.$$

Use the Peter–Weyl basis to reduce H_{rot} to finite-dimensional blocks. For a symmetric top, derive

$$E_{JK} = \frac{\hbar^2 J(J+1)}{2I_\perp} + \frac{\hbar^2 K^2}{2}\left(\frac{1}{I_\parallel} - \frac{1}{I_\perp}\right).$$

Derive the electric-dipole selection rules and the splitting in a static electric field. Compare the predicted components with microwave measurements of the symmetric-top molecule s-trioxane reported in NBS Special Publication 343 (1971).

A rigid rotor has compact configuration space $SO(3)$, so Peter–Weyl gives discrete angular modes. By contrast, the hyperbolic plane is the noncompact homogeneous space $SL_2(\mathbb{R})/SO(2)$, and freely propagating waves on it have a continuous spectral parameter. Harish-Chandra theory explains what replaces the Peter–Weyl sum for this and, more generally, for noncompact semisimple symmetry groups.

5.4 Semisimple groups and Harish–Chandra theory

Definition 5.7 (Cartan and Iwasawa decompositions)**.** Let G be a connected real semisimple Lie group with finite center and let $\mathfrak{g}$ be its Lie algebra. A *Cartan involution* gives a vector space decomposition

$$\mathfrak{g} = \mathfrak{k} \oplus \mathfrak{p}$$

into its $+1$ and -1 eigenspaces; the subgroup K with Lie algebra $\mathfrak{k}$ is maximal compact, and G/K is a noncompact Riemannian symmetric space. Choose a maximal abelian subspace $\mathfrak{a} \subseteq \mathfrak{p}$, put $A = \exp \mathfrak{a}$, choose positive restricted roots, and let N be the subgroup generated by their root spaces. The *Iwasawa decomposition* is the diffeomorphism

$$K \times A \times N \longrightarrow G, \qquad (k, a, n) \longmapsto kan.$$

If $\mathfrak{a}^+$ is a positive Weyl chamber, the *Cartan decomposition* is $G = K\overline{A^+}K$. Write $M = Z_K(A)$ and $W = N_K(A)/M$, and let ρ be half the sum of the positive restricted roots, counted with multiplicity.

Exercise 5.11 ($SL_2(\mathbb{R})$ and the hyperbolic plane)**.** For $G = SL_2(\mathbb{R})$, set

$$K = SO(2), \qquad a_t = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix}, \qquad n_x = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}.$$

Use Gram–Schmidt or QR decomposition to prove the Iwasawa decomposition, and prove that $W \cong \mathbb{Z}/2\mathbb{Z}$ acts on t by reflection. Under the fractional-linear action on the upper half-plane

$$\mathbb{H}^2 = \{x + iy : y > 0\}, \qquad g \cdot z = \frac{az + b}{cz + d},$$

show that K stabilizes i and hence $SL_2(\mathbb{R})/SO(2) \cong \mathbb{H}^2$. Derive the invariant metric and measure

$$ds^2 = \frac{dx^2 + dy^2}{y^2}, \qquad d\mu = \frac{dx\, dy}{y^2},$$

and show that a_r moves i a distance $|r|$. Deduce from KAK that a K-invariant function on G/K is a radial function of this distance.

Definition 5.8 (Spherical principal series and spherical transform)**.** For $\lambda \in \mathfrak{a}^*$, the normalized *spherical principal series* is

$$\pi_\lambda = \mathrm{Ind}_{MAN}^G (1_M \otimes e^{i\lambda} \otimes 1_N),$$

where normalized induction incorporates the modular factor determined by ρ. It contains a unit K-fixed vector v_λ, and its *zonal spherical function* is

$$\varphi_\lambda(g) = \langle v_\lambda, \pi_\lambda(g) v_\lambda \rangle.$$

It is K-bi-invariant, satisfies $\varphi_\lambda(e) = 1$, and is a joint eigenfunction of the G-invariant differential operators on G/K.

For $f \in C_c^\infty(K\backslash G/K)$, its *spherical transform* is

$$\widetilde{f}(\lambda) = \int_G f(g) \varphi_{-\lambda}(g)\, dg.$$

Harish-Chandra's c-function is the coefficient governing the leading asymptotics of $\varphi_\lambda(\exp H)$ as $H \to \infty$ in a positive chamber.

Theorem 5.9 (Spherical Plancherel and inversion; used without proof)**.** *With compatible Haar and Lebesgue normalizations, the spherical transform satisfies*

$$\|f\|_2^2 = C_G \int_{\mathfrak{a}^*/W} |\widetilde{f}(\lambda)|^2 |c(\lambda)|^{-2}\, d\lambda,$$

with the corresponding inverse integral. Thus $|c(\lambda)|^{-2}$ is the curvature- and root-dependent density replacing the dimension weights in the Peter–Weyl sum.

Exercise 5.12 (Harish-Chandra transform on $\mathbb{H}^2$)**.** For the sign convention in which the Laplace–Beltrami operator is

$$\Delta_{\mathbb{H}^2} = y^2(\partial_x^2 + \partial_y^2),$$

derive its radial part

$$\Delta_{\mathrm{rad}} = \frac{d^2}{dr^2} + \coth r \frac{d}{dr}.$$

Show that the normalized radial eigenfunction satisfying

$$\Delta\varphi_\lambda = -(\lambda^2 + \tfrac{1}{4})\varphi_\lambda, \qquad \varphi_\lambda(0) = 1,$$

is the conical Legendre function

$$\varphi_\lambda(r) = P_{-1/2+i\lambda}(\cosh r).$$

For a smooth compactly supported radial function, define

$$\widetilde{f}(\lambda) = 2\pi \int_0^\infty f(r)\varphi_\lambda(r) \sinh r\, dr.$$

Taking the Mehler–Fock inversion theorem as input, prove

$$f(r) = \frac{1}{2\pi} \int_0^\infty \widetilde{f}(\lambda)\varphi_\lambda(r)\lambda \tanh(\pi\lambda)\, d\lambda$$

and

$$\|f\|_{L^2(\mathbb{H}^2)}^2 = \frac{1}{2\pi} \int_0^\infty |\widetilde{f}(\lambda)|^2 \lambda \tanh(\pi\lambda)\, d\lambda.$$

Deduce that the radial sector has continuous spectrum $[1/4, \infty)$; using the full Helgason transform, conclude the same for $-\Delta$ on $L^2(\mathbb{H}^2)$. Compare the high-frequency density $\lambda \tanh(\pi\lambda) \sim \lambda$ with the radial Fourier density in the Euclidean plane. Use the transform to solve the radial wave equation from compactly supported initial data.

Exercise 5.13 (A finite hyperbolic drum)**.** Discretize the Laplace–Beltrami operator by a real symmetric weighted graph Laplacian on a finite regular hyperbolic tiling with Dirichlet boundary. Prove that the finite operator has a discrete orthogonal eigenbasis, and explain why those eigenvalues approximate a bounded domain with a boundary rather than the continuous spectrum of all $\mathbb{H}^2$. Compare the predicted mode ordering and propagation along discrete geodesics with the circuit measurements in Lenggenhager et al., *Nature Communications* **13** (2022). State which observations support the effective hyperbolic metric and which do not by themselves establish the infinite-space Plancherel theorem.

Definition 5.10 (Harish-Chandra characters and discrete series)**.** Let π be an irreducible admissible unitary representation of a connected real reductive group. Although the infinite-dimensional operator $\pi(g)$ usually has no trace, for $f \in C_c^\infty(G)$ the integrated operator

$$\pi(f) = \int_G f(g)\pi(g)\, dg$$

is trace class. The *Harish-Chandra character* is the conjugation-invariant distribution

$$\Theta_\pi(f) = \mathrm{Tr}(\pi(f)).$$

The regularity theorem represents it by a locally integrable function, real analytic on the regular semisimple set.

An irreducible unitary representation belongs to the *discrete series* if it has a nonzero matrix coefficient square-integrable modulo the center; then the corresponding coefficients satisfy the discrete-series orthogonality relations. Such representations contribute discrete summands, with formal-degree weights, to the Plancherel decomposition; continuous tempered families contribute integrals.

Exercise 5.14 (From compact characters to the semisimple Plancherel formula)**.** For a compact group, prove

$$\Theta_\pi(f) = \int_G f(g)\,\mathrm{Tr}(\pi(g))\, dg$$

and recover the ordinary character used in Peter–Weyl theory. Show that a unitary operator on an infinite-dimensional Hilbert space is not trace class, which explains the distributional definition above. Prove directly that Θ_π is invariant under conjugation. Taking Harish-Chandra's regularity theorem as input, explain why it nevertheless has pointwise values almost everywhere.

For $SL_2(\mathbb{R})$, identify the K-types of the spherical principal series and show that they contain a K-fixed vector. Contrast them with the holomorphic and antiholomorphic discrete series, whose nonzero one-sided K-types imply that they have no K-fixed vector. Deduce that spherical analysis on G/K sees the class-one principal series but not the discrete series, while the full Plancherel formula for $L^2(G)$ contains both the appropriate principal-series integrals and discrete-series sums, up to measure-zero limits. Consult Harish-Chandra, *Proceedings of the National Academy of Sciences* **43** (1957) for spherical functions and Harish-Chandra, *Transactions of the American Mathematical Society* **119** (1965) for invariant eigendistributions.

5.5 Noncommutative Plancherel theory

Definition 5.11 (Noncommutative Plancherel transform)**.** A locally compact group is *unimodular* when its left and right Haar measures agree. It is *type I* when its unitary representations have essentially unique direct-integral decompositions into irreducibles. Let G be second-countable, unimodular, and type I. For $f \in L^1(G)$ and an irreducible unitary representation $\pi : G \to U(H_\pi)$, define the operator-valued Fourier transform

$$\widehat{f}(\pi) = \int_G f(g)\pi(g)^*\, dg.$$

The noncommutative Plancherel theorem supplies a measure ν_G on the unitary dual $\widehat{G}$ such that, initially for $f \in L^1(G) \cap L^2(G)$,

$$\|f\|_2^2 = \int_{\widehat{G}} \|\widehat{f}(\pi)\|_{\mathrm{HS}}^2\, d\nu_G(\pi), \qquad \widehat{f * g}(\pi) = \widehat{g}(\pi)\widehat{f}(\pi).$$

For abelian groups the representation spaces are one-dimensional; for compact groups the integral becomes the dimension-weighted Peter–Weyl sum.

The standard symplectic form on classical phase space determines a central extension of its translation group. Harmonic analysis of this noncommutative group is the bridge from classical phase space to quantization.

Definition 5.12 (Heisenberg group and Schrödinger representations)**.** On the standard symplectic space $\mathbb{R}^{2n} = \mathbb{R}^n_q \oplus \mathbb{R}^n_p$, set

$$\sigma((q,p),(q',p')) = p \cdot q' - q \cdot p'.$$

The *Heisenberg group* is $\mathbb{H}_n = \mathbb{R}^{2n} \times \mathbb{R}$ with multiplication

$$(z,s)(z',s') = (z+z', s+s' + \tfrac{1}{2}\sigma(z,z')).$$

For $\lambda \neq 0$, its *Schrödinger representation* on $L^2(\mathbb{R}^n)$ is

$$(\pi_\lambda(q,p,s)\psi)(x) = e^{i\lambda(s+p\cdot(x-q/2))}\psi(x-q).$$

Here $\sigma = -\omega_0$ for the standard symplectic-form convention used earlier; this sign is paired with the translation convention in the displayed representation. The physical central character is $\lambda = 1/\hbar$.

Exercise 5.15 (Harmonic analysis of the Heisenberg group)**.** Verify the group law, identify the center, and prove that Lebesgue measure is Haar measure on $\mathbb{H}_n$. Prove that each π_λ is an irreducible unitary representation and derive its Weyl commutation relations. At $\lambda = 1/\hbar$, differentiate the position and momentum subgroups and recover Q_j, P_j, and $[Q_j, P_k] = i\hbar\delta_{jk}$. Taking the Stone–von Neumann theorem as input, show that every irreducible representation with central character $e^{i\lambda s}$ is unitarily equivalent to π_λ.

Compute the operator-valued group Fourier transform in these representations and derive the Plancherel decomposition with measure $c_n|\lambda|^n\, d\lambda$, determining c_n from the Haar and Fourier normalizations above. Show that the one-dimensional representations with trivial central character have zero Plancherel measure. Derive the distributional orthogonality relation for Weyl operators and the resulting Hilbert–Schmidt Plancherel and inversion formulas. Prove that

$$S \longmapsto \chi_S(u,v) = \mathrm{Tr}\Big(Se^{i(u\cdot X+v\cdot Y)}\Big)$$

is injective on trace-class operators, where $[X_j, Y_k] = i\delta_{jk}$. Interpret this as noncommutative Fourier analysis turning functions on a group into operators rather than scalar frequency amplitudes.

Part II

Probability

Probability assigns expectations to observables. Dynamics describes repeated observations, algebraic products describe how systems combine, and positive maps describe how information is transformed or lost. The theory progresses from commuting observables through time to noncommutative states, free and tensor independence, and finally the matrix amplification produced by arbitrary ancillary systems.

6 Ergodic Theory

Experimental observations.

- A dilute gas obeys the ideal-gas law and exhibits a temperature-dependent distribution of molecular speeds.

Statistical mechanics begins with commuting observables on phase space and a measure-preserving evolution. Conditional expectation extracts invariant information, ergodic theorems relate time averages to ensemble averages, and entropy selects equilibrium states.

6.1 Probability, conditioning, and entropy

Definition 6.1 (Probability and conditioning)**.** A *probability space* is a measure space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{P}(\Omega) = 1$. A *random variable* is a measurable map, and its *expectation* is its integral. For $X \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ and a sub-σ-algebra $\mathcal{G} \subseteq \mathcal{F}$, a *conditional expectation* $\mathbb{E}[X \mid \mathcal{G}]$ is a $\mathcal{G}$-measurable integrable random variable satisfying

$$\int_G \mathbb{E}[X \mid \mathcal{G}]\, d\mathbb{P} = \int_G X\, d\mathbb{P} \qquad (G \in \mathcal{G}).$$

Definition 6.2 (Shannon entropy)**.** For a discrete probability vector p, its *Shannon entropy* is

$$H(p) = -\sum_i p_i \log p_i.$$

The corresponding physical entropy is $S(p) = k_B H(p)$. For probability vectors p, q with $p_i = 0$ whenever $q_i = 0$, their *relative entropy* is

$$D(p\|q) = \sum_i p_i \log \frac{p_i}{q_i}.$$

For a joint distribution p_{XY}, the *mutual information* is $I(X : Y) = D(p_{XY} \| p_X p_Y)$. For a phase-space probability density ρ relative to a fixed dimensionless normalization of Liouville measure μ_L (equivalently, to a chosen phase-space cell), its *statistical entropy* is $S(\rho) = -k_B \int \rho \log \rho \, d\mu_L$; for densities ρ, σ, set $D(\rho\|\sigma) = \int \rho \log(\rho/\sigma)\, d\mu_L$ when defined. Rescaling the reference measure changes $S(\rho)$ by an additive constant, whereas $D(\rho\|\sigma)$ is independent of that normalization.

6.2 Ergodic dynamics and equilibrium states

Definition 6.3 (Ergodic dynamics and dynamical entropy)**.** A measurable transformation T of a probability space is *measure preserving* if $\mu(T^{-1}A) = \mu(A)$. Its *Koopman operator* is the isometry $U_T f = f \circ T$ on $L^2(\mu)$. The system is *ergodic* if every invariant measurable set has measure zero or one, and *mixing* if

$$\mu(T^{-n}A \cap B) \longrightarrow \mu(A)\mu(B)$$

for all measurable A, B. An *equilibrium state* is an invariant probability measure for the dynamics. For a finite measurable partition $\mathcal{P}$, set

$$H_\mu(\mathcal{P}) = -\sum_{A \in \mathcal{P}} \mu(A) \log \mu(A), \qquad h_\mu(T, \mathcal{P}) = \lim_{n\to\infty} \frac{1}{n} H_\mu\left(\bigvee_{j=0}^{n-1} T^{-j}\mathcal{P}\right).$$

The *Kolmogorov–Sinai entropy* is $h_\mu(T) = \sup_{\mathcal{P}} h_\mu(T, \mathcal{P})$. It measures the asymptotic information per time step revealed by successively refined observations.

Exercise 6.1 (Ergodic averages, mixing, and entropy rate)**.** Prove that the fixed vectors of U_T are precisely the invariant L^2-observables, and that ergodicity is equivalent to all such observables being constant almost everywhere. Prove the von Neumann mean ergodic theorem

$$\frac{1}{N}\sum_{n=0}^{N-1} U_T^n f \longrightarrow P_{\mathrm{inv}} f \quad \text{in } L^2(\mu),$$

where P_{inv} is orthogonal projection onto the invariant observables. Taking Birkhoff's pointwise ergodic theorem as input, show for $f \in L^1(\mu)$ that these averages converge almost everywhere to the conditional expectation onto the invariant σ-algebra, and hence to $\int f\, d\mu$ in an ergodic system. Prove that mixing implies ergodicity and that it gives decay of correlations $\int (f \circ T^n) g\, d\mu - \int f\, d\mu \int g\, d\mu \to 0$ for bounded observables.

For every finite partition $\mathcal{P}$, prove by subadditivity that the limit defining $h_\mu(T, \mathcal{P})$ exists. For the two-sided Bernoulli shift on $\{1, \ldots, r\}^{\mathbb{Z}}$ with product probabilities $p_1, \ldots, p_r$, prove mixing first for cylinder sets and then for all measurable sets. Show that the time-zero coordinate partition is generating and, taking the Kolmogorov–Sinai generator theorem as input, compute

$$h_\mu(T) = -\sum_{i=1}^{r} p_i \log p_i = H(p).$$

Thus the Shannon entropy of one symbol equals the information production per time step of the Bernoulli shift.

Definition 6.4 (Gibbs states)**.** For a classical Hamiltonian H on a phase space with Liouville measure μ_L, the *canonical Gibbs state* at inverse temperature β is

$$d\mathbb{P}_\beta = Z(\beta)^{-1} e^{-\beta H}\, d\mu_L, \qquad Z(\beta) = \int e^{-\beta H}\, d\mu_L,$$

provided $0 < Z(\beta) < \infty$. It is stationary under the Hamiltonian dynamics generated by H.

Exercise 6.2 (Entropy and equilibrium)**.** For every density ρ with finite mean energy, prove

$$D(\rho \| \rho_\beta) = -\frac{S(\rho)}{k_B} + \beta \langle H \rangle_\rho + \log Z(\beta).$$

Deduce that the Gibbs ensemble uniquely maximizes entropy at fixed mean energy, equivalently minimizes free energy at fixed temperature. Prove $\langle H \rangle = -\partial_\beta \log Z$ and $\mathrm{Var}(H) = \partial_\beta^2 \log Z$. Prove invariance of Gibbs measure under Hamiltonian flow. If the flow is ergodic on an invariant energy surface, use Birkhoff's theorem to identify long-time averages of integrable observables with their microcanonical ensemble averages.

Exercise 6.3 (Ideal-gas thermodynamics)**.** Compute the classical canonical partition function of N indistinguishable noninteracting particles in a box, including the phase-space normalization $h^{3N} N!$. Derive the ideal-gas law, internal energy, and entropy, and the one-particle Maxwell–Boltzmann speed distribution. Derive its temperature scaling and compare both predictions with dilute-gas measurements in the classical regime.

For a wall of area A, express the pressure measured during time τ as the time-averaged normal momentum transfer

$$P_\tau = \frac{1}{A\tau} \sum_{\substack{\text{collisions with the wall} \\ 0<t_j\le\tau}} 2m|v_{j,\perp}|.$$

Average the collision flux using the Maxwell–Boltzmann distribution and recover the same pressure $P = Nk_BT/V$. Identify the stationarity and ergodic assumptions that relate the long-time reading of one gas sample to the ensemble calculation. Explain why agreement of pressure, velocity statistics, and relaxation rates for selected observables and finite times tests consequences of those assumptions but does not prove that the full microscopic dynamics is mathematically ergodic.

7 Operator Algebras

Experimental observations.

- When identically prepared spin-1/2 particles pass through a Stern–Gerlach analyzer, they arrive at one of two detector regions whose relative frequencies depend on the analyzer direction.
- Below its Curie temperature, a ferromagnet can retain nonzero magnetization after a weak aligning field is removed; reversing the field selects the opposite macroscopic magnetization.

Replacing commuting functions by an algebra of noncommuting observables changes both probability and dynamics. States still assign expectations, the GNS construction produces their Hilbert-space realizations, and von Neumann algebras supply the limits, conditional expectations, and modular flows needed for infinite quantum systems.

7.1 C^*-algebras, states, and GNS representations

Definition 7.1 (C^*-algebras and states)**.** A *unital Banach algebra* is a complete normed algebra with identity and $\|ab\| \le \|a\|\|b\|$. An *involution* satisfies $(ab)^* = b^*a^*$ and $(a^*)^* = a$. A *C^*-algebra* is an involutive Banach algebra satisfying $\|a^*a\| = \|a\|^2$. An element is *positive* if it is b^*b for some b. A *state* is a positive linear functional φ with $\varphi(1) = 1$; positivity then implies $\|\varphi\| = 1$.

Definition 7.2 (Noncommutative probability)**.** A *noncommutative probability space* is a unital $*$-algebra A together with a linear functional $\varphi : A \to \mathbb{C}$ satisfying $\varphi(a^*a) \ge 0$ and $\varphi(1) = 1$. Its random variables are elements of A, and the joint $*$-distribution of $X_1, \dots, X_r$ is the functional

$$p \longmapsto \varphi(p(X_1, \dots, X_r, X_1^*, \dots, X_r^*))$$

on noncommutative $*$-polynomials. For a self-adjoint variable X, its moments are $\varphi(X^n)$. A *tracial probability space* also satisfies $\varphi(ab) = \varphi(ba)$. Random Hermitian matrices form such spaces after combining expectation with the normalized trace.

Definition 7.3 (Representations and GNS construction)**.** A *representation* of a C^*-algebra A is a $*$-homomorphism $\pi : A \to B(H)$. For a state φ, define $N_\varphi = \{a : \varphi(a^*a) = 0\}$ and $\langle [a], [b] \rangle = \varphi(a^*b)$ on A/N_φ.

Exercise 7.1 (GNS theorem)**.** Complete A/N_φ and prove that left multiplication induces a cyclic representation $(\pi_\varphi, H_\varphi, \Omega_\varphi)$ satisfying $\varphi(a) = \langle \Omega_\varphi, \pi_\varphi(a)\Omega_\varphi \rangle$. Prove uniqueness up to a cyclic unitary equivalence.

7.2 Quantum mechanics and canonical quantization

Definition 7.4 (Quantum states and observables)**.** A positive operator S has the coordinate-free trace

$$\operatorname{Tr} S = \sup_P \operatorname{tr}\left((PSP)|_{PH}\right),$$

where P ranges over orthogonal projections with finite-dimensional image. A bounded operator T is *trace class* when $\operatorname{Tr}|T| < \infty$, where $|T| = (T^*T)^{1/2}$; its trace is obtained by linearity from the positive and negative parts of its real and imaginary parts. A *density operator* is positive and trace class with $\operatorname{Tr}\rho = 1$. An *observable* is a self-adjoint operator A with spectral measure E_A. The *Born probability* of an event B is $\operatorname{Tr}(\rho E_A(B))$. A closed-system *unitary evolution* is $\rho(t) = U_t\rho(0)U_t^*$, where $(U_t)_{t\in\mathbb{R}}$ is a strongly continuous one-parameter unitary group. For unbounded observables and generators, spectral measures, expectations, and exponentials use the earlier unbounded spectral calculus and its natural domains.

Exercise 7.2 (Classical and quantum state spaces)**.** Let X be compact Hausdorff and let H be finite-dimensional. Show that a point $x \in X$, a probability measure μ on X, a unit vector $\psi \in H$, and a density operator ρ determine states by

$$\delta_x(f) = f(x), \qquad \omega_\mu(f) = \int_X f\, d\mu, \qquad \omega_\psi(A) = \langle\psi, A\psi\rangle, \qquad \omega_\rho(A) = \operatorname{Tr}(\rho A).$$

Prove that the extreme points of the classical state space are the point evaluations and that the extreme points of the quantum state space are the rank-one density operators. Explain why mixtures are convex combinations in both theories and why the structural distinction is that $C(X)$ is commutative whereas $B(H)$ generally is not.

Exercise 7.3 (Stern–Gerlach projections and the Born rule)**.** For a spin-1/2 system and unit vector n, use $(n\cdot\sigma)^2 = \mathrm{id}$ to construct the complete spectral projections of $S_n = (\hbar/2)n\cdot\sigma$ as $P_\pm(n) = (1\pm n\cdot\sigma)/2$. For $\rho = (1+r\cdot\sigma)/2$, derive $p_\pm(n) = \operatorname{Tr}(\rho P_\pm(n)) = (1\pm r\cdot n)/2$. Model an inhomogeneous magnetic field by $H_{\text{int}} = -\boldsymbol{\mu}\cdot\mathbf{B}$ and explain the separation into discrete beams. For successive analyzers along axes n and m, derive the transition probability $\operatorname{Tr}(P_+(n)P_+(m)) = (1+n\cdot m)/2$ and interpret the intermediate projection. Compare the two-way splitting with Gerlach–Stern, *Zeitschrift für Physik* **9** (1922). Distinguish the original observation of space quantization in silver atoms from its later spin interpretation, and state which discrete outcomes and frequency predictions test the projection postulate and Born rule.

Exercise 7.4 (Projective quantum states by reduction)**.** Let K be a finite-dimensional complex Hilbert space and regard its underlying real vector space as symplectic with

$$\omega(u,v) = -\operatorname{Im}\langle u,v\rangle.$$

For the scalar $U(1)$-action, identify its Lie algebra with $\mathbb{R}$ acting infinitesimally by $z \mapsto iz$, and prove that $J(z) = \|z\|^2/2$ is a momentum map. Show that reduction at 1/2 gives

$$K/\!/_{1/2}U(1) = S(K)/U(1) = \mathbb{P}(K),$$

the space of complex lines in K. Identify $[z] \in \mathbb{P}(K)$ with the rank-one density operator

$$P_z : v \longmapsto z\langle z, v\rangle$$

for a unit representative z, and hence with the pure state $A \mapsto \mathrm{Tr}(P_z A)$. Prove that the reduced symplectic form is the Fubini–Study form, characterized on horizontal representatives $u, v \in z^\perp$ by

$$(\omega_{\mathrm{FS}})_{[z]}([u],[v]) = -\operatorname{Im}\langle u, v\rangle.$$

Thus normalization and quotient by global phase produce the pure quantum state space as a symplectic reduction.

Exercise 7.5 (Canonical quantization and Weyl relations)**.** On $\mathcal{S}(\mathbb{R}^n) \subset L^2(\mathbb{R}^n)$, set $Q_j\psi = x_j\psi$ and $P_j\psi = -i\hbar\partial_j\psi$. Prove $[Q_j, P_k] = i\hbar\delta_{jk}$ and derive the Weyl relations for their unitary groups. Using symmetric ordering, verify the Poisson-bracket–commutator rule for affine and quadratic observables. Compare different orderings of q^2p^2 and explain the Groenewold–van Hove obstruction to quantizing all observables by this rule.

Exercise 7.6 (Quantized harmonic oscillator)**.** For $H = P^2/(2m) + m\omega^2Q^2/2$, define creation and annihilation operators on the Schwartz core $\mathcal{S}(\mathbb{R})$ and prove there that $[a, a^*] = 1$ and $H = \hbar\omega(a^*a + 1/2)$. Construct the eigenstates with $E_n = \hbar\omega(n + 1/2)$ and use the spectral-lines exercise to obtain the frequency spacing ω. Prove that their span is a core for the self-adjoint closure of H. For a two-level internal transition of frequency Ω_0 coupled linearly to Q, compute the oscillator matrix elements, obtain motional sidebands at $\Omega_0 \pm \omega$, and compare their spacing with trapped-ion spectra.

Exercise 7.7 (Liouville and von Neumann dynamics)**.** Prove that the Born rule defines a probability measure and that its expected value is $\mathrm{Tr}(\rho A)$ when defined. Derive the following classical and quantum evolution equations.

Let $\mu_t = (\phi_t)_*\mu_0$ have smooth density p_t relative to the Liouville measure μ_L of a time-independent Hamiltonian system. Using Exercise 3.7, prove

$$\partial_t p_t + \{p_t, H\} = 0, \qquad \frac{d}{dt}\int_M f\, d\mu_t = \int_M \{f, H\}\, d\mu_t.$$

For $U_t = e^{-itH/\hbar}$, set $\rho_t = U_t\rho_0 U_t^*$ and $A_t = U_t^* A_0 U_t$. Prove

$$\dot\rho_t = -\frac{i}{\hbar}[H, \rho_t], \qquad \dot A_t = \frac{i}{\hbar}[H, A_t].$$

Prove

$$\int_M f\, d\mu_t = \int_M (f \circ \phi_t)\, d\mu_0, \qquad \mathrm{Tr}(\rho_t A_0) = \mathrm{Tr}(\rho_0 A_t),$$

and identify these as the equivalence of Schrödinger and Heisenberg pictures. Compare the derivations $f \mapsto \{f, H\}$ and $A \mapsto (i\hbar)^{-1}[A, H]$, and prove conservation of expected energy in both theories.

Exercise 7.8 (Blackbody radiation from quantized Fourier modes)**.** For electromagnetic radiation in a large conducting cavity, use its Fourier modes and two polarizations to derive the mode density $g(\omega) = \omega^2/(\pi^2c^3)$ per unit volume. Show that classical Gibbs equipartition assigns mean energy k_BT to each mode and produces the ultraviolet divergence. Using the oscillator levels derived in Exercise 7.6, compute the quantum Gibbs partition function and the thermal mean energy $\hbar\omega/(e^{\beta\hbar\omega} - 1)$. Deduce Planck's law

$$u(\omega, T) = \frac{\hbar\omega^3}{\pi^2c^3}\frac{1}{e^{\hbar\omega/(k_BT)} - 1},$$

the Stefan–Boltzmann T^4 law, and Wien scaling $\omega_{\max} \propto T$. Compare the low- and high-frequency limits with the cavity spectra discussed by Planck, *Annalen der Physik* **309** (1901), and identify precisely where quantization removes the classical divergence.

7.3 Fermionic statistics and the CAR algebra

Definition 7.5 (CAR algebra and fermionic Fock space)**.** For a complex Hilbert space K, the *canonical anticommutation relations algebra* $\mathrm{CAR}(K)$ is the unital C^*-algebra generated by conjugate-linear symbols $a(f)$ satisfying

$$\{a(f), a(g)\} = 0, \qquad \{a(f), a(g)^*\} = \langle f, g\rangle 1.$$

Let $\Lambda_2^n K$ be the Hilbert-space completion of the nth antisymmetric tensor power of K, with $\Lambda_2^0 K = \mathbb{C}$. The *antisymmetric Fock space* is the Hilbert direct sum

$$\mathcal{F}_-(K) = \bigoplus_{n\geq 0} \Lambda_2^n K.$$

Its vacuum is $\Omega = 1 \in \Lambda_2^0 K$; creation by f is exterior multiplication, and annihilation by f is its Hilbert-space adjoint.

Exercise 7.9 (CAR and GNS predict fermionic antibunching)**.** Prove that creation and annihilation on $\mathcal{F}_-(K)$ satisfy the CAR and that $\omega_0(x) = \langle \Omega, x\Omega\rangle$ is a state. Apply the GNS theorem to identify its cyclic representation with the Fock representation. Let ω_C be a gauge-invariant quasifree state with one-particle density $0 \leq C \leq 1$, and use its determinantal Wick rule to prove, for orthonormal detector modes f, g, that

$$\omega_C(N_f N_g) = \omega_C(N_f)\omega_C(N_g) - |\langle f, Cg\rangle|^2, \qquad N_f = a(f)^* a(f).$$

Deduce $N_f^2 = N_f$ and fermionic antibunching. Compare the sign and separation dependence of the correlation deficit with the neutral-atom measurements in Fölling et al., *Nature* **444** (2006).

Finite systems use matrix algebras, but a thermodynamic phase contains effectively infinitely many degrees of freedom. Its norm-closed algebra records all observables localized in finite regions. Macroscopic averages need not converge in norm, because the norm tests their action in every state, including configurations that differ arbitrarily far away. Once a physical state is chosen, however, its GNS representation supplies a weaker notion of convergence: in an up- or down-magnetized phase, the spatially averaged spin can converge strongly to the measured magnetization. A mixture of phases instead produces a central limiting observable whose spectral projections distinguish the macroscopic alternatives. This is the physical reason to pass from the local C^*-algebra to its strong or weak operator closure.

7.4 Von Neumann algebras and noncommutative conditioning

Definition 7.6 (Operator topologies and von Neumann algebras)**.** The *weak operator topology* is generated by $T \mapsto \langle x, Ty\rangle$; the *strong operator topology* is generated by $T \mapsto \|Tx\|$. For $S \subseteq B(H)$, its *commutant* is $S' = \{T : TS = ST \text{ for all } S \in S\}$. A *von Neumann algebra* is a unital $*$-subalgebra $M \subseteq B(H)$ with $M = M''$.

Exercise 7.10 (Bicommutant theorem)**.** Prove that for a unital $*$-subalgebra $A \subseteq B(H)$, the strong and weak operator closures of A equal A''. Deduce that von Neumann algebras have sufficient projections to approximate their self-adjoint elements spectrally.

Exercise 7.11 (Macroscopic magnetization and von Neumann closure)**.** Let $\mathcal{A}_N = M_2(\mathbb{C})^{\otimes N}$, include $\mathcal{A}_N$ in $\mathcal{A}_{N+1}$ by $a \mapsto a \otimes 1$, and form the quasilocal spin algebra

$$\mathcal{A} = \overline{\bigcup_{N\geq 1} \mathcal{A}_N}^{\|\cdot\|}.$$

For $Z = \mathrm{diag}(1, -1)$, let Z_j act as Z at site j and set

$$M_N = \frac{1}{N}\sum_{j=1}^{N} Z_j.$$

Prove $\|M_{2N} - M_N\| = 1$, so (M_N) has no norm limit in $\mathcal{A}$.

Let $|+\rangle, |-\rangle$ be the Z-eigenvectors and define product states on local elementary tensors by

$$\omega_\pm(a_1 \otimes \cdots \otimes a_N) = \prod_{j=1}^{N} \langle \pm | a_j | \pm \rangle.$$

Prove that these extend to states on $\mathcal{A}$. In their GNS triples $(H_\pm, \pi_\pm, \Omega_\pm)$, prove

$$\pi_+(M_N) \longrightarrow I_{H_+}, \qquad \pi_-(M_N) \longrightarrow -I_{H_-}$$

in the strong operator topology. First prove the convergence on vectors $\pi_\pm(a)\Omega_\pm$ with $a \in \mathcal{A}_k$, using

$$\|[M_N, a]\| \leq \frac{2k}{N}\|a\|,$$

and then extend it to all vectors using $\|M_N\| \leq 1$.

On $H_+ \oplus H_-$, let $\pi = \pi_+ \oplus \pi_-$ and $\Omega = 2^{-1/2}(\Omega_+ \oplus \Omega_-)$. Prove that Ω represents the mixed state $\omega = (\omega_+ + \omega_-)/2$ and is cyclic. Show that

$$\pi(M_N) \longrightarrow C = I_{H_+} \oplus (-I_{H_-})$$

strongly, that $C \in \pi(\mathcal{A})'' \cap \pi(\mathcal{A})'$, and that the spectral projections $(I \pm C)/2$ select the two magnetized phases. Thus the C^*-algebra describes the common local experiments, while its von Neumann closure in a chosen state also contains the macroscopic phase observable. Interpret this as an idealized zero-temperature ferromagnet; in an extremal low-temperature phase of an interacting ferromagnet, the corresponding limit is $\pm m_\beta I$ under the appropriate ergodic or clustering hypotheses.

The passage to a von Neumann closure introduces limits of observables, so a physical state should assign compatible limits to their probabilities. For example, if projections p_n describe an increasing sequence of detector acceptance regions with limit p, one expects $\Pr(p_n) \uparrow \Pr(p)$. Normality is precisely this continuity requirement, extended from sequences to the nets needed for general von Neumann algebras.

Definition 7.7 (Normality)**.** A positive functional φ on a von Neumann algebra is *normal* if $a_\lambda \uparrow a$ implies $\varphi(a_\lambda) \uparrow \varphi(a)$ for every bounded increasing net of positive elements. A representation is *normal* if it preserves such suprema.

Independent quantum systems still combine on $H \otimes K$, but their observable algebras must now contain the weak limits of finite sums of product observables $a \otimes b$. Taking that closure gives the tensor product appropriate to normal states and measurements.

Definition 7.8 (Spatial tensor products)**.** For concrete von Neumann algebras $M \subseteq B(H)$ and $N \subseteq B(K)$, their *spatial tensor product* is

$$M \mathbin{\overline{\otimes}} N = (M \odot N)'' \subseteq B(H \otimes K).$$

Conditioning is the complementary operation: it replaces the full algebra by the information retained in a chosen subalgebra while preserving the probabilities of the events already visible there.

Definition 7.9 (Noncommutative conditional expectation)**.** For a von Neumann subalgebra $N \subseteq M$, a *conditional expectation* is a normal positive unital projection $E : M \to N$ satisfying

$$E(n_1 a n_2) = n_1 E(a) n_2 \qquad (a \in M,\ n_1, n_2 \in N).$$

It is *φ-preserving* for a normal state φ when $\varphi \circ E = \varphi$. When $M = L^\infty(\Omega, \mathcal{F}, \mathbb{P})$ and $N = L^\infty(\Omega, \mathcal{G}, \mathbb{P})$, this recovers classical conditional expectation onto $\mathcal{G}$.

Exercise 7.12 (Conditioning on subalgebras)**.** Show that classical conditional expectation is positive, unital, contractive, and satisfies the bimodule identity above. For a normal state ψ on N, prove that

$$\mathrm{id}_M \mathbin{\overline{\otimes}} \psi : M \mathbin{\overline{\otimes}} N \longrightarrow M$$

is a normal conditional expectation onto $M \otimes 1$ after the canonical identification, and determine which product states it preserves. If M is a finite von Neumann algebra with normalized trace τ and $N \subseteq M$, show that the orthogonal projection $L^2(M, \tau) \to L^2(N, \tau)$ restricts to the unique τ-preserving conditional expectation $M \to N$. Derive the tower property for nested von Neumann subalgebras.

Even factors can describe very different infinite systems. The operational distinction is how finely a yes–no experiment, represented by a projection, can be subdivided. Matrix algebras have atomic projections; type-II factors allow a continuous notion of projection size; and in type-III factors every nonzero projection is infinite in the Murray–von Neumann sense. This last behavior is typical of local observable algebras in relativistic quantum field theory and explains why an intrinsic density matrix and trace need not exist for a local region.

7.5 Factor types, modular theory, and relative entropy

Definition 7.10 (Traces and factor types)**.** A *finite trace* on a von Neumann algebra is a normal positive functional τ satisfying $\tau(ab) = \tau(ba)$. A possibly unbounded *normal tracial weight* is an additive, positively homogeneous map $\tau : M_+ \to [0, \infty]$ satisfying $\tau(x^*x) = \tau(xx^*)$ and $\tau(x_\lambda) \uparrow \tau(x)$ whenever $x_\lambda \uparrow x$. It is *semifinite* if every nonzero $x \in M_+$ majorizes a nonzero $y \in M_+$ with $\tau(y) < \infty$. Thus the standard traces on infinite type-I and type-II$_\infty$ factors are normal semifinite weights, not finite functionals.

Projections $p, q \in M$ are *Murray–von Neumann equivalent*, written $p \sim q$, if some $v \in M$ satisfies $v^*v = p$ and $vv^* = q$. A projection p is *finite* if $q \le p$ and $q \sim p$ imply $q = p$, and it is *minimal* if its only subprojections are 0 and p.

A *factor* is a von Neumann algebra with center $\mathbb{C}1$. A factor is *type I* if it has a nonzero minimal projection, equivalently if it is isomorphic to some $B(K)$; it is *type II* if it has nonzero finite projections but no minimal projections; and it is *type III* if it has no nonzero finite projection. Thus a type-III factor has no normal tracial state and no intrinsic trace-density representation of its normal states. A factor is *hyperfinite* if it is the weak closure of an increasing union of finite-dimensional $*$-subalgebras.

Exercise 7.13 (Projection dimension and factors)**.** For a finite factor with normalized trace τ, prove additivity and unitary invariance of τ on projections. Relate the possible projection values to matrix factors and to diffuse finite factors. Prove that $B(K)$ is type I and that a factor with a faithful normal tracial state cannot be type III. Explain why finite-dimensional quantum systems see only type-I factors, whereas the classification permits genuinely different infinite systems.

For a finite system in a faithful state ρ, the operator $K_\rho = -\log\rho$ packages the state into the canonical flow $a \mapsto \rho^{it} a \rho^{-it}$. An infinite system or a local region in quantum field theory may have no trace and no density operator belonging to its observable algebra, so this formula cannot be the definition. The replacement problem is therefore: can the pair consisting of an observable algebra and a faithful state determine its own equilibrium flow? Modular theory answers yes; cyclicity and separation are the vector-state form of the needed faithfulness.

Definition 7.11 (Cyclic and separating vectors)**.** For $M \subseteq B(H)$, a vector Ω is *cyclic* if $\overline{M\Omega} = H$ and *separating* if $a\Omega = 0$ implies $a = 0$. For such a pair (M, Ω), define the densely defined antilinear operator $S_0(a\Omega) = a^*\Omega$.

Definition 7.12 (KMS states)**.** With the present sign convention, a state φ is *β-KMS* for a flow α_t if, for analytic a, b,

$$\varphi(a\alpha_{t-i\beta}(b)) = \varphi(\alpha_t(b)a).$$

Theorem 7.13 (Tomita–Takesaki; used without proof)**.** *The operator S_0 is closable. Write the polar decomposition of its closure as $S = J\Delta^{1/2}$. Then*

$$\Delta^{it} M \Delta^{-it} = M, \qquad \sigma_t^{\Omega}(a) = \Delta^{it} a \Delta^{-it}$$

defines an automorphism group of M. The positive operator Δ is the modular operator*, and σ^{Ω} is the* modular flow*. The vector state of Ω is 1-KMS for this flow.*

For a faithful density operator ρ on H, modular flow on $B(H)$ is $\sigma_t^{\rho}(a) = \rho^{it} a \rho^{-it}$, and $K_\rho = -\log\rho$ is its *modular Hamiltonian.*

Exercise 7.14 (Bisognano–Wichmann flow and its experimental limit)**.** Let $W_R = \{x : x^1 > |x^0|\}$ be the right Rindler wedge of Minkowski space, and let Ω be the vacuum of a relativistic local net with stress-energy density T_{00}. Assume the Bisognano–Wichmann theorem

$$\Delta_{W_R}^{it} = U(\Lambda_{W_R}(-2\pi t)),$$

which identifies vacuum modular flow with the Lorentz boosts preserving W_R; see Bisognano–Wichmann, *Journal of Mathematical Physics* **16** (1975). In units $\hbar = c = k_B = 1$, derive on the time-zero half-space the modular Hamiltonian

$$K_{W_R} = 2\pi \int_{x^1>0} x^1 T_{00}(0, \mathbf{x})\, d^{d-1}x$$

up to an additive constant. Use its KMS property to obtain the Unruh temperature $T = a/(2\pi)$ for proper acceleration a.

Discretize the integral for a half-chain with local energy terms h_i and derive the lattice Bisognano–Wichmann ansatz in which the coefficient of h_i grows with its distance from the entanglement cut. Compare the predicted locality and spatial temperature profile with the entanglement Hamiltonians learned in the 51-ion XXZ-chain experiment of Joshi et al., *Nature* **624** (2023). Explain why this tests a finite-lattice consequence of modular QFT but cannot by itself establish that a continuum local algebra is type III. Identify the covariance, positive-energy, and vacuum hypotheses used by the modular-flow prediction that are not consequences of the factor type alone.

Definition 7.14 (Relative entropy of operator-algebraic states)**.** Let φ and ψ be faithful normal states on M, represented by vectors Ω_φ and Ω_ψ in standard form. Their relative Tomita operator is the closure of

$$S_{\varphi|\psi,0}(a\Omega_\psi) = a^*\Omega_\varphi,$$

and their *relative modular operator* is

$$\Delta_{\varphi|\psi} = S_{\varphi|\psi}^* S_{\varphi|\psi}.$$

The *Araki relative entropy* is the extended nonnegative number

$$D_M(\varphi\|\psi) = \langle \Omega_\varphi, \log \Delta_{\varphi|\psi}\, \Omega_\varphi \rangle.$$

For $M = B(H)$ and faithful density operators ρ, σ, the standard Hilbert–Schmidt representation gives $\Delta_{\rho|\sigma}(X) = \rho X \sigma^{-1}$ and hence

$$D(\rho\|\sigma) = \mathrm{Tr}(\rho(\log\rho - \log\sigma))\,.$$

Unlike the entropy of an individual density matrix, relative entropy and modular flow are intrinsic to an algebra and its states and remain meaningful for type-III factors.

8 Free Probability

Experimental observations.

- Resonance spectra of compound nuclei and chaotic microwave cavities exhibit level repulsion and universal correlations determined mainly by symmetry rather than by the microscopic details of the system.

A noncommutative probability space admits several inequivalent notions of independence. Free products govern normalized traces of large random matrices: free cumulants determine their global laws, while Dyson's symmetry classes govern the local eigenvalue correlations seen in resonance experiments.

8.1 Free independence and additive convolution

Definition 8.1 (Free independence and free cumulants)**.** Unital subalgebras $(A_i)_i$ of (A, φ) are *freely independent* if

$$\varphi(a_1 \cdots a_m) = 0$$

whenever $a_j \in A_{i_j}$, $\varphi(a_j) = 0$, and neighboring indices are distinct. Variables are free when their generated unital subalgebras are.

Let $NC(n)$ be the lattice of noncrossing partitions of $\{1, \ldots, n\}$. The *free cumulants* κ_n are the multilinear functionals determined recursively by

$$\varphi(a_1 \cdots a_n) = \sum_{\pi \in NC(n)} \kappa_\pi(a_1, \ldots, a_n),$$

where κ_π is the product of the cumulants associated to the blocks of π. Freeness is equivalently the vanishing of every mixed free cumulant.

Definition 8.2 (Semicircular variables)**.** A self-adjoint variable S is *semicircular* with variance σ^2 if its distribution is

$$d\mu_{\mathrm{sc},\sigma}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2 - x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x)\,dx.$$

Equivalently, its only nonzero free cumulant is $\kappa_2(S,S) = \sigma^2$.

Exercise 8.1 (Free central limit theorem)**.** Use noncrossing pairings to prove

$$\varphi(S^{2m+1}) = 0, \qquad \varphi(S^{2m}) = C_m\sigma^{2m}, \qquad C_m = \frac{1}{m+1}\binom{2m}{m}.$$

Let $X_1, X_2, \ldots$ be freely independent, identically distributed, centered self-adjoint variables with $\varphi(X_i^2) = \sigma^2$ and moments of every order. Use multilinearity and vanishing mixed cumulants to prove that the moments of $n^{-1/2}(X_1 + \cdots + X_n)$ converge to these semicircular moments. Compare this free central limit theorem with the Gaussian limit for classically independent variables.

Definition 8.3 (Cauchy and R-transforms)**.** For a compactly supported probability measure μ on $\mathbb{R}$, its *Cauchy transform* is

$$G_\mu(z) = \int_{\mathbb{R}} \frac{d\mu(t)}{z-t}, \qquad z \in \mathbb{C} \setminus \operatorname{supp}\mu.$$

Near infinity it has a local compositional inverse $K_\mu(w) = G_\mu^{\langle -1\rangle}(w)$ near $w = 0$. The *R-transform* is defined by

$$K_\mu(w) = \frac{1}{w} + R_\mu(w), \qquad R_\mu(w) = \sum_{n\geq 1}\kappa_n(\mu)w^{n-1}.$$

If freely independent self-adjoint variables X, Y have laws μ, ν, then the law of $X+Y$ is their *free additive convolution* $\mu \boxplus \nu$, characterized by

$$R_{\mu\boxplus\nu} = R_\mu + R_\nu.$$

Exercise 8.2 (Free convolution and the semicircle equation)**.** Derive the additivity of the R-transform from the vanishing of mixed free cumulants. Show that a centered semicircular law of variance σ^2 has $R(w) = \sigma^2 w$ and hence

$$\sigma^2 G(z)^2 - zG(z) + 1 = 0.$$

Select the solution satisfying $zG(z) \to 1$ at infinity and recover the semicircle density from the Stieltjes inversion formula. Deduce that the free sum of semicircular variables of variances σ_1^2 and σ_2^2 is semicircular of variance $\sigma_1^2 + \sigma_2^2$, and rederive the free central limit theorem at the level of transforms.

8.2 Random matrices and spectral statistics

Definition 8.4 (Wigner and Gaussian matrix ensembles)**.** A *Wigner matrix* H_N is an $N \times N$ Hermitian random matrix whose upper-triangular entries are independent and centered, with $\mathbb{E}|H_{ij}|^2 = \sigma^2/N$ off the diagonal and, for each fixed m,

$$\sup_{N,i,j} \mathbb{E}|\sqrt{N}H_{ij}|^m < \infty.$$

Its *empirical spectral measure* is

$$L_{H_N} = \frac{1}{N}\sum_{j=1}^{N} \delta_{\lambda_j(H_N)}, \qquad \int x^k \, dL_{H_N}(x) = \frac{1}{N}\operatorname{Tr}(H_N^k).$$

The *Gaussian unitary ensemble* is the unitarily invariant Wigner ensemble with density proportional to $\exp[-N\operatorname{Tr}(H^2)/(2\sigma^2)]$ on Hermitian matrices. Sequences of random matrices are *asymptotically free* when their normalized joint trace moments converge to those of freely independent variables.

Exercise 8.3 (Wigner semicircle law and asymptotic freeness)**.** Expand $N^{-1}\mathbb{E}\operatorname{Tr}(H_N^k)$ as a sum over closed index walks. Show that only pair-matched walks survive as $N \to \infty$, that crossing pairings are lower order, and that the limiting moments are the semicircular moments. Prove that the variance tends to zero and hence that L_{H_N} converges in probability, in moments, to $\mu_{\mathrm{sc},\sigma}$.

For independent GUE matrices $H_N^{(1)}, \dots, H_N^{(r)}$, apply Wick's rule to normalized mixed traces of centered polynomials and prove that the family converges in joint moments to freely independent semicircular variables. Explain why ordinary independence of matrix entries becomes free independence of the limiting noncommuting matrices.

Definition 8.5 (Dyson ensembles and unfolded spectra)**.** The Gaussian ensembles with Dyson index $\beta = 1, 2, 4$ consist respectively of real symmetric, complex Hermitian, and quaternionic self-dual matrices. They are the Gaussian orthogonal (GOE), unitary (GUE), and symplectic (GSE) ensembles. They model an antiunitary symmetry squaring to $+1$, no antiunitary symmetry, and an antiunitary symmetry squaring to -1 after Kramers pairs are removed. Their ordered eigenvalues have joint density

$$p_{N,\beta}(\lambda_1, \dots, \lambda_N) = \frac{1}{Z_{N,\beta}} \exp\left[-\frac{\beta N}{4\sigma^2}\sum_{j=1}^{N} \lambda_j^2\right] \prod_{i<j} |\lambda_i - \lambda_j|^{\beta}.$$

The Vandermonde factor suppresses coincident eigenvalues and produces *level repulsion.*

To compare spectra with nonconstant mean density, let $\overline{N}(E)$ be a smooth approximation to the cumulative level count. The *unfolded levels* and nearest-neighbor spacings are

$$x_j = \overline{N}(E_j), \qquad s_j = x_{j+1} - x_j,$$

so their local mean spacing is one. Exact symmetry sectors must be separated before unfolding, since levels from independent sectors do not repel.

Exercise 8.4 (Level repulsion and resonance spectra)**.** Use the eigenvalue–eigenvector change of variables to derive the Vandermonde factor in the Gaussian eigenvalue density. For 2×2 GOE and GUE matrices, separate the trace from the traceless coordinates and compute the radial distribution of the eigenvalue gap. After rescaling to unit mean spacing, obtain the Wigner surmises

$$P_1(s) = \frac{\pi}{2} s e^{-\pi s^2/4}, \qquad P_2(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}.$$

Explain from the joint density why the full ensembles satisfy $P_\beta(s) \sim c_\beta s^\beta$ as $s \downarrow 0$, in contrast with the Poisson spacing law e^{-s} for uncorrelated levels.

For the GUE, take the orthogonal-polynomial determinant formula for the k-point correlation functions and the bulk asymptotics of the Hermite kernel as input. Show that unfolding gives the sine kernel

$$K_{\sin}(x, y) = \frac{\sin \pi(x-y)}{\pi(x-y)}, \qquad R_2(s) = 1 - \left(\frac{\sin \pi s}{\pi s}\right)^2.$$

This local limit is universal even though the global density depends on the matrix ensemble.

Finally, describe how one must separate nuclear levels by spin and parity, or microwave resonances by exact cavity symmetry, fit $\overline{N}(E)$, unfold, and then compare spacing and correlation statistics with the appropriate Dyson class. Time-reversal-invariant compound nuclei and chaotic microwave billiards without spin are predicted by the GOE. Compare with Haq–Pandey–Bohigas, *Physical Review Letters* **48** (1982) and Stöckmann–Stein, *Physical Review Letters* **64** (1990). Identify missing or misassigned levels and an inaccurate unfolding function as experimental effects that can imitate deviations from universality.

8.3 Bulk and edge universality

The GUE calculation above is exact because unitary invariance converts its eigenvalues into a determinantal point process. Universality is the much stronger assertion that the same microscopic limits survive after Gaussian entries and unitary invariance are removed. The matrix does not become Gaussian: only selected local eigenvalue statistics forget the entry distribution.

Definition 8.6 (Global, bulk, and edge scales)**.** Order the eigenvalues of a Wigner matrix as $\lambda_1 \le \cdots \le \lambda_N$, and let

$$\rho_{\mathrm{sc}}(x) = \frac{1}{2\pi\sigma^2}\sqrt{4\sigma^2 - x^2}\,\mathbf{1}_{[-2\sigma,2\sigma]}(x).$$

The *classical location* γ_j is defined by

$$\frac{j}{N} = \int_{-2\sigma}^{\gamma_j} \rho_{\mathrm{sc}}(x)\,dx.$$

The semicircle law is a *global law*: it concerns intervals containing order N eigenvalues. At a bulk energy $E \in (-2\sigma, 2\sigma)$, the microscopic coordinate is

$$u = N\rho_{\mathrm{sc}}(E)(\lambda - E),$$

so the mean spacing is normalized to one. At the upper edge the density vanishes as a square root and the microscopic coordinate is

$$\xi = \frac{N^{2/3}}{\sigma}(\lambda - 2\sigma).$$

Bulk and edge universality concern the limiting point processes in these coordinates, not merely convergence of the empirical measure.

Exercise 8.5 (Deriving the microscopic scales)**.** At a bulk point E, use the expected count $N\rho_{\mathrm{sc}}(E)\,dE$ to derive the spacing scale $[N\rho_{\mathrm{sc}}(E)]^{-1}$. Next put $x = 2\sigma - s$ and show

$$\rho_{\mathrm{sc}}(2\sigma - s) \sim \frac{1}{\pi\sigma^{3/2}}\sqrt{s} \qquad (s \downarrow 0).$$

Estimate the expected number of eigenvalues in $[2\sigma - s, 2\sigma]$ and show that it becomes order one when s is of order $\sigma N^{-2/3}$. Explain why a single rescaling cannot describe both the bulk and the edge.

Theorem 8.7 (Bulk GUE universality; used without proof)**.** *Let H_N be a genuinely complex Hermitian Wigner ensemble satisfying the moment assumptions above and the standard nondegeneracy hypotheses. Fix $E \in (-2\sigma, 2\sigma)$. Its local eigenvalue process, rescaled by $N\rho_{\mathrm{sc}}(E)$, converges in the* *sense of smooth compactly supported test functions to the GUE sine process. When the k-point correlation density $\rho_N^{(k)}$ exists, this says*

$$\begin{aligned}\frac{1}{[N\rho_{\mathrm{sc}}(E)]^k}\rho_N^{(k)}\left(E+\frac{u_1}{N\rho_{\mathrm{sc}}(E)},\dots,E+\frac{u_k}{N\rho_{\mathrm{sc}}(E)}\right)\\ \longrightarrow \det[K_{\sin}(u_a,u_b)]_{a,b=1}^k\,.\end{aligned}\tag{BU}$$

For discrete entry laws, the same statement is formulated using correlation measures or observables of consecutive gaps. The corresponding real symmetric limit is GOE rather than GUE. Thus the symmetry index β, not Gaussianity of the entries, determines the bulk limit.

This theorem is a deep input. A broad version is proved in Erdős–Yau–Yin, Probability Theory and Related Fields ***154*** *(2012).*

The proof has three conceptual layers. First, a *local semicircle law* controls

$$G_N(z)=\frac{1}{N}\operatorname{Tr}(z-H_N)^{-1}$$

even when $\operatorname{Im} z$ is only slightly larger than N^{-1}; this implies eigenvalue rigidity near the classical locations. Second, adding a tiny Gaussian matrix evolves the eigenvalues by Dyson Brownian motion, whose local statistics rapidly relax to the appropriate Gaussian ensemble. Third, a resolvent comparison removes the small Gaussian component. This explains the mechanism without pretending that the estimates needed for a proof follow from the elementary moment calculation of the semicircle law.

Definition 8.8 (Airy kernel and the Tracy–Widom law)**.** Let Ai be the solution of $y''=xy$ that decays as $x\to+\infty$. The *Airy kernel* is

$$K_{\mathrm{Ai}}(x,y)=\frac{\mathrm{Ai}(x)\,\mathrm{Ai}'(y)-\mathrm{Ai}'(x)\,\mathrm{Ai}(y)}{x-y},$$

with the diagonal defined by continuity. It determines the Airy determinantal point process at the soft edge. The GUE Tracy–Widom distribution is

$$F_2(s)=\det(I-K_{\mathrm{Ai}})_{L^2(s,\infty)}\,,$$

where the right side is the Fredholm determinant of the Airy integral operator. This formula is taken as a definition here; its analysis requires more operator theory than the present section develops.

Theorem 8.9 (Edge universality; used without proof)**.** *Under the Wigner hypotheses above,*

$$\frac{N^{2/3}}{\sigma}(\lambda_N-2\sigma)\ \xrightarrow{d}\ \chi_2,\qquad \Pr(\chi_2\le s)=F_2(s)\tag{EU}$$

for the complex Hermitian symmetry class. More strongly, the joint process of the top eigenvalues converges to the Airy point process. The lower edge is analogous, while the real and quaternionic classes have the Tracy–Widom laws F_1 and F_4. The Airy-kernel formula was obtained by Tracy and Widom, Communications in Mathematical Physics ***159*** *(1994); edge universality for Wigner matrices was established, under successively weakened hypotheses, beginning with Soshnikov,* Communications in Mathematical Physics ***207*** *(1999).*

Exercise 8.6 (The Airy diagonal). Use the Airy equation and l'Hôpital's rule to show

$$K_{\mathrm{Ai}}(x,x) = \mathrm{Ai}'(x)^2 - x\,\mathrm{Ai}(x)^2.$$

For a determinantal point process, the integral of the diagonal kernel over an interval is the expected number of points there. Interpret

$$\int_s^\infty K_{\mathrm{Ai}}(x,x)\,dx$$

as the expected number of rescaled edge eigenvalues above s. Explain why this contains information that the semicircle density, which vanishes at the edge, cannot resolve.

Exercise 8.7 (A global law does not determine local statistics). Let D_N be the deterministic diagonal matrix with entries $\gamma_1, \dots, \gamma_N$. Prove from Riemann sums that its empirical spectral measure converges to the same semicircle law as a Wigner matrix. For indices $j/N \to \alpha \in (0,1)$, use the inverse function theorem to show

$$N\rho_{\mathrm{sc}}(\gamma_j)(\gamma_{j+1} - \gamma_j) \longrightarrow 1.$$

Thus the unfolded spectrum of D_N approaches a rigid clock, whereas GUE has the random sine process and level repulsion. Identify precisely why the moment proof of the global semicircle law cannot distinguish these local behaviors.

Universality therefore has a definite domain, not a claim that every large Hamiltonian is described by GUE. Exact symmetry sectors must still be separated; time-reversal symmetry changes β; integrable or localized systems commonly have Poisson rather than Wigner–Dyson statistics; and sparse, banded, or heavy-tailed ensembles can require different hypotheses or exhibit crossovers. Within the mean-field Wigner class, however, (BU) and (EU) make precise the physical statement that microscopic spectral fluctuations depend chiefly on symmetry and spectral location, not on the detailed distribution of matrix entries.

9 Quantum Information Theory

Experimental observations.

- Tomography of prepared entangled pairs reconstructs a nearly pure joint state while either subsystem alone has nearly equal two-outcome frequencies.

Physical subsystems combine by tensor products rather than free products. Measurements turn noncommutative states into classical distributions, relative entropy quantifies distinguishability, and completely positive maps describe the processing, loss, and protection of quantum information.

9.1 Measurements, entropy, and tomography

Definition 9.1 (Measurements). A *positive operator-valued measure* is a map E from measurable events to positive operators with $E(X) = 1$ that is countably additive in the weak operator topology. In state ρ, its outcome probability is $\mathrm{Tr}(\rho E(B))$.

Exercise 9.1 (POVMs and dilation). Prove that projection-valued measures are POVMs. Prove Naimark's dilation theorem in finite dimensions, realizing every POVM's outcome probabilities by a projective measurement on a larger Hilbert space. Show that state discrimination can require a nonprojective POVM.

Definition 9.2 (Von Neumann entropy)**.** For a density operator with discrete spectral resolution $\rho = \sum_{\lambda>0} \lambda P_\lambda$, its *von Neumann entropy* is

$$S(\rho) = -k_B \operatorname{Tr}(\rho \log \rho) = -k_B \sum_{\lambda>0} (\operatorname{rank} P_\lambda) \lambda \log \lambda,$$

with $0 \log 0 = 0$; in infinite dimension the value may be $+\infty$. It is k_B times the Shannon entropy of the outcomes of a projective measurement in an eigenbasis of ρ, and it vanishes exactly for pure states.

Exercise 9.2 (Entropy of projective measurements)**.** For a rank-one projective measurement (P_i) in state ρ, let $p_i = \operatorname{Tr}(\rho P_i)$. Prove

$$S(p) \geq S(\rho),$$

with equality when the measurement basis diagonalizes ρ. Interpret the excess as uncertainty introduced by measuring in an incompatible basis. Show that unitary evolution preserves $S(\rho)$ even though it can change the outcome entropy of a fixed measurement.

A detector cannot measure Q and P jointly with arbitrary precision, but an optical local oscillator can select any rotated quadrature $Q \cos\theta + P \sin\theta$. This raises a tomography question: do the ordinary probability distributions from many such one-dimensional measurements determine the unknown continuous-variable state? The answer passes through the Wigner distribution: each quadrature histogram is one of its line projections, so varying the angle supplies the data for an inverse Radon transform.

Definition 9.3 (Quadratures, Wigner distributions, and homodyne data)**.** For one canonical degree of freedom, choose $q_0, p_0 > 0$ with $q_0 p_0 = \hbar$ and set

$$X = Q/q_0, \qquad Y = P/p_0, \qquad X_\theta = X \cos\theta + Y \sin\theta,$$

so that $[X, Y] = i$. For a density operator ρ, its *Weyl characteristic function* is

$$\chi_\rho(u, v) = \operatorname{Tr}\Big(\rho e^{i(uX+vY)}\Big),$$

and its *Wigner distribution* is the inverse Fourier transform

$$W_\rho(x, y) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \chi_\rho(u, v) e^{-i(ux+vy)} \, du \, dv,$$

interpreted as a tempered distribution when necessary. If E_θ is the spectral measure of X_θ, the *homodyne quadrature distribution* is

$$p_\rho(r \mid \theta) \, dr = \operatorname{Tr}\big(\rho E_\theta(dr)\big).$$

In balanced optical homodyne detection, the local-oscillator phase selects θ and repeated measurements estimate this distribution.

Exercise 9.3 (Optical homodyne tomography)**.** Use Exercise 5.15 and the spectral theorem to prove the Fourier-slice identity

$$\widetilde{p}_\rho(k \mid \theta) = \int_{\mathbb{R}} p_\rho(r \mid \theta) e^{ikr} \, dr = \chi_\rho(k \cos\theta, k \sin\theta).$$

Deduce, in the distributional sense, that the measured quadrature density is the Radon transform of the Wigner distribution:

$$p_\rho(r \mid \theta) = \int_{\mathbb{R}^2} W_\rho(x,y)\delta(r - x\cos\theta - y\sin\theta)\,dx\,dy.$$

Prove that the data for $0 \le \theta < \pi$ determine χ_ρ, hence W_ρ and ρ, and derive the filtered-backprojection formula

$$W_\rho(x,y) = \frac{1}{(2\pi)^2}\int_0^\pi \int_{\mathbb{R}} |k|\,\widetilde{p}_\rho(k \mid \theta)e^{-ik(x\cos\theta + y\sin\theta)}\,dk\,d\theta.$$

Show that every $p_\rho(\,\cdot \mid \theta)$ is a genuine probability density although W_ρ may be negative. Compute the data and reconstruction for vacuum, coherent, squeezed, and one-photon states. Explain why the factor $|k|$ amplifies high-frequency noise and why finite-angle experimental data require regularization or constrained density-matrix estimation. Compare the predicted reconstruction of vacuum and squeezed states with Smithey et al., *Physical Review Letters* **70** (1993).

Definition 9.4 (Quantum relative entropy)**.** For finite-dimensional density operators with $\operatorname{supp}\rho \le \operatorname{supp}\sigma$, their *quantum relative entropy* is

$$D(\rho\|\sigma) = \operatorname{Tr}(\rho(\log\rho - \log\sigma))\,,$$

and it is $+\infty$ otherwise. It reduces to classical relative entropy when ρ and σ commute.

Exercise 9.4 (Entropy and quantum Gibbs states)**.** Let K be n-dimensional, let $P_1,\dots,P_n$ be mutually orthogonal rank-one projections with sum 1_K, and let $\mathcal{D} = \operatorname{span}\{P_1,\dots,P_n\} \cong \mathbb{C}^n$. Identify a state on $\mathcal{D}$ with a probability vector $p = (p_i)$ and the density operator $\rho_p = \sum_i p_i P_i$. Prove

$$S(\rho_p) = -k_B\sum_i p_i \log p_i,$$

so von Neumann entropy restricts to classical entropy on a commutative observable algebra. Prove more generally that $S(U\rho U^*) = S(\rho)$ for every unitary U.

Let $H = H^* \in B(K)$ have at least two distinct eigenvalues, and let the prescribed energy E lie strictly between its smallest and largest eigenvalues. Prove that there is a unique $\beta \in \mathbb{R}$ for which

$$\rho_\beta = \frac{e^{-\beta H}}{Z(\beta)}, \qquad Z(\beta) = \operatorname{Tr}(e^{-\beta H}), \qquad \operatorname{Tr}(\rho_\beta H) = E.$$

Prove

$$D(\rho\|\rho_\beta) = -\frac{S(\rho)}{k_B} + \beta\operatorname{Tr}(\rho H) + \log Z(\beta),$$

and use nonnegativity to show that ρ_β uniquely maximizes entropy at fixed mean energy, equivalently minimizes free energy at fixed temperature. Derive

$$\operatorname{Tr}(\rho_\beta H) = -\partial_\beta \log Z, \qquad \operatorname{Var}_{\rho_\beta}(H) = \partial_\beta^2 \log Z.$$

Finally, restrict ρ_β to the commutative algebra generated by the spectral projections of H and recover the classical Gibbs probabilities, including their energy-level degeneracies. If g_i is the degeneracy of E_i and p_i is the total population of that energy level, derive

$$\frac{p_i}{p_j} = \frac{g_i}{g_j}e^{-\beta(E_i - E_j)}.$$

Explain why the degeneracy factor is absent for the ratio of populations of individual eigenstates, and compare both formulas with measured level-population ratios.

9.2 Composite systems and entanglement

Definition 9.5 (Composite and reduced states)**.** The Hilbert space of a composite system is $H_A \otimes H_B$, and its normal states are the density operators on this tensor product. A density operator is *separable* if it lies in the trace-norm closed convex hull of product density operators and *entangled* otherwise. The *reduced state* $\rho_A = \operatorname{Tr}_B \rho$ is determined by $\operatorname{Tr}(\rho_A A) = \operatorname{Tr}(\rho(A \otimes \mathrm{id}))$ for all bounded A.

Definition 9.6 (Classical and tensor independence)**.** Families of σ-algebras are *independent* when the probability of every finite intersection of events, one from each family, is the product of their probabilities. Equivalently, bounded random variables measurable with respect to distinct families have factorizing mixed expectations.

Commuting unital subalgebras $M_1, M_2 \subseteq M$ are independent in the state φ when $\varphi(ab) = \varphi(a)\varphi(b)$ for $a \in M_1$ and $b \in M_2$. They are *tensor independent* when the algebra they generate, together with the state, is a spatial tensor product with a product state.

Exercise 9.5 (Normal states and independent subsystems)**.** Show that every density operator ρ gives the normal state $a \mapsto \operatorname{Tr}(\rho a)$, and, using trace-class duality, prove the converse on $B(H)$. Prove $B(H) \mathbin{\overline{\otimes}} B(K) = B(H \otimes K)$ and identify product states and partial traces in this language. Explain why local normality—normality on every local algebra—is the infinite-system analogue of requiring ordinary reduced density matrices for all finite laboratories.

Exercise 9.6 (Independent noncommutative subsystems)**.** Characterize classical independence by factorization of expectations and derive variance addition for independent square-integrable random variables. For commuting von Neumann subalgebras $M_1, M_2 \subseteq M$, compare factorization in a state φ with a tensor-product state when $M_1 \vee M_2$ is spatially isomorphic to $M_1 \mathbin{\overline{\otimes}} M_2$, and explain why the split property is the relevant replacement for sharply adjacent local regions in continuum QFT. Contrast this with free independence: show that freely independent centered variables satisfy vanishing alternating mixed moments without commuting or determining a joint classical law.

Exercise 9.7 (Wishart matrices and typical entanglement)**.** Let G be a $d_A \times d_B$ matrix of independent standard complex Gaussian entries and set

$$W = GG^*, \qquad \rho_A = \frac{W}{\operatorname{Tr} W}.$$

Show that ρ_A has the same distribution as the reduced state of a Haar-random unit vector in $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$. Using Gaussian or Haar integration, prove

$$\mathbb{E} \operatorname{Tr}(\rho_A^2) = \frac{d_A + d_B}{d_A d_B + 1}.$$

Deduce that when $d_B \gg d_A$, a typical reduced state is close to maximally mixed in Hilbert–Schmidt norm. Interpret this as a random-matrix explanation of why a typical pure state of a large composite system is highly entangled.

Definition 9.7 (Entanglement entropy and mutual information)**.** For a pure state $\rho = |\psi\rangle\langle\psi|$ on $H_A \otimes H_B$, its *entanglement entropy* is $S_A(\psi) = S(\rho_A)$; the Schmidt spectral data will show that this also equals $S(\rho_B)$. For an arbitrary bipartite state, its *quantum mutual information* is

$$I(A:B)_\rho = D(\rho \| \rho_A \otimes \rho_B) = \frac{S(\rho_A) + S(\rho_B) - S(\rho)}{k_B}.$$

It measures total correlation and vanishes exactly for product states.

Exercise 9.8 (Schmidt projections and entanglement)**.** Let H_A and H_B be finite-dimensional. For a pure vector $\psi \in H_A \otimes H_B$, apply the projection form of the singular-value decomposition to the contraction map $C_\psi : H_A^* \to H_B$. If Q_s and R_s are its domain and range projections, decompose ψ into mutually orthogonal spectral blocks supported on $(Q_s H_A^*)^* \otimes R_s H_B$. Explain why a rank-one refinement within a block of multiplicity greater than one is not canonical. Prove that a pure state is separable exactly when its reduced state has rank one, and that the two reduced states have the same nonzero spectral values with multiplicities $\operatorname{rank} Q_s = \operatorname{rank} R_s$. Derive

$$S_A(\psi) = -k_B \sum_{s>0} (\operatorname{rank} Q_s) s^2 \log s^2$$

and prove that it vanishes exactly for separable pure states. Compute the joint and reduced states, local outcome frequencies, and entanglement entropy of a Bell state, and compare them with entangled-pair tomography.

Exercise 9.9 (Finite modular theory and entanglement tomography)**.** For finite-dimensional H, let $B(H)$ act on itself by left multiplication with inner product $\langle X, Y\rangle = \operatorname{Tr}(X^*Y)$, and set $\Omega_\rho = \rho^{1/2}$ for faithful ρ. Prove that Ω_ρ is cyclic and separating. For $S_0(a\Omega_\rho) = a^*\Omega_\rho$, compute the polar decomposition and show that its modular operator is $\Delta(X) = \rho X \rho^{-1}$, yielding the flow above. If ρ is a reduced state, recover its entanglement spectrum from K_ρ. Apply this reconstruction at successive times after a quench and compare the resulting spectrum with the trapped-ion data analyzed by Kokail et al., *Nature Physics* **17** (2021), and explain why finite-subsystem tomography does not by itself establish continuum modular theory.

Exercise 9.10 (Type-III localization and the split property)**.** Assume the structural theorem that, under the usual nontriviality, phase-space, and scaling hypotheses, local observable algebras in relativistic continuum quantum field theory are hyperfinite type III_1 factors, the full-modular-spectrum case of the type-III classification; see Yngvason, *Reports on Mathematical Physics* **55** (2005). Deduce that the restriction of the vacuum to a sharply bounded region has no intrinsic reduced density operator and that $-\operatorname{Tr}(\rho_A \log \rho_A)$ is therefore not an intrinsic continuum entropy. Explain how a split inclusion

$$\mathcal{A}(O_1) \subset F \subset \mathcal{A}(O_2), \qquad \overline{O_1} \subset O_2,$$

with F type I supplies a density-matrix regulator whose entropy depends on the collar $O_2 \setminus \overline{O_1}$ and on the choice of F. Contrast this regulator dependence and the usual ultraviolet divergence as the collar shrinks with the intrinsic relative entropy $D_{\mathcal{A}(O_1)}(\varphi\|\psi)$.

Exercise 9.11 (Bell correlations and CHSH)**.** For a polarization-entangled two-photon state, derive the CHSH bound 2 for local hidden-variable models. Choose four polarization angles for which the quantum correlations attain $2\sqrt{2}$, and derive the predicted coincidence frequencies. As experimental input, use the adjusted local-realistic p-value 2.3×10^{-7} reported by Shalm et al., *Physical Review Letters* **115** (2015). Determine at the 5%, 1%, and five-standard-deviation significance levels, using the one-sided Gaussian five-sigma threshold 2.87×10^{-7}, whether the local-realistic null is excluded. State the tested freedom-of-choice and spacelike-separation assumptions and explain why the result does not exclude arbitrary superdeterministic or signaling models.

9.3 Quantum channels and data processing

Definition 9.8 (Positive and completely positive maps)**.** A linear map $\Phi : A \to B$ between C^*-algebras is *positive* if it preserves positive elements and *completely positive* if every $\mathrm{id}_{M_n} \otimes \Phi$ is positive. A *quantum channel* is a completely positive trace-preserving map on trace-class operators, equivalently a normal unital completely positive map on observables in the adjoint picture.

Exercise 9.12 (Stinespring and Kraus representations)**.** Prove that a unital completely positive map $\Phi : A \to B(H)$ has a representation $\Phi(a) = V^*\pi(a)V$. In finite dimensions, when $\Phi = \mathcal{N}^*$ is the Heisenberg adjoint of a Schrödinger-picture channel $\mathcal{N}$, derive the paired Kraus formulas

$$\mathcal{N}(\rho) = \sum_i K_i \rho K_i^*, \qquad \Phi(a) = \sum_i K_i^* a K_i, \qquad \sum_i K_i^* K_i = 1.$$

Prove that every channel is unitary evolution on a larger system followed by discarding an environment. Use the bimodule identity to prove that every conditional expectation introduced earlier is completely positive. For a finite POVM $(E_x)_x$, prove that

$$\mathcal{M}(\rho) = \sum_x \mathrm{Tr}(\rho E_x) |x\rangle\langle x|$$

is a quantum-to-classical channel, compute its adjoint, and identify its Stinespring dilation with the Naimark dilation constructed earlier.

Exercise 9.13 (Quantum data processing)**.** In finite dimensions, prove unitary invariance and joint convexity of quantum relative entropy. Average over a unitary error basis on H_B to prove

$$D(\mathrm{Tr}_B\, \rho \| \mathrm{Tr}_B\, \sigma) \le D(\rho\|\sigma).$$

Use a Stinespring dilation to deduce, for every channel $\mathcal{N}$,

$$D(\mathcal{N}(\rho)\|\mathcal{N}(\sigma)) \le D(\rho\|\sigma).$$

Interpret this as the impossibility of increasing distinguishability by a physical process. If one recovery channel reverses $\mathcal{N}$ on both ρ and σ, prove that equality holds. Apply data processing to a measurement channel and recover the classical inequality for its outcome distributions.

Exercise 9.14 (Relative entropy and data processing)**.** Prove the log-sum inequality and deduce $D(p\|q) \ge 0$, with equality exactly when $p = q$. For a stochastic map $K(y \mid x)$, prove

$$D(Kp\|Kq) \le D(p\|q).$$

Interpret a random variable, finite-resolution detector, or forgotten label as a stochastic map and explain why none can increase distinguishability. Prove $I(X : Y) = S(p_X)/k_B + S(p_Y)/k_B - S(p_{XY})/k_B$, and show that it vanishes exactly when X and Y are independent.

Exercise 9.15 (Amplitude damping reproduces spontaneous emission)**.** For a qubit with ground state $|0\rangle$, excited state $|1\rangle$, and environment vacuum $|0\rangle_E$, let

$$\begin{aligned} &V_t|0\rangle = |0\rangle|0\rangle_E, \\ &V_t|1\rangle = \sqrt{\eta_t}\, |1\rangle|0\rangle_E + \sqrt{1-\eta_t}\, |0\rangle|1\rangle_E, \qquad \eta_t = e^{-t/T_1}. \end{aligned}$$

In the Schrödinger picture, trace out the environment and derive

$$K_0 = |0\rangle\langle 0| + \sqrt{\eta_t}|1\rangle\langle 1|, \qquad K_1 = \sqrt{1-\eta_t}|0\rangle\langle 1|, \qquad \mathcal{N}_t(\rho) = K_0\rho K_0^* + K_1 \rho K_1^*.$$

Prove complete positivity, trace preservation, and the semigroup law. Derive $\rho_{11}(t) = e^{-t/T_1}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-t/(2T_1)}\rho_{01}(0)$. Compare the predicted decay and the environmental dependence of T_1 with the transmon measurements in Houck et al., *Physical Review Letters* **101** (2008).

9.4 Quantum error correction and operator systems

Definition 9.9 (Quantum codes and correctable errors)**.** Let $V : K \to H$ be an isometry between finite-dimensional Hilbert spaces. Its image $C = \operatorname{im} V$, with projection $P = VV^*$, is a *quantum code*; K is the logical system and H the physical system. Write $\mathcal{T}(H)$ for the trace-class operators on H. A channel $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ is *correctable on* C if there is a recovery channel $\mathcal{R}$ such that

$$(\mathcal{R} \circ \mathcal{N})(V\rho V^*) = V\rho V^*$$

for every state ρ on K. A *stabilizer code* is the common $+1$ spectral subspace of commuting self-adjoint unitaries. When $H = \bigotimes_{j=1}^n H_j$, the *distance* of C is the least number of tensor factors supporting an operator E for which PEP is not a scalar multiple of P, or $+\infty$ if there is no such operator.

Exercise 9.16 (Knill–Laflamme correction and decoupling)**.** Let $\mathcal{N}(\rho) = \sum_a E_a \rho E_a^*$ and let P project onto a code C. Prove that $\mathcal{N}$ is correctable on C exactly when there is a positive semidefinite matrix (λ_{ab}) such that

$$PE_a^* E_b P = \lambda_{ab} P \qquad \text{for every } a, b.$$

Show that this condition is independent of the chosen Kraus representation, and construct a recovery from the polar decomposition of the combined error map

$$W : C \otimes \ell^2(\{a\}) \longrightarrow H', \qquad W(\psi \otimes \delta_a) = E_a \psi,$$

together with the spectral projections of (λ_{ab}). Deduce that a code of distance d corrects every error supported on at most $\lfloor (d-1)/2 \rfloor$ tensor factors.

For an encoding $V : K \to H_A \otimes H_B$, prove that erasure of B is correctable exactly when the complementary channel $\rho \mapsto \operatorname{Tr}_A(V\rho V^*)$ is constant on logical states. Interpret this as the equivalence between recovery from A and absence of encoded information in B; see Knill–Laflamme, *Physical Review A* **55** (1997).

The Knill–Laflamme condition never uses arbitrary products of noise operators. It uses only the linear span of the pairwise overlaps $E_a^* E_b$, which records which input states the environment can confuse. This span is self-adjoint and contains the identity, but it need not be an algebra. Operator systems isolate exactly this amount of structure, while their matrix-order cones retain positivity when an arbitrary ancilla is attached.

Definition 9.10 (Operator systems)**.** An *operator system* is a closed self-adjoint linear subspace $S \subseteq B(H)$ containing the identity. Its matrix order is the family of cones

$$M_n(S)_+ = M_n(S) \cap B(H^{\oplus n})_+.$$

Completely positive maps are precisely the maps that preserve these cones at every matrix level.

Definition 9.11 (The confusability operator system of a channel)**.** Let H, H' be finite-dimensional and let $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ be a channel with Kraus operators $E_a : H \to H'$. Its *confusability operator system* is

$$S_{\mathcal{N}} = \operatorname{span}\{E_a^* E_b : a, b\} \subseteq B(H).$$

It is self-adjoint, and it contains the identity because $\sum_a E_a^* E_a = 1_H$.

Exercise 9.17 (Kraus independence of the confusability system)**.** If $\{E_a\}$ and $\{F_\mu\}$ are two Kraus representations of $\mathcal{N}$, pad the smaller family by zero operators and prove that there is a unitary matrix $(u_{\mu a})$ such that $F_\mu = \sum_a u_{\mu a} E_a$. Use this relation in both directions to prove

$$\mathrm{span}\{F_\mu^* F_\nu\} = \mathrm{span}\{E_a^* E_b\}.$$

Thus $S_\mathcal{N}$ depends only on the channel, not on a choice of environment coordinates. Interpret it as the noncommutative analogue of a classical channel's confusability graph; see Weaver, *Quantum Graphs as Quantum Relations*, Journal of Geometric Analysis **31** (2021).

Definition 9.12 (Protected observable algebras)**.** Let P project onto a code $C \subseteq H$. A unital $*$-subalgebra $\mathcal{A} \subseteq PB(H)P$, whose identity is P, is *protected* or *correctable* for $\mathcal{N}$ if there is a recovery channel $\mathcal{R} : \mathcal{T}(H') \to \mathcal{T}(H)$ such that

$$P(\mathcal{R} \circ \mathcal{N})^*(A)P = A \qquad (A \in \mathcal{A}).$$

Thus every observable in $\mathcal{A}$, rather than necessarily every encoded state, can be recovered. Write $\mathcal{A}' = \{X \in PB(H)P : [X, A] = 0 \text{ for all } A \in \mathcal{A}\}$ for the commutant in the code corner.

Exercise 9.18 (Exact operator-algebra quantum error correction)**.** Prove that $\mathcal{A}$ is correctable for $\mathcal{N}$ if and only if

$$[PE_a^* E_b P, A] = 0 \qquad (a, b,\ A \in \mathcal{A}),$$

or, equivalently,

$$PS_\mathcal{N}P \subseteq \mathcal{A}'.$$

Derive necessity by applying a Stinespring dilation to the recovery identity, and prove sufficiency by decomposing the finite-dimensional algebra and constructing a recovery on each summand. For $\mathcal{A} = PB(H)P \cong B(C)$, use $\mathcal{A}' = \mathbb{C}P$ to recover the ordinary Knill–Laflamme equations $PE_a^* E_b P = \lambda_{ab} P$. Show that a commutative algebra protects a classical label, while a decomposition

$$\mathcal{A} \cong \bigoplus_\alpha (B(H_\alpha^A) \otimes 1_{H_\alpha^B})$$

protects hybrid information: the central label α is classical, the H_α^A factor is quantum, and H_α^B is an unprotected gauge subsystem. Compare with Bény–Kempf–Kribs, *Quantum Error Correction of Observables*, Physical Review A **76** (2007).

Exercise 9.19 (Normal channels and von Neumann protected algebras)**.** Let H, H' be arbitrary Hilbert spaces, let $\mathcal{N} : \mathcal{T}(H) \to \mathcal{T}(H')$ be a channel whose adjoint is normal, and choose a Stinespring isometry $V : H \to H' \otimes K$. Define the complementary channel and its adjoint by

$$\mathcal{N}^c(\rho) = \mathrm{Tr}_{H'}(V\rho V^*), \qquad (\mathcal{N}^c)^*(Y) = V^*(1_{H'} \otimes Y)V.$$

For a projection P and a von Neumann algebra $\mathcal{A} \subseteq PB(H)P$, define correctability using a recovery channel with normal adjoint and the same observable identity as above. Prove the infinite-dimensional exact criterion

$$P(\mathcal{N}^c)^*(B(K))P \subseteq \mathcal{A}'.$$

Show from uniqueness of minimal Stinespring dilations that this condition is independent of the chosen complementary channel. Explain how it replaces a Kraus-index calculation when the environment is not finite-dimensional and allows protected algebras of every von Neumann type; see Bény–Kempf–Kribs, *Quantum Error Correction on Infinite-Dimensional Hilbert Spaces*, Journal of Mathematical Physics **50** (2009).

10 Operator Space Theory

Experimental observations.

- Entanglement-assisted probes distinguish certain measurement devices with a lower inconclusive probability than any unentangled single-system probe.

Tensoring a map with an ancillary identity exposes behavior invisible at the scalar level. Operator spaces encode every such matrix amplification; their tensor norms and completely bounded maps quantify assisted discrimination and the strength of nonlocal correlations.

After establishing this operational meaning of matrix norms, the section follows complete boundedness in two directions. One studies Fourier multipliers and then compact quantum groups as generalized symmetries; its simplest classical example also admits finite symmetry-preserving compressions. The other uses extension theorems and tensor norms to return to the comparison between classical and entangled correlations.

10.1 Matrix norms and channel discrimination

Definition 10.1 (Operator spaces)**.** An *operator space* is a closed subspace $E \subseteq B(H)$ with matrix norms inherited from $M_n(E) \subseteq B(H^{\oplus n})$. A map $u : E \to F$ is *completely bounded* if

$$\|u\|_{cb} = \sup_n \|\operatorname{id}_{M_n} \otimes u\| < \infty.$$

Thus an operator space records not only the norm of an observable but all norms obtained after coupling it to matrix-valued ancillary systems.

Definition 10.2 (Ancilla-assisted channel discrimination)**.** A *one-use discrimination strategy* for channels $\Phi_1, \ldots, \Phi_m$ prepares a state on $H \otimes K$, applies $\Phi_j \otimes \operatorname{id}_K$ to the unknown channel input, and measures the output with a POVM. The strategy is *ancilla assisted* when $\dim K > 1$. It is *unambiguous* when every conclusive outcome identifies the channel without error. Its *inconclusive probability* is denoted P_I.

Exercise 10.1 (An ancilla improves measurement-channel discrimination)**.** For $0 < \theta < \pi/4$, define

$$|\phi\rangle = \cos\theta|0\rangle + \sin\theta|1\rangle, \quad |\psi\rangle = \cos\theta|0\rangle - \sin\theta|1\rangle,$$

and let $\mathcal{M}, \mathcal{N}$ be the two projective measurement channels with rank-one projection families $\{P_\phi, P_{\phi^\perp}\}$ and $\{P_\psi, P_{\psi^\perp}\}$. Use a singlet probe, retain the second qubit as an ancilla, and condition on the classical measurement outcome. Reduce the task to unambiguous discrimination of $|\phi\rangle$ and $|\psi\rangle$ and prove

$$P_I^{\text{anc}} = \cos(2\theta), \qquad P_I^{\text{sep}} = \frac{1 + \cos^2(2\theta)}{2}$$

for the optimal ancilla-assisted and single-qubit strategies. Deduce the strict advantage for $0 < \theta < \pi/4$ and express it as a distinction between the first and amplified matrix levels of the measurement channels. Compare the predicted success and inconclusive probabilities with Miková et al., *Physical Review A* **90** (2014).

Exercise 10.2 (Concrete matrix-norm axioms)**.** Prove that concrete operator-space norms satisfy

$$\|x \oplus y\| = \max(\|x\|, \|y\|), \qquad \|axb\| \leq \|a\|\|x\|\|b\|.$$

Track the matrix sizes and prove that completely bounded maps preserve both direct sums and multiplication by scalar matrices at every level.

Exercise 10.3 (Hahn–Banach separation)**.** Let Y be a linear subspace of a complex normed space X. Use Zorn's lemma, first over the underlying real spaces and then after complexification, to prove that every bounded linear functional on Y extends to X without increasing its norm. Deduce that each nonzero $x \in X$ has a norm-one functional $f \in X^*$ with $f(x) = \|x\|$.

Exercise 10.4 (Matrix Hahn–Banach separation)**.** Let E carry matrix norms satisfying the two concrete matrix-norm axioms, and let $F \in M_n(E)^*$ have norm at most one. Apply geometric Hahn–Banach separation to the matrix unit ball and prove that there are states p, q on M_n such that

$$|F(ayb)| \le p(aa^*)^{1/2} \, \|y\|_n \, q(b^*b)^{1/2} \qquad (a, b \in M_n,\ y \in M_n(E)).$$

Represent p and q by unit vectors in amplifications of $\mathbb{C}^n$. Use the displayed inequality to construct a bounded sesquilinear form on their cyclic subspaces and hence a contraction $T : M_n(E) \to M_n(M_n)$ satisfying

$$T(ayb) = (a \otimes 1)T(y)(b \otimes 1), \qquad F(y) = \langle T(y)\xi, \eta \rangle.$$

Prove from the bimodule identity that $T = \phi^{(n)}$ for a linear map $\phi : E \to M_n$, and use the matrix-norm axioms to show that $\|\phi\|_{cb} = \|\phi^{(n)}\| \le 1$. Finally, for each $x \in M_n(E)$ choose a norming F and deduce

$$\|\phi^{(n)}(x)\| = \|x\|_n.$$

Exercise 10.5 (Ruan's representation theorem)**.** Prove conversely that every vector space with matrix norms satisfying these axioms admits a complete isometry into $B(H)$. Apply matrix Hahn–Banach separation to a norming family of matrices and take the direct sum of the resulting maps.

Operator spaces become especially concrete for degrees of freedom labelled by a symmetry group G. Left translations λ_s generate the regular representation, while their matrix coefficients are the associated Fourier amplitudes. Noise that is diagonal in these group modes multiplies them by a function on G; requiring the same bound after coupling an ancilla is exactly a completely bounded multiplier condition. This returns harmonic analysis to the operator-space setting for a physical reason, rather than as a separate change of subject.

10.2 Fourier algebras and completely bounded multipliers

Definition 10.3 (Group von Neumann and Fourier algebras)**.** For a locally compact group G with left regular representation λ on $L^2(G)$, its *group von Neumann algebra* is

$$VN(G) = \{\lambda_s : s \in G\}''.$$

The *Fourier algebra* $A(G)$ consists of the coefficient functions

$$u(s) = \langle \xi, \lambda_s \eta \rangle, \qquad \xi, \eta \in L^2(G),$$

with norm

$$\|u\|_{A(G)} = \inf\{\|\xi\| \, \|\eta\| : u(s) = \langle \xi, \lambda_s \eta \rangle\}.$$

The pairing $\langle u, \lambda_s \rangle = u(s)$ identifies $A(G) = VN(G)_*$ and equips $A(G)$ with its canonical predual operator-space structure. The normal unital $*$-homomorphism

$$\Delta : VN(G) \longrightarrow VN(G) \mathbin{\overline{\otimes}} VN(G), \qquad \Delta(\lambda_s) = \lambda_s \otimes \lambda_s,$$

is the *group coproduct*; its preadjoint gives pointwise multiplication on $A(G)$. This is the operator-algebraic Fourier duality introduced by Eymard, *Bulletin de la Société Mathématique de France* **92** (1964).

Exercise 10.6 (Fourier algebras as operator-space preduals)**.** For a discrete group G, prove that the canonical trace on $VN(G)$ satisfies $\tau(\lambda_s) = \delta_{s,e}$ and that the coefficient functions above separate the generators λ_s. Verify the coproduct identities

$$(\Delta \mathbin{\overline{\otimes}} \mathrm{id})\Delta = (\mathrm{id} \mathbin{\overline{\otimes}} \Delta)\Delta, \qquad \Delta(\lambda_s\lambda_t) = \Delta(\lambda_s)\Delta(\lambda_t).$$

Show from the predual matrix norms that pointwise multiplication on $A(G)$ is completely contractive at every matrix level. For abelian G, use Pontryagin duality to identify $A(G)$ with $L^1(\widehat{G})$ under the inverse Fourier transform and show that this pointwise product corresponds to convolution on $L^1(\widehat{G})$. Compare this coefficient-space viewpoint with the operator-valued Plancherel transform developed in Harmonic Analysis.

Definition 10.4 (Completely bounded Fourier multipliers)**.** A function $\phi : G \to \mathbb{C}$ is a *Fourier multiplier* when $m_\phi(u) = \phi u$ maps $A(G)$ to itself. It is *completely bounded* when m_ϕ is completely bounded for the predual operator-space structure. Its adjoint is the normal map

$$M_\phi : VN(G) \to VN(G), \qquad M_\phi(\lambda_s) = \phi(s)\lambda_s.$$

The Bożejko–Fendler transference theorem says that ϕ is completely bounded exactly when the invariant kernel

$$k_\phi(x, y) = \phi(xy^{-1})$$

defines a Schur multiplier on $B(L^2(G))$. Equivalently, there are a Hilbert space K and bounded continuous maps $\xi, \eta : G \to K$ such that

$$\phi(xy^{-1}) = \langle \xi(x), \eta(y)\rangle, \qquad \|m_\phi\|_{cb} = \inf \|\xi\|_\infty \|\eta\|_\infty.$$

Such functions are called *Herz–Schur multipliers*.

Exercise 10.7 (Fourier–Schur transference and quantum channels)**.** Let G be discrete. On the matrix units of $B(\ell^2(G))$, define

$$S_\phi(E_{x,y}) = \phi(xy^{-1})E_{x,y}.$$

Verify directly that $S_\phi(\lambda_s) = \phi(s)\lambda_s$, so its restriction to $VN(G)$ is M_ϕ. Use the factorization in the preceding definition to prove that S_ϕ and M_ϕ are completely bounded and have the stated norm; taking the converse factorization theorem as input, obtain the full transference equivalence.

A function ϕ is *positive definite* if $[\phi(s_i^{-1}s_j)]_{i,j} \geq 0$ for every finite family (s_i). Prove that if ϕ is positive definite and $\phi(e) = 1$, then S_ϕ and M_ϕ are normal unital completely positive maps. For finite G, prove that M_ϕ preserves the canonical trace and hence is a quantum channel in the Schrödinger adjoint picture. Show

$$M_\phi M_\psi = M_{\phi\psi}.$$

Taking Schoenberg's theorem as input, let $\ell : G \to [0, \infty)$ be conditionally negative definite with $\ell(e) = 0$ and prove that $M_t(\lambda_s) = e^{-t\ell(s)}\lambda_s$ is a completely positive Markov semigroup. Interpret it as symmetry-diagonal decoherence: Fourier modes with larger $\ell(s)$ decay faster while the identity mode is preserved.

The coproduct, rather than the existence of individual group elements, is the structure that tells a symmetry how to act on a composite system. This suggests keeping the operator algebra and coproduct while allowing the algebra itself to become noncommutative. Harmonic analysis survives this passage: Haar measure becomes a state, functions become operators, and Fourier coefficients become matrix coefficients of corepresentations.

10.3 Compact quantum groups and quantum Peter–Weyl theory

Definition 10.5 (Compact quantum groups and their Haar states)**.** A *compact quantum group* $\mathbb{G}$ consists of a unital C^*-algebra $C(\mathbb{G})$ and a unital $*$-homomorphism

$$\Delta : C(\mathbb{G}) \longrightarrow C(\mathbb{G}) \otimes_{\min} C(\mathbb{G})$$

where $\otimes_{\min}$ is the spatial C^*-tensor product obtained from faithful Hilbert-space representations, such that

$$(\Delta \otimes \mathrm{id})\Delta = (\mathrm{id} \otimes \Delta)\Delta$$

and the linear spans of $\Delta(C(\mathbb{G}))(1 \otimes C(\mathbb{G}))$ and $\Delta(C(\mathbb{G}))(C(\mathbb{G}) \otimes 1)$ are dense in the minimal tensor product. The density conditions are the cancellation laws without points.

There is a unique *Haar state* h satisfying

$$(h \otimes \mathrm{id})\Delta(a) = h(a)1 = (\mathrm{id} \otimes h)\Delta(a) \qquad (a \in C(\mathbb{G})).$$

In the GNS representation of the reduced compact quantum group, set

$$L^\infty(\mathbb{G}) = \pi_h(C(\mathbb{G}))'', \qquad L^1(\mathbb{G}) = L^\infty(\mathbb{G})_*.$$

The coproduct extends normally to $L^\infty(\mathbb{G})$, and its preadjoint defines the completely contractive convolution

$$(\omega * \eta)(a) = (\omega \otimes \eta)(\Delta a) \qquad (\omega, \eta \in L^1(\mathbb{G})).$$

Thus $L^1(\mathbb{G})$ is simultaneously a Banach algebra and a canonical operator space. This analytic formulation of compact quantum groups is due to Woronowicz, *Communications in Mathematical Physics* **111** (1987).

Exercise 10.8 (The two classical faces of a compact quantum group)**.** For a compact group G, put $C(\mathbb{G}) = C(G)$ and

$$(\Delta f)(s, t) = f(st).$$

Verify coassociativity and the cancellation density conditions, identify h with Haar integration, and show that the predual product on $L^1(G)$ is ordinary convolution.

For a discrete group Γ, put $C(\mathbb{G}) = C^*_r(\Gamma)$ and

$$\Delta(\lambda_s) = \lambda_s \otimes \lambda_s, \qquad h(\lambda_s) = \delta_{s,e}.$$

Prove the same axioms and identify $L^\infty(\mathbb{G}) = VN(\Gamma)$ and $L^1(\mathbb{G}) = A(\Gamma)$. Show that convolution of its predual functionals is pointwise multiplication of their coefficient functions. Explain why these examples are respectively commutative and cocommutative, and why a genuine quantum group may be neither. Finally, use complete contractivity of the normal coproduct and operator-space duality to verify at matrix levels that its preadjoint product on $L^1(\mathbb{G})$ is completely contractive.

Definition 10.6 (Quantum Peter–Weyl theory)**.** A finite-dimensional *unitary corepresentation* of $\mathbb{G}$ on H_U is a unitary

$$U \in B(H_U) \otimes C(\mathbb{G})$$

satisfying

$$(\mathrm{id} \otimes \Delta)(U) = U_{12}U_{13}.$$

If $U = (u_{ij})$ in an orthonormal basis, this says $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$. Tensor products are formed using the coproduct; their order can matter because Δ need not equal its opposite.

The quantum Peter–Weyl theorem states that the coefficients of irreducible unitary corepresentations span a dense Hopf $*$-subalgebra $\mathrm{Pol}(\mathbb{G}) \subset C(\mathbb{G})$. Haar orthogonality decomposes $L^2(\mathbb{G}, h)$ into finite-dimensional coefficient spaces. For an irreducible corepresentation U^α, the failure of h to be a trace is measured by a positive matrix Q_α. Normalizing it so that

$$\mathrm{Tr}(Q_\alpha) = \mathrm{Tr}(Q_\alpha^{-1}),$$

this common value is the *quantum dimension* $d_q(\alpha)$. In the tracial, classical compact-group case, $Q_\alpha = 1$ and $d_q(\alpha) = \dim H_\alpha$.

Exercise 10.9 ($SU_q(2)$ and arithmetic dimensions)**.** Fix $0 < q < 1$. Let $C(SU_q(2))$ be generated by α, γ so that

$$u = \begin{pmatrix} \alpha & -q\gamma^* \\ \gamma & \alpha^* \end{pmatrix}$$

is unitary, equivalently

$$\alpha^*\alpha + \gamma^*\gamma = 1, \qquad \alpha\alpha^* + q^2\gamma\gamma^* = 1, \qquad \gamma^*\gamma = \gamma\gamma^*,$$
$$\alpha\gamma = q\gamma\alpha, \qquad \alpha\gamma^* = q\gamma^*\alpha.$$

Define $\Delta(u_{ij}) = \sum_k u_{ik} \otimes u_{kj}$. Check on generators that Δ preserves the relations and is coassociative. Explain why the noncommuting coordinate functions describe a noncommutative compact space, while the matrix u still transforms like a fundamental representation.

Taking the classification of irreducible corepresentations as supplied, write them as U^ℓ for $\ell = 0, \frac{1}{2}, 1, \ldots$, with fusion rule

$$U^{1/2} \boxtimes U^\ell \cong U^{\ell+1/2} \oplus U^{\ell-1/2}.$$

Use $d_q(1/2) = q + q^{-1}$ and multiplicativity of quantum dimension to derive

$$d_q(\ell) = [2\ell + 1]_q = \frac{q^{2\ell+1} - q^{-(2\ell+1)}}{q - q^{-1}}.$$

Recover the ordinary dimensions as $q \to 1$. Distinguish the Hilbert-space dimension $2\ell + 1$ from the arithmetic Laurent polynomial $[2\ell + 1]_q$.

The dense coordinate algebra $\mathrm{Pol}(\mathbb{G})$ has an algebraic Hopf dual, typified by the quantized enveloping algebra $U_q(\mathfrak{g})$. The function-algebra side leads to C^*- and von Neumann completions, Haar states, preduals, and completely bounded maps. The enveloping-algebra side controls tensor products, R-matrices, and q-difference equations. Their pairing is the noncommutative harmonic analysis that will reappear in Arithmetic: quantum spherical matrix coefficients become Macdonald polynomials, while their radial operators are organized by affine and double affine Hecke algebras.

10.4 Equivariant compression and Toeplitz operator systems

A generalized symmetry is encoded by its algebraic and analytic structure, but an experiment may resolve only finitely many modes of a representation. Already for an ordinary compact group, compressing observables to an invariant finite-dimensional subspace can preserve the group action while destroying closure under multiplication. The operator systems introduced in quantum error correction provide exactly the remaining matrix-ordered structure.

Definition 10.7 (Equivariant compression)**.** Let a compact group G act on a Hilbert space H by a unitary representation U, and let a unital C^*-algebra $A \subseteq B(H)$ be invariant under conjugation by U_g. If a finite-rank orthogonal projection P commutes with every U_g, define

$$S_P = PAP = \{PaP : a \in A\} \subseteq B(PH).$$

The restricted representation $U_g^P = U_g|_{PH}$ acts on S_P by conjugation. We call S_P an *equivariant compression* of A.

Exercise 10.10 (Compression preserves symmetry and matrix order)**.** Prove that S_P is a finite-dimensional operator system with identity P and that

$$U_g^P(PaP)(U_g^P)^* = P(U_g a U_g^*)P.$$

Thus the G-action survives the compression. On the other hand, compute

$$(PaP)(PbP) = PabP - Pa(1-P)bP.$$

Show that the second term can obstruct closure under products, whereas S_P is an algebra if P commutes with every $a \in A$. Explain why positivity, states, and completely positive maps still make sense on S_P without an internal product.

Exercise 10.11 (Rotation-covariant Toeplitz compression)**.** Let $A = C(S^1)$ act by multiplication on $H = L^2(S^1)$, with normalized Fourier basis $e_k(\theta) = (2\pi)^{-1/2}e^{ik\theta}$. Let P_N project onto $H_N = \operatorname{span}\{e_k : -N \le k \le N\}$ and set $S_N = P_N A P_N$. Prove

$$\langle e_j, P_N M_f P_N e_k\rangle = \widehat{f}(j-k), \qquad -N \le j,k \le N,$$

and deduce that S_N is exactly the operator system of $(2N+1)$-dimensional Toeplitz matrices.

For $z(\theta) = e^{i\theta}$, put $T_N = P_N M_z P_N$. Show that T_N is a truncated shift and that $T_N T_N^*$ has nonconstant main diagonal. Conclude that S_N is not an algebra, although the C^*-algebra generated by S_N is the full matrix algebra $B(H_N)$.

Finally, let rotations act by $(U_\phi \xi)(\theta) = \xi(\theta - \phi)$. Prove that P_N commutes with U_ϕ and that conjugation by $U_\phi|_{H_N}$ preserves S_N. Thus this Fourier cutoff retains the continuous rotation symmetry even though the compressed observables no longer form an algebra. This is a classical compact-group example; equivariant compressions for quantum-group actions require compatible coaction data and are not assumed here.

Equivariant compression gives a concrete reason to study maps defined only on operator systems: finite resolution may retain symmetry, positivity, and matrix order after multiplication has been lost. The next question is whether completely positive and completely bounded maps on such subspaces extend to the surrounding operator algebras. The resulting extension theorems lead in turn to tensor norms and their application to nonlocal correlations.

10.5 Extension theorems, tensor norms, and Grothendieck inequalities

Exercise 10.12 (Finite-dimensional Arveson–Wittstock extension)**.** Prove in finite dimensions that a completely positive map from an operator system extends completely positively to the containing C^*-algebra. Deduce, by the 2×2 operator-system construction, that completely bounded maps between finite-dimensional operator spaces admit norm-preserving extensions.

Definition 10.8 (Operator-space tensor products)**.** The *minimal operator-space tensor norm* on $E \otimes F$ is inherited from concrete inclusions in $B(H \otimes K)$. The *Haagerup norm* is

$$\|z\|_h = \inf\{\|[x_1 \ \cdots \ x_r]\| \, \|[y_1 \ \cdots \ y_r]^T\| : z = \sum_i x_i \otimes y_i\}.$$

A bilinear form $u : E \times F \to \mathbb{C}$ is *jointly completely bounded* when the maps

$$u_n([x_{ij}], [y_{kl}]) = [u(x_{ij}, y_{kl})]_{(i,k),(j,l)}$$

from $M_n(E) \times M_n(F)$ to M_{n^2} are uniformly bounded; the supremum of their norms is $\|u\|_{jcb}$.

Exercise 10.13 (Tensor norms linearize bilinear maps)**.** Prove the universal property of the projective Banach-space tensor norm. Prove that the Haagerup tensor product linearizes completely bounded factorizations and compare the minimal, Haagerup, and projective norms on finite-dimensional matrix spaces.

Exercise 10.14 (Noncommutative Grothendieck inequality)**.** Assume that for every bounded bilinear form $u : A \times B \to \mathbb{C}$ on unital C^*-algebras there are states f_1, f_2 on A and g_1, g_2 on B such that

$$|u(a, b)| \leq \|u\| \big(f_1(a^*a) + f_2(aa^*)\big)^{1/2} \big(g_1(b^*b) + g_2(bb^*)\big)^{1/2}.$$

Specialize this inequality to commutative C^*-algebras and recover the classical Grothendieck domination of bounded bilinear forms. Apply it to matrix algebras and express the two quadratic terms as row and column norms. Use Haagerup, *The Grothendieck inequality for bilinear forms on C^*-algebras.*

Exercise 10.15 (Operator-space Grothendieck inequality)**.** Assume that for every jointly completely bounded bilinear form $u : A \times B \to \mathbb{C}$ on unital C^*-algebras there are states f_1, f_2 on A and g_1, g_2 on B such that

$$|u(a, b)| \leq \|u\|_{jcb} \left(f_1(aa^*)^{1/2} g_1(b^*b)^{1/2} + f_2(a^*a)^{1/2} g_2(bb^*)^{1/2} \right),$$

as proved by Haagerup–Musat, *The Effros–Ruan conjecture for bilinear forms on C^*-algebras.* Use matrix amplification to explain why $\|u\|_{jcb}$, rather than only $\|u\|$, controls coupling to arbitrary ancillas. Give examples showing that both row and column terms are necessary.

10.6 Nonlocal games as operator-space tensors

Definition 10.9 (Bell functionals and nonlocal games)**.** A *Bell functional* is an array $M = (M_{xy}^{ab})$ evaluated on conditional distributions by $\langle M, p \rangle = \sum_{a,b,x,y} M_{xy}^{ab} p(a, b \mid x, y)$. A two-player *nonlocal game* $G = (\pi, V)$ consists of a question distribution $\pi(x, y)$ and a payoff $0 \leq V(a, b \mid x, y) \leq 1$. Its classical and entangled values $\omega(G)$ and $\omega^*(G)$ are the optimal expected payoffs using, respectively, shared randomness and local POVMs $\{E_a^x\}_a, \{F_b^y\}_b$ on a shared finite-dimensional state ρ, with $p(a, b \mid x, y) = \mathrm{Tr}(\rho(E_a^x \otimes F_b^y))$. An *XOR game* has binary answers and scores only their parity, equivalently a correlation $\sum_{x,y} M_{xy} \mathbb{E}[a_x b_y]$ for signs a_x, b_y.

Exercise 10.16 (Magic-square pseudotelepathy)**.** The referee chooses $(x, y) \in \{1, 2, 3\}^2$ uniformly. Alice returns a row $a \in \{\pm 1\}^3$ with product $+1$, Bob returns a column $b \in \{\pm 1\}^3$ with product -1, and they win when $a_y = b_x$. Prove by multiplying all row and column constraints that no deterministic classical strategy wins every question, construct one losing on only one question, and hence show $\omega(G_{\mathrm{MS}}) = 8/9$. Using two Bell pairs and a Mermin–Peres square of commuting two-qubit Pauli observables, construct a strategy with $\omega^*(G_{\mathrm{MS}}) = 1$.

Compare the ideal query-by-query predictions with the hyperentangled-photon experiment of Xu et al., *Physical Review Letters* **129** (2022). Explain separately the locality and fair-sampling assumptions in that experiment. Show why a score above 8/9 is a device-independent witness under the stated assumptions; then, assuming a robust magic-square rigidity theorem, explain how a score $1-\varepsilon$ certifies two Bell pairs up to local isometries and errors controlled by ε.

Definition 10.10 (The tensor of a game)**.** Set

$$E_{X,A} = \ell_1^X(\ell_\infty^A), \qquad \|z\|_{E_{X,A}} = \sum_x \max_a |z_{x,a}|,$$

with its standard dual operator-space structure. The tensor associated to $G = (\pi, V)$ is

$$\widehat{G} = \sum_{x,y,a,b} \pi(x,y)V(a,b \mid x,y)(e_x \otimes e_a) \otimes (e_y \otimes e_b) \in E_{X,A} \otimes E_{Y,B}.$$

Exercise 10.17 (Game values as Banach and operator-space norms)**.** Identify deterministic strategies with extreme points of the relevant unit balls and prove

$$\omega(G) = \|\widehat{G}\|_{E_{X,A} \otimes_\varepsilon E_{Y,B}}, \qquad \omega^*(G) = \|\widehat{G}\|_{E_{X,A} \otimes_{\min} E_{Y,B}}.$$

Equivalently, show that these are the bounded and completely bounded norms of the bilinear payoff form on the dual mixed-norm spaces. Locate in the proof the use of positivity of a game tensor, and explain why the same simple formula need not hold for an arbitrary signed Bell functional. Use Palazuelos–Vidick, *Survey on Nonlocal Games and Operator Space Theory*, *Journal of Mathematical Physics* **57** (2016) as a guide.

Exercise 10.18 (Grothendieck bounds for XOR games)**.** For an XOR game, prove that its classical and entangled biases are

$$\beta(G) = \sup_{a_x, b_y = \pm 1} \left| \sum M_{xy} a_x b_y \right|, \quad \beta^*(G) = \sup_{\|u_x\|, \|v_y\| \le 1} \left| \sum M_{xy} \langle u_x, v_y \rangle \right|.$$

Use Tsirelson's vector representation and the real Grothendieck inequality to prove $\beta^*(G) \le K_G^{\mathbb{R}} \beta(G)$. Relate the CHSH values derived earlier to this general comparison.

Exercise 10.19 (Unbounded advantages for multi-output games)**.** For general multi-output games, take as supplied the Khot–Vishnoi estimates $\omega(G_{\mathrm{KV}}) \le C/n$ and $\omega^*(G_{\mathrm{KV}}) \ge c/(\log n)^2$, achieved with local entanglement dimension n. Use the preceding norm dictionary to deduce

$$\frac{\|\widehat{G}_{\mathrm{KV}}\|_{\min}}{\|\widehat{G}_{\mathrm{KV}}\|_\varepsilon} \ge c' \frac{n}{(\log n)^2}.$$

Explain why the XOR comparison remains dimension independent while general game tensors require operator-space matrix levels. Consult Buhrman–Regev–Scarpa–de Wolf, *Theory of Computing* **8** (2012) for the supplied estimates.

Part III

Topology

Probability and operator-space theory distinguish states, channels, and correlations by quantities that generally vary under perturbation. Some physical responses instead remain quantized while the Hamiltonian changes continuously, provided a gap does not close. Topology supplies invariants adapted to this stability. This part passes from homotopy and characteristic classes to index theory, then from group-valued invariants of invertible phases to the fusion and braiding data of topological order.

11 Index Theory

Experimental observations.

- In the integer quantum Hall regime, transverse conductance forms plateaus at integer multiples of e^2/h while longitudinal resistance is strongly suppressed.

In an idealized gapped Hall system, small changes of fields, hopping strengths, or disorder do not change the quantized conductance as long as the relevant spectral or mobility gap remains open. Thus the first mathematical question is not the precise Hamiltonian but whether two Hamiltonians can be continuously deformed into one another without leaving the gapped regime. Homotopy formalizes such deformations; winding, cohomology, vector bundles, K-theory, and finally the Fredholm index supply successively richer invariants that cannot change along them.

11.1 Homotopy, homology, and generalized cohomology

Definition 11.1 (Homotopy and fundamental group)**.** A *homotopy* from $f : X \to Y$ to $g : X \to Y$ is a continuous map $H : X \times [0, 1] \to Y$ with endpoints f and g. The *fundamental group* $\pi_1(X, x_0)$ is the group of endpoint-fixed homotopy classes of loops based at x_0, with concatenation.

Exercise 11.1 (Fundamental groups and winding)**.** Prove invariance of π_1 under pointed homotopy equivalence and show that an unpointed homotopy changes the induced map only by the appropriate change-of-basepoint isomorphism. Prove the Seifert–van Kampen theorem for a union with path-connected intersection. Compute $\pi_1(S^1) \cong \mathbb{Z}$ by lifting loops to $\mathbb{R}$, and prove that the resulting integer is the winding number.

Definition 11.2 (Homology and cohomology)**.** Fix a commutative ring R with identity. The *singular chain group* $C_n(X; R)$ is the free R-module on continuous maps $\Delta^n \to X$, with alternating face boundary ∂. The *homology* is $H_n(X; R) = \ker \partial_n / \operatorname{im} \partial_{n+1}$. The *cochain complex* is $C^n(X; R) = \operatorname{Hom}_R(C_n(X; R), R)$ with coboundary $\delta\phi = \phi\partial$, and its cohomology is $H^n(X; R)$.

For a smooth manifold M, integration over smooth singular simplices sends a form ω to the cochain $\sigma \mapsto \int_{\Delta^k} \sigma^*\omega$. Stokes' theorem makes this a cochain map.

Theorem 11.3 (de Rham; used without proof)**.** *Integration over smooth singular simplices induces a natural isomorphism of graded rings*

$$H^*_{\mathrm{dR}}(M) \cong H^*(M; \mathbb{R}).$$

Ordinary cohomology detects fluxes and characteristic classes, but stable band theory asks a more specific question: what remains after trivial bands may be added, and how does a bulk invariant become a boundary invariant one dimension lower? The needed invariants must survive homotopy, decompose over independent pieces, satisfy excision, and shift degree under suspension. A generalized cohomology theory packages these requirements. Complex K-theory, introduced below from stable vector bundles, is the example that will classify the stable noninteracting symmetry-free band problem.

Definition 11.4 (Cofibers, finite CW complexes, and cohomology theories)**.** A based inclusion $A \hookrightarrow X$ is a *cofibration* if every homotopy on A that extends at its initial time to X extends over the whole homotopy. Its *cofiber* is X/A. More generally, the cofiber of a based map $f : A \to X$ is the pushout $X \cup_f CA$, where CA is the reduced cone; the reduced suspension is $\Sigma A = CA/A$.

A *finite CW complex* is obtained from the empty space by finitely many pushouts

$$\begin{array}{ccc} \coprod_\alpha S^{n-1} & \longrightarrow & X^{n-1} \\ \downarrow & & \downarrow \\ \coprod_\alpha D^n & \longrightarrow & X^n, \end{array}$$

where X^n is the nth skeleton. A *reduced cohomology theory* on based finite CW complexes is a graded contravariant functor $\widetilde{E}^*$ that is homotopy invariant, vanishes on a point, takes finite wedges to direct sums, has natural suspension isomorphisms $\widetilde{E}^{q+1}(\Sigma X) \cong \widetilde{E}^q(X)$, and makes

$$\widetilde{E}^q(X/A) \longrightarrow \widetilde{E}^q(X) \longrightarrow \widetilde{E}^q(A)$$

exact for every cofibration $A \hookrightarrow X$.

Definition 11.5 (Products and pairings)**.** The *Kronecker pairing* is $H^n(X; R) \times H_n(X; R) \to R$, $([\phi], [c]) \mapsto \phi(c)$. The *cup product* is the product on cohomology induced by the front–back face formula on cochains.

Exercise 11.2 (Homology, cohomology, and degree)**.** Prove $\partial^2 = 0$, $\delta^2 = 0$, and homotopy invariance. For a CW pair (X, A), define $C_*(X, A; R) = C_*(X; R)/C_*(A; R)$, construct barycentric subdivision, and use it to prove excision and $H^q(X, A; R) \cong \widetilde{H}^q(X/A; R)$. Construct the Puppe sequence of a cofibration and derive the long exact sequence of a pair and the Mayer–Vietoris sequence. Deduce that reduced singular cohomology is a reduced cohomology theory.

Prove that $X^{n-1} \hookrightarrow X^n$ is a cofibration and X^n/X^{n-1} is a finite wedge of n-spheres. By induction over the skeleta, prove that a natural transformation between reduced cohomology theories that is an isomorphism on S^0 in every degree is an isomorphism on every finite CW complex; prove the required five-term diagram chase. Compute the homology and cohomology rings of spheres and tori. For a continuous map $f : S^n \to S^n$, define its degree by $f_*[S^n] = \deg(f)[S^n]$ and prove multiplicativity and agreement with winding number for $n = 1$.

11.2 Bundles, holonomy, and characteristic classes

Exercise 11.3 (Topological bundles from transition functions)**.** Forget the smooth structure on the bundles introduced with gauge fields and retain their continuous transition maps. Prove that transition functions satisfying the cocycle condition construct a topological bundle and that changes of local trivialization give isomorphic bundles. Prove that a principal bundle is trivial exactly when it has a global section. Construct associated vector bundles from representations of G.

Definition 11.6 (Global connections and holonomy)**.** A *connection* on a vector bundle $E \to M$ is a map $\nabla : \Gamma(E) \to \Omega^1(M; E)$ satisfying $\nabla(fs) = df \otimes s + f\nabla s$. Its curvature is $F_\nabla = \nabla^2 \in \Omega^2(M; \operatorname{End} E)$. Its *holonomy* along a loop is the automorphism of the initial fiber given by parallel transport.

Exercise 11.4 (Curvature and holonomy)**.** Derive local connection and curvature forms and their gauge-transformation laws. Prove the Bianchi identity. Show that infinitesimal holonomy is curvature and compute the $U(1)$ holonomy around the boundary of a surface as $\exp(i \int F)$ when a global potential is available.

Exercise 11.5 (Aharonov–Bohm holonomy)**.** On $X = \mathbb{R}^2 \setminus \{0\}$, consider the $U(1)$ gauge potential $A = (\Phi/2\pi)d\theta$. Verify that $F = dA = 0$ on X but $\oint_\gamma A = \Phi$ for a loop of winding number one. Show that a particle of charge q acquires the gauge-invariant phase $\exp(iq\Phi/\hbar)$, derive the relative phase of two paths enclosing the excluded flux, and obtain the flux period $h/|q|$. Prove that A cannot be removed by a single-valued gauge transformation unless its holonomy is trivial. Compare the predicted fringe displacement with the shielded-flux electron-holography experiment of Tonomura et al., *Physical Review Letters* **56** (1986). Explain why the result establishes the physical significance of holonomy, not of a gauge-dependent potential by itself, even though curvature vanishes along the electron paths.

Definition 11.7 (Grassmannians and universal bundles)**.** For a finite-dimensional Hermitian space V, the *Grassmannian* $\mathrm{Gr}_n(V)$ is the space of n-dimensional subspaces of V. Its *tautological bundle* has fiber W over $W \in \mathrm{Gr}_n(V)$. Isometric inclusions of Hermitian spaces induce compatible inclusions of their Grassmannians and tautological bundles, and

$$BU(n) = \varinjlim_N \mathrm{Gr}_n(\mathbb{C}^N), \qquad \gamma_n = \varinjlim_N \gamma_{n,N}.$$

Exercise 11.6 (Classifying maps)**.** Let $E \to X$ be a rank-n complex vector bundle over a compact Hausdorff space. Construct a finite trivializing cover, a subordinate partition of unity, a Hermitian metric, and a fiberwise isometric embedding $E \hookrightarrow X \times \mathbb{C}^N$. Show that its ranges define a continuous map $f_E : X \to \mathrm{Gr}_n(\mathbb{C}^N)$ with $E \cong f_E^* \gamma_{n,N}$. Prove that two choices of embedding become homotopic after stabilization by placing them in orthogonal summands, and prove that homotopic maps have isomorphic pullbacks. Deduce the bijection

$$[X, BU(n)] \longleftrightarrow \{\text{rank-}n \text{ complex bundles over } X\}/\cong .$$

Exercise 11.7 (Universal Chern classes)**.** Identify $BU(1)$ with $\mathbb{P}(\mathbb{C}^\infty)$ and use its even-dimensional cell decomposition to compute $H^*(BU(1); \mathbb{Z}) = \mathbb{Z}[x]$, where x evaluates as -1 on the projective line carrying the tautological line bundle. Let

$$s : BU(1)^n \longrightarrow BU(n)$$

classify the direct sum of the n tautological lines, and set $x_i = \mathrm{pr}_i^* x$. Construct the Schubert cell decompositions of the finite Grassmannians and their flag bundles. Use the cellular filtration and the splitting map s to prove that s^* is injective with image

$$\mathbb{Z}[x_1, \ldots, x_n]^{\mathfrak{S}_n} = \mathbb{Z}[e_1(x), \ldots, e_n(x)].$$

Define $c_j(\gamma_n) \in H^{2j}(BU(n); \mathbb{Z})$ by $s^* c_j(\gamma_n) = e_j(x_1, \ldots, x_n)$. For the block-sum map $BU(n) \times BU(m) \to BU(n+m)$, prove

$$c(\gamma_n \oplus \gamma_m) = c(\gamma_n)c(\gamma_m), \qquad c = 1 + c_1 + c_2 + \cdots .$$

Definition 11.8 (Chern classes and character)**.** If a rank-n complex vector bundle $E \to X$ is classified by $f : X \to BU(n)$, its *Chern classes* are $c_j(E) = f^*c_j(\gamma_n) \in H^{2j}(X;\mathbb{Z})$. If the formal roots of $1 + c_1(E) + \cdots + c_n(E)$ are $y_1, \ldots, y_n$, the *Chern character* is the degreewise class

$$\mathrm{ch}(E) = \sum_{i=1}^{n} e^{y_i} \in \prod_{k\geq 0} H^{2k}(X;\mathbb{Q}),$$

where each symmetric polynomial in the y_i is expressed in the Chern classes by the Newton identities and the definition extends additively to $K^0(X)$.

Exercise 11.8 (Naturality, sums, and the Chern character)**.** Prove that the classifying-map definition is independent of every choice and that $c_j(g^*E) = g^*c_j(E)$. Use the block-sum calculation above to prove the Whitney formula $c(E \oplus F) = c(E)c(F)$. Prove that the Chern character is a natural ring map and that $\mathrm{ch}(L) = e^{c_1(L)}$ for a line bundle.

Exercise 11.9 (Curvature representatives)**.** For a Hermitian bundle with unitary connection, prove that

$$\det\left(1 + \frac{i}{2\pi}F_\nabla\right) \quad \text{and} \quad \operatorname{Tr}\exp\left(\frac{i}{2\pi}F_\nabla\right)$$

are closed forms whose cohomology classes are independent of the connection. Identify their de Rham classes with the images of the Chern classes and Chern character, and prove their Whitney-sum formulas.

11.3 Berry geometry and topological K-theory

Definition 11.9 (Berry geometry)**.** For a smooth family of one-dimensional spectral projections P_x on a Hilbert space, the ranges form a line bundle. The *Berry connection* is $\nabla = P \circ d$ and its curvature is the *Berry curvature*. Its holonomy is the *Berry phase*. The integral $\langle c_1(L), [\Sigma]\rangle$ is the *Chern number* over an oriented closed surface Σ.

Exercise 11.10 (Berry phase and quantized response)**.** Compute the Berry curvature of a two-level Hamiltonian $H(n) = n \cdot \sigma$ over S^2 for both the lower and upper eigenline bundles. With the standard orientation on S^2, compute their opposite Chern numbers and state the sign convention used. Identify the two eigenlines with classifying maps $f_\pm : S^2 \to \mathbb{P}(\mathbb{C}^2)$ and, for the normalization in Exercise 7.4, prove

$$F^{\pm}_{\text{Berry}} = -2if^*_\pm\omega_{\text{FS}}.$$

For a gapped two-dimensional periodic Hamiltonian, derive the Kubo formula expressing Hall conductance as e^2/h times the occupied-band Chern number. Deduce integer quantization under gap-preserving perturbations and compare with integer quantum Hall plateaus. For an adiabatic closed parameter loop, compute the Berry holonomy, show that it is independent of traversal rate, and identify the corresponding interference-fringe shift.

Cohomology assigns additive classes to bundles. K-theory first makes bundle addition invertible and then connects stable topology to the analytic index of differential operators.

Definition 11.10 (Grothendieck completion and stable equivalence)**.** The *Grothendieck group* of a commutative monoid M is the universal abelian group receiving a monoid map from M. Vector bundles E, F are *stably equivalent* if $E \oplus \underline{\mathbb{C}}^m \cong F \oplus \underline{\mathbb{C}}^n$ for some m, n; this relation forgets rank and therefore describes reduced stable classes. The group $K^0(X)$ is the Grothendieck group of complex vector bundles on compact X. For based X, let $\widetilde{K}^0(X)$ be the kernel of restriction to the basepoint and, for every $j \geq 0$, define

$$\widetilde{K}^j(X) = \widetilde{K}^0(\Sigma^j X).$$

For unbased X, write $X_+ = X \sqcup \{*\}$ and set $K^j(X) = \widetilde{K}^j(X_+)$ for $j \geq 0$.

Definition 11.11 (Relative and compactly supported K-theory)**.** For a compact pair (X, A) and a locally compact space Y, define

$$K^j(X, A) = \widetilde{K}^j(X/A), \qquad K_c^j(Y) = \widetilde{K}^j(Y^+),$$

where Y^+ is the one-point compactification. A triple (F^0, F^1, σ) of vector bundles on Y, with $\sigma : F^0 \to F^1$ invertible off a compact set, represents a class in $K_c^0(Y)$ up to homotopy and stabilization.

Exercise 11.11 (Exactness of K-theory)**.** Prove homotopy invariance and finite-wedge additivity for reduced K-theory. For a cofibration $A \hookrightarrow X$, use clutching of bundles to prove exactness of the cofiber sequence in all nonnegative degrees defined above. Derive the relative and Mayer–Vietoris sequences from the general cofiber construction, and prove that external product is compatible with them.

Definition 11.12 (The K-theory Thom class)**.** Let $\pi : E \to X$ be a rank-r Hermitian complex vector bundle. Exterior multiplication by v and its adjoint give the odd endomorphism

$$c(v) = \varepsilon(v) - \iota(v) : \Lambda^* E_{\pi(v)} \longrightarrow \Lambda^* E_{\pi(v)}, \qquad c(v)^2 = -\|v\|^2.$$

Writing $c^+(v)$ for its even-to-odd part, the *K-theory Thom class* is

$$u_E = [\pi^* \Lambda^{\mathrm{even}} E, \pi^* \Lambda^{\mathrm{odd}} E, c^+] \in K_c^0(E).$$

Exercise 11.12 (Clutching and the Bott class)**.** Prove the clutching description of complex vector bundles on suspensions and spheres. For the complex line over a point, identify $u_{\mathbb{C}}$ under $\mathbb{C}^+ \cong S^2$ with

$$\beta = [H] - [\underline{\mathbb{C}}] \in \widetilde{K}^0(S^2),$$

where H is the Hopf line. Prove that β generates $\widetilde{K}^0(S^2)$ and that $\widetilde{K}^0(S^3) = 0$. Deduce that the Thom map for a point is an isomorphism in degrees zero and one.

Exercise 11.13 (Thom isomorphism and Bott periodicity)**.** Apply K-theory exactness and cellular induction to the preceding point calculation and prove the trivial-line result over finite CW complexes. Iterate it for trivial bundles and then use Mayer–Vietoris over bundle trivializations to prove the complex Thom isomorphism

$$K^j(X) \longrightarrow K_c^j(E), \qquad a \longmapsto \pi^* a \, u_E, \qquad j = 0, 1,$$

for finite CW complexes X. For the trivial line over a compact based finite CW complex, identify its relative Thom space with $S^2 \wedge X$ and deduce complex Bott periodicity:

$$\widetilde{K}^j(X) \xrightarrow{\boxtimes \beta} \widetilde{K}^j(S^2 \wedge X)$$

is an isomorphism.

Exercise 11.14 (K-theory of spheres)**.** Use Bott periodicity and the clutching calculation to compute the complex K-groups of spheres. Extend K^j to negative degrees by two-periodicity and use K-theory exactness to prove that it is a reduced cohomology theory on finite CW complexes.

Exercise 11.15 (K-theory of tori and class-A phases)**.** Derive from the cofibration for $X \times S^1$ and Bott periodicity the split formula $K^j(X \times S^1) \cong K^j(X) \oplus K^{j-1}(X)$, with indices taken modulo two, and compute the complex K-groups of tori. Show that a gapped family of finite-rank Hamiltonians defines the stable class of its occupied bundle and identify the strong and weak class-A invariants on Brillouin tori in dimensions one through three.

Definition 11.13 (Operator K-theory)**.** For a unital C^*-algebra A, $K_0(A)$ is the Grothendieck group of stable homotopy classes of projections in matrix algebras over A, where orthogonal sum gives the underlying monoid, and $K_1(A)$ is the stable homotopy group of unitaries in matrix algebras over A.

Exercise 11.16 (Topological and operator K-theory)**.** Construct the Serre–Swan correspondence between vector bundles over compact X and finitely generated projective $C(X)$-modules. Deduce $K^j(X) \cong K_j(C(X))$ and formulate Bott periodicity for C^*-algebras.

11.4 Fredholm operators and elliptic index theory

Definition 11.14 (Compact and Fredholm operators)**.** An operator on a Hilbert space is *compact* if it is a norm limit of finite-rank operators. A bounded operator $T : H_0 \to H_1$ is *Fredholm* if its kernel and cokernel are finite-dimensional and its range is closed; its *index* is $\operatorname{ind} T = \dim\ker T - \dim\operatorname{coker} T$. The *Calkin algebra* is $\mathcal{Q}(H) = B(H)/\mathcal{K}(H)$.

Exercise 11.17 (Fredholm parametrices and the Calkin algebra)**.** For $T : H_0 \to H_1$, prove that T is Fredholm exactly when there is a bounded $S : H_1 \to H_0$ such that $ST - \mathrm{id}_{H_0}$ and $TS - \mathrm{id}_{H_1}$ are compact. When $H_0 = H_1 = H$, identify this condition with invertibility of the image of T in $\mathcal{Q}(H)$. Prove homotopy invariance, compact-perturbation invariance, and additivity of the index.

Exercise 11.18 (Shift index and the boundary map)**.** On an infinite-dimensional separable Hilbert space, compute the index of the unilateral shift and identify the connecting map $K_1(\mathcal{Q}(H)) \to K_0(\mathcal{K}(H))$ from the Calkin extension with the Fredholm index.

Definition 11.15 (Elliptic symbols)**.** For vector bundles E, F over a compact smooth manifold, the *principal symbol* of an order-m differential operator $D : \Gamma(E) \to \Gamma(F)$ is the homogeneous bundle map $\sigma_D : \pi^*E \to \pi^*F$ on T^*M. The operator is *elliptic* if $\sigma_D(x, \xi)$ is invertible for every $\xi \neq 0$. Its *analytic index* is its Fredholm index on Sobolev completions; its *topological index* is the K-theoretic pushforward of its symbol class.

Exercise 11.19 (Elliptic regularity and Fredholmness)**.** Assume the construction of Sobolev completions, the Rellich compactness theorem, elliptic regularity, and, for an elliptic operator D of order m, the estimates

$$\|s\|_{H^{r+m}} \leq C_r(\|Ds\|_{H^r} + \|s\|_{H^r})$$

for D and its formal adjoint. Prove that $D : H^{r+m}(E) \to H^r(F)$ is Fredholm and that its index is independent of r.

Exercise 11.20 (Atiyah–Singer index theorem)**.** Use the Thom isomorphism above, and assume the construction of the K-theoretic pushforward and uniqueness of an index transformation satisfying excision, multiplicativity, and Bott normalization. Prove that analytic index has these three properties. Verify them for topological index and conclude $\operatorname{ind}_{an} D = \operatorname{ind}_{top}(\sigma_D)$.

11.5 Dirac operators, anomalies, and topological phases

Definition 11.16 (Spin structures and Dirac operators)**.** A *spin structure* on an oriented Riemannian n-manifold is a lift of its oriented orthonormal frame bundle through $\mathrm{Spin}(n) \to SO(n)$. Its complex spin representation defines the *spinor bundle* S, with Clifford action c and spin connection. The *Dirac operator* is $D = c \circ \nabla^S$. In even dimensions the chirality operator splits $S = S^+ \oplus S^-$. In formal curvature roots,

$$\widehat{A}(TM) = \prod_j \frac{x_j/2}{\sinh(x_j/2)}.$$

Exercise 11.21 (Hirzebruch–Riemann–Roch)**.** Assume that the topological index of the Dolbeault symbol on a compact complex manifold is $\int_M \mathrm{ch}(E)\,\mathrm{Td}(TM)$, where $\mathrm{Td}(TM) = \prod_j x_j/(1-e^{-x_j})$ in formal Chern roots. Apply Atiyah–Singer to derive the Hirzebruch–Riemann–Roch formula and recover Riemann–Roch for a line bundle on a compact Riemann surface.

Exercise 11.22 (Dirac index formula)**.** Assume that the topological index of the coupled chiral Dirac symbol is $\int_M \widehat{A}(TM)\,\mathrm{ch}(E)$. Apply Atiyah–Singer to derive

$$\mathrm{ind}(D^+ \otimes E) = \int_M \widehat{A}(TM)\,\mathrm{ch}(E).$$

Definition 11.17 (Chirality and instanton number)**.** Let M be an oriented Riemannian spin four-manifold with the spinor splitting above. For a Hermitian vector bundle E with unitary connection A, the coupled Dirac operator is odd and has a *chiral part*

$$D_A^+ : \Gamma(S^+ \otimes E) \longrightarrow \Gamma(S^- \otimes E).$$

For an instanton or anti-instanton from Exercise 4.9, its *instanton number* is

$$k = -\frac{1}{8\pi^2} \int_M \mathrm{Tr}(F_A \wedge F_A) \in \mathbb{Z}.$$

With the Chern–Weil convention $\mathrm{ch}(E) = \mathrm{Tr}\exp(iF_A/2\pi)$ used above, $k = \int_M \mathrm{ch}_2(E)$. Thus for an $SU(r)$ bundle, $\mathrm{ch}_2(E) = -c_2(E)$; conventions that call $\int_M c_2(E)$ the instanton charge use the opposite sign.

Exercise 11.23 (Fermion chirality detects instanton number)**.** Identify the characteristic number $k(A)$ from Exercise 4.9 with $\int_M \mathrm{ch}_2(E)$. Then apply the Dirac-index formula from the preceding exercise to prove

$$\mathrm{ind}\, D_A^+ = \dim\ker D_A^+ - \dim\ker (D_A^+)^* = \int_M \big[\widehat{A}(TM)\,\mathrm{ch}(E)\big]_4.$$

For the fundamental $SU(2)$ bundle over S^4, show that this index is k with the orientation convention in the definition. Use the Weitzenböck formula for a self-dual or anti-self-dual connection to determine which chirality has no zero modes and deduce that the other has exactly $|k|$ zero modes. For a fermion in a representation R, derive the factor $2T(R)k$. Finally, compute the fermionic measure under the axial rotation $\psi \mapsto e^{i\alpha\gamma_5}\psi$ and obtain the integrated anomaly $\Delta Q_5 = 2\,\mathrm{ind}\, D_A^+$, identifying the chirality change carried by an instanton transition.

Exercise 11.24 (Index theory and invertible free-fermion phases)**.** Let $H(k)$ be a continuous family of finite-dimensional self-adjoint free-fermion Hamiltonians over a compact momentum space X, with a uniform gap at zero. Flatten its spectrum to

$$Q(k) = \mathrm{sgn}\, H(k) = H(k)|H(k)|^{-1}, \qquad P(k) = \frac{1 - Q(k)}{2},$$

and prove that the occupied spaces $E_k = \operatorname{im} P(k)$ form a vector bundle $E \to X$. Show that its stable class $[E] \in K^0(X)$ is invariant under gap-preserving deformations.

When X is an even-dimensional spin manifold, pair this class with its Dirac operator and use Atiyah–Singer to obtain

$$\langle [E], [D_X] \rangle = \operatorname{ind}(D_X^+ \otimes E) = \int_X \widehat{A}(TX)\operatorname{ch}(E).$$

For a two-dimensional Brillouin torus, identify this integer with the first Chern number and derive the class-A Hall response $\sigma_H = (e^2/h)\langle c_1(E), [X] \rangle$.

Prove that stacking Hamiltonians gives direct sum and addition in $K^0(X)$. After passing to reduced K-theory, show that spectral reversal replaces E by $E^\perp$ and gives the additive inverse because $E \oplus E^\perp$ is trivial. Explain why an invertible phase can have protected boundary modes or symmetry defects but no nontrivial intrinsic bulk fusion sectors. Contrast this group-valued classification with the sector-valued noninvertible phases in the next section.

Exercise 11.25 (The free-fermion periodic table of topological insulators and superconductors)**.** Let $H(k)$ be a spectrally flattened Bloch or Bogoliubov–de Gennes Hamiltonian on the compactified momentum space S^d. Time-reversal, particle-hole, and chiral symmetries act by

$$\mathsf{T}H(k)\mathsf{T}^{-1} = H(-k), \qquad \mathsf{C}H(k)\mathsf{C}^{-1} = -H(-k), \qquad \mathsf{S}H(k)\mathsf{S}^{-1} = -H(k),$$

where T and C are antiunitary, with square $+1$ or -1 when present, and S is unitary, with its phase chosen so that $\mathsf{S}^2 = 1$. Organize the complex classes A, AIII and the real classes in the order

s	0	1	2	3	4	5	6	7
class	AI	BDI	D	DIII	AII	CII	C	CI
T^2	$+1$	$+1$	0	-1	-1	-1	0	$+1$
C^2	0	$+1$	$+1$	$+1$	0	-1	-1	-1

and verify that when both antiunitary symmetries occur their product gives a chiral symmetry, up to an irrelevant phase.

Assume the Atiyah–Bott–Shapiro identification of stable symmetry-compatible Hamiltonians with Clifford-extension spaces, as used by Kitaev, *Periodic table for topological insulators and superconductors*. In the complex classes the strong classification groups in dimension d are

$$\pi_0(C_{-d}) \quad \text{for A}, \qquad \pi_0(C_{1-d}) \quad \text{for AIII},$$

and in real class s the group is

$$\pi_0(R_{s-d}) \cong KO^{d-s}(\mathrm{pt}).$$

Indices are periodic, and their coefficient groups are

$$\pi_0(C_q) = \begin{cases} \mathbb{Z} & q \equiv 0 \pmod 2, \\ 0 & q \equiv 1 \pmod 2, \end{cases}$$

$q \bmod 8$	0	1	2	3	4	5	6	7
$\pi_0(R_q)$	$\mathbb{Z}$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0.

Use these data, rather than a memorized chart, to fill every entry of the periodic table:

class	T^2	C^2	S	$d = 0$	1	2	3	4	5	6	7
A	0	0	0								
AIII	0	0	1								
AI	+1	0	0								
BDI	+1	+1	1								
D	0	+1	0								
DIII	−1	+1	1								
AII	−1	0	0								
CII	−1	−1	1								
C	0	−1	0								
CI	+1	−1	1								

Recover in particular the integer quantum Hall phase in class A, the one-dimensional Majorana chain in class D, the two-dimensional quantum spin Hall phase in class AII, and the three-dimensional class-DIII topological superconductor. Explain why an abstract $\mathbb{Z}$ entry coming from R_4 is often printed as $2\mathbb{Z}$ when its generator maps to twice the corresponding complex invariant; in dimension zero this is the even complex dimension forced by Kramers degeneracy. Finally, explain why this is only the table of *strong* phases: a lattice Brillouin torus can also carry weak invariants. Relate the symmetry-free case to the earlier decomposition of $K^*(\mathbb{T}^d)$, and state why antiunitary symmetries require real, equivariant, or twisted K-theory, following Freed–Moore, *Annales Henri Poincaré* **14** (2013). State why this stable noninteracting classification need not remain unchanged in the presence of strong interactions.

11.6 Spectral triples, ultraviolet cutoffs, and operator systems

The preceding uses of Dirac operators extracted stable topological indices. Their spectrum and commutators also answer a different question: how much metric information is accessible at a finite energy resolution?

On a compact spin manifold, smooth functions act on square-integrable spinors while the Dirac operator measures their variation. This replaces coordinates by spectral data: the algebra records observables, the Hilbert space records their representation, and the spectrum of the Dirac operator sets the available length scales. Restricting that spectrum models the finite energy range of an experiment. Equivariant Toeplitz compression earlier isolated the resulting operator-system structure; a spectral triple now adds metric information through commutators with the Dirac operator.

Definition 11.18 (Spectral triples and spectral distance)**.** A *spectral triple* (A, H, D) consists of a unital $*$-algebra A represented faithfully by bounded operators on a Hilbert space H and a self-adjoint operator D with dense domain such that $(1 + D^2)^{-1/2}$ is compact, and each commutator $[D, a]$, initially defined on $\mathrm{Dom}(D)$, extends to a bounded operator. Its Lipschitz seminorm is

$$L_D(a) = \|[D, a]\|.$$

On the state space of the C^*-closure of A, the associated extended *spectral distance* is

$$d_D(\varphi, \psi) = \sup\{|\varphi(a) - \psi(a)| : a = a^*,\ L_D(a) \le 1\}.$$

The value may be infinite if the commutator fails to distinguish enough observables.

Exercise 11.26 (Recovering Riemannian distance)**.** Let M be a compact connected Riemannian spin manifold, let $A = C^\infty(M)$ act by multiplication on $H = L^2(M, S)$, and let D be its Dirac operator. Prove

$$[D, f] = c(df), \qquad \|[D, f]\| = \sup_{x \in M} \|df_x\|.$$

Using the characterization of geodesic distance by 1-Lipschitz functions, show for the evaluation states δ_x, δ_y that

$$d_D(\delta_x, \delta_y) = d_M(x, y).$$

Thus the metric can be recovered from how the Dirac operator fails to commute with the observable algebra.

Definition 11.19 (Spectral truncation)**.** For $\Lambda > 0$, let

$$P_\Lambda = \mathbf{1}_{[-\Lambda,\Lambda]}(D), \qquad H_\Lambda = P_\Lambda H, \qquad D_\Lambda = P_\Lambda D P_\Lambda.$$

Compactness of the resolvent makes H_Λ finite-dimensional. The compressed observables form

$$S_\Lambda = P_\Lambda A P_\Lambda = \{P_\Lambda a P_\Lambda : a \in A\} \subseteq B(H_\Lambda).$$

As in equivariant compression, this is an operator system with identity P_Λ, inherited adjoint, and inherited matrix-order cones. A state on S_Λ is a positive linear functional ω with $\omega(P_\Lambda) = 1$. The truncated commutator seminorm and spectral distance are

$$L_\Lambda(x) = \|[D_\Lambda, x]\|, \qquad d_\Lambda(\omega, \eta) = \sup_{\substack{x = x^* \\ L_\Lambda(x) \le 1}} |\omega(x) - \eta(x)|.$$

Exercise 11.27 (Compressed commutators)**.** Write $P = P_\Lambda$. Since P is a spectral projection of D, prove on PH that

$$[PDP, PaP] = P[D, a]P$$

for every $a \in A$, and hence

$$L_\Lambda(PaP) \le L_D(a).$$

In the circle example, show that $P_N M_{z^{2N+1}} P_N = 0$ even though $z^{2N+1} \neq 0$. Thus compression need not be injective. Explain why the metric problem after a cutoff must be formulated on states of PAP, rather than by simply restricting the original point states.

Exercise 11.28 (Metric geometry of the truncated circle)**.** Return to the Toeplitz operator system S_N from Exercise 10.11, restrict the algebra to $A = C^\infty(S^1)$, and set

$$D = -i\frac{d}{d\theta}, \qquad D_N = P_N D P_N.$$

Verify

$$[D_N, P_N M_f P_N]_{jk} = (j - k)\widehat{f}(j - k)$$

and deduce that the kernel of L_N on S_N consists only of scalars.

Show that rotations commute with D_N and hence preserve L_N and the spectral distance on the state space of S_N. For

$$v_{N,\theta} = (2N + 1)^{-1/2} \sum_{k=-N}^{N} e^{-ik\theta} e_k,$$

identify the expectation $\langle v_{N,\theta}, P_N M_f P_N v_{N,\theta}\rangle$ with convolution by a Fejér kernel and show that it tends to $f(\theta)$ as $N \to \infty$. Interpret these vector states as points blurred by finite spectral resolution; see Connes–van Suijlekom, *Spectral Truncations in Noncommutative Geometry and Operator Systems*, Communications in Mathematical Physics **383** (2021).

There is a parallel position-space description. For a finite metric space X and resolution $\varepsilon > 0$, define

$$E_\varepsilon = \operatorname{span}\{|x\rangle\langle y| : d(x,y) < \varepsilon\} \subseteq B(\ell^2(X)).$$

Reflexivity and symmetry of the relation $d(x,y) < \varepsilon$ make E_ε an operator system. Matrix-unit multiplication shows that it is an algebra exactly when the relation is also transitive. The usual failure of transitivity therefore records finite resolution algebraically: two steps individually shorter than ε may connect points that the chosen resolution does not identify in one step.

Spectral truncation concerns metric information retained after a cutoff. We now return to topological phases, passing from the group-valued classification of invertible phases to noninvertible phases whose localized excitations have fusion and braiding.

12 Quantum Topology

Experimental observations.

- At filling $\nu = 1/3$, shot-noise measurements give quasiparticle charge near $e/3$, and interferometry detects phase evolution associated with fractional statistics.

Fractional charge poses an algebraic puzzle. Neutral local observables can create charge–anticharge pairs, but they cannot create a single isolated topological charge from the vacuum. If N is the algebra generated by observables in separated regions and M is the larger algebra of everything compatible with measurements outside those regions, the inclusion $N \subseteq M$ records precisely this missing charged information. A conditional expectation, when available, forgets it, and subfactor index measures how much was forgotten.

This is an algebraic prototype rather than a direct model of the Hall sample. In a completely rational conformal-net description of a chiral edge, the two-interval index is $\sum_a d_a^2$, the total quantum dimension squared of the localized charge sectors. Their transport and exchange then produce fusion and braid data. The sequence below develops that bridge from inclusions, through sectors, to anyons and topological field theory.

12.1 Subfactors, Temperley–Lieb algebras, and the Jones polynomial

Definition 12.1 (Subfactors, conditional expectations, and index)**.** A *subfactor* is an inclusion $N \subseteq M$ of factors with common unit. A *normal conditional expectation* $E : M \to N$ is a normal positive unital N-bimodule projection. For finite factors with normalized trace, the *Jones index* $[M : N]$ is the Murray–von Neumann dimension of $L^2(M)$ as a left N-module, and $E_N : M \to N$ denotes the unique trace-preserving normal conditional expectation.

For arbitrary factors and a faithful normal conditional expectation E, set

$$[M : N]_E = \lambda_E^{-1}, \qquad \lambda_E = \sup\bigl(\{0\} \cup \{\lambda > 0 : E(x) \geq \lambda x \text{ for every } x \in M_+\}\bigr).$$

The *minimal index* is $[M : N]_{\min} = \inf_E [M : N]_E$, with value $+\infty$ if no finite-index expectation exists. For the type-III inclusions below, $[M : N]$ denotes this minimal index.

Definition 12.2 (Basic construction)**.** For finite factors $N \subseteq M$, the *Jones projection* e_N on $L^2(M)$ is the orthogonal projection onto $L^2(N)$. The *basic construction* is $M_1 = \langle M, e_N\rangle''$. Iteration gives the *Jones tower* $N \subseteq M \subseteq M_1 \subseteq M_2 \subseteq \cdots$.

Exercise 12.1 (Jones tower and Temperley–Lieb relations)**.** For a finite-index inclusion $N \subseteq M$ of finite factors, derive $e_N x e_N = E_N(x) e_N$ and construct the canonical trace on the basic construction. Prove that the normalized Jones projections in the tower satisfy

$$e_i^2 = e_i, \qquad e_i e_j = e_j e_i \ (|i-j| > 1), \qquad e_i e_{i\pm 1} e_i = [M:N]^{-1} e_i.$$

Exercise 12.2 (Diagrammatic calculus and index quantization)**.** Construct the Temperley–Lieb diagrammatic category, with composition by stacking and each closed loop valued at $\delta = [M:N]^{1/2}$. Use positivity of its Markov trace to prove that index values below 4 are $4\cos^2(\pi/n)$ for integers $n \geq 3$.

Definition 12.3 (Braid groups and Markov traces)**.** The *braid group B_n* has generators σ_i with relations $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ and $\sigma_i\sigma_j = \sigma_j\sigma_i$ for $|i-j| > 1$. A *Markov trace* on a tower of braid-group quotients is a compatible trace satisfying a fixed stabilization rule.

Exercise 12.3 (Jones polynomial)**.** Construct a braid-group representation from the Temperley–Lieb generators. Use the Markov trace and Markov moves to obtain an invariant of oriented links, normalize it to the Jones polynomial, and derive its skein relation. Compute it for the unknot, trefoil, and Hopf link.

12.2 Conformal nets and superselection sectors

Definition 12.4 (Local conformal nets)**.** Let $\mathcal{I}$ be the proper open intervals of S^1. A *local conformal net* assigns a von Neumann algebra $\mathcal{A}(I) \subset B(H)$ to each $I \in \mathcal{I}$ such that

$$I \subset J \Longrightarrow \mathcal{A}(I) \subset \mathcal{A}(J), \qquad I \cap J = \varnothing \Longrightarrow [\mathcal{A}(I), \mathcal{A}(J)] = 0.$$

The assignment is covariant under orientation-preserving diffeomorphisms of S^1, represented projectively on H, and the rotation generator is positive. Its vacuum vector Ω is invariant under the Möbius subgroup and cyclic for the local algebras, and $\bigvee_I \mathcal{A}(I) = B(H)$. Physically, S^1 is the compactified chiral light ray and $\mathcal{A}(I)$ consists of observables localized in I. The norm-closed C^*-algebra

$$\mathfrak{A} = C^*\left(\bigcup_{I\in\mathcal{I}} \mathcal{A}(I)\right)$$

generated by the local algebras is the *quasilocal algebra*. For the regular nontrivial conformal nets used below, the local algebras are hyperfinite type-III$_1$ factors; the physical content lies in their inclusions and covariance, not in their abstract isomorphism type alone.

Definition 12.5 (Duality and regularity for conformal nets)**.** Write I' for the interior of the complement of I. A net has *Haag duality* when $\mathcal{A}(I) = \mathcal{A}(I')'$. It has the *split property* when $\overline{I} \subset J$ implies $\mathcal{A}(I) \subset F \subset \mathcal{A}(J)$ for some type-I factor F. It is *strongly additive* when removing an interior point from I to obtain $I_1 \cup I_2$ gives $\mathcal{A}(I) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. A representation of the quasilocal algebra is *locally normal* when its restriction to every $\mathcal{A}(I)$ is normal.

Exercise 12.4 (The two-interval subfactor)**.** For $E = I_1 \cup I_2$ a union of disjoint intervals, set $\mathcal{A}(E) = \mathcal{A}(I_1) \vee \mathcal{A}(I_2)$. First use locality and vacuum cyclicity for $\mathcal{A}(I')$ to prove that the vacuum is separating for $\mathcal{A}(I)$. Use the split property and the spatial tensor product to identify separated local algebras as independent subsystems. Then use locality to obtain the inclusion

$$\mathcal{A}(E) \subseteq \mathcal{A}(E')'.$$

Under the split and irreducibility hypotheses, show that this is a subfactor. Define the *μ-index* $\mu(\mathcal{A}) = [\mathcal{A}(E')' : \mathcal{A}(E)]$, and prove that it is independent of the two-interval configuration by conformal covariance. Contrast single-interval Haag duality $\mathcal{A}(I) = \mathcal{A}(I')'$ with the possible failure of duality for E, measured by this Jones index.

Definition 12.6 (DHR sectors)**.** A *DHR sector* of $\mathcal{A}$ is represented by a locally normal, transportable endomorphism ρ of the quasilocal algebra that acts identically on $\mathcal{A}(I')$ for some interval I; transportability means that an equivalent representative can be localized in any interval. Composition is the tensor product. For a finite-index irreducible sector localized in I, its statistical dimension satisfies

$$d(\rho)^2 = [\mathcal{A}(I) : \rho(\mathcal{A}(I))].$$

Exercise 12.5 (Critical Ising chain, Rydberg spectroscopy, and the Ising net)**.** For $J, h > 0$, consider the transverse-field Ising chain

$$H_{\mathrm{I}} = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i.$$

Use the Jordan–Wigner transformation and a Bogoliubov rotation to obtain

$$\varepsilon(k) = 2\sqrt{J^2 + h^2 - 2Jh\cos k}.$$

Show that the gap is $2|J - h|$ and closes at $h = J$, where the long-wavelength modes are massless Majorana fermions. Taking the standard fermionic determinant asymptotics as input, obtain $\langle Z_i Z_{i+r} \rangle \sim Cr^{-1/4}$ and a connected energy-density correlator proportional to r^{-2}. Identify the bulk Ising fields $1, \sigma, \varepsilon$ with scaling dimensions $0, 1/8, 1$, respectively.

The Rydberg experiment realizes a different microscopic chain,

$$H_{\mathrm{R}} = \sum_i \left(\frac{\Omega}{2} P_{i-1} X_i P_{i+1} - \Delta n_i \right) + \sum_{j-i \geq 2} V_{ij} n_i n_j,$$

in the nearest-neighbor blockade regime. Explain why equality of microscopic Hamiltonians is unnecessary for the two critical points to share the Ising universality class; here n_i projects onto the Rydberg state and $P_i = 1 - n_i$. Take as input the open-chain CFT spectrum, including its conformal towers and $E_a(L) - E_0(L) = \pi v x_a / L + o(L^{-1})$, and recover the universal even-parity ratios $2 : 4 : 6 : 8$ and odd-parity ratios $3 : 5 : 7$. Compare them with the reported modulation-spectroscopy peaks of Sun et al., *Experimental observation of conformal field theory spectra* (2026), stating the finite-size and experimental limitations.

Identify the chiral continuum theory with the $c = 1/2$ Ising conformal net. Its DHR sectors $1, \psi, \sigma$ have conformal weights $0, 1/2, 1/16$; relate the bulk energy field to $\psi\bar{\psi}$ and distinguish these chiral weights from the bulk scaling dimensions above. Explain why the experiment supports the Ising CFT spectrum but does not directly measure the net or all of its superselection sectors.

The sector operations abstract to a braided fusion category. Its diagrammatic data in turn define topological field theories and link invariants.

12.3 Fusion, braiding, and modular tensor categories

Definition 12.7 (Monoidal and rigid categories)**.** A *monoidal category* is a category $\mathcal{C}$ with a tensor functor, unit object, and coherent associativity and unit isomorphisms. A *dual* of X is an object X^* with evaluation and coevaluation morphisms satisfying the snake identities. A monoidal category is *rigid* if every object has left and right duals.

Definition 12.8 (Fusion, braided, and modular categories)**.** A *fusion category* over $\mathbb{C}$ is a finite semisimple rigid $\mathbb{C}$-linear monoidal category with simple unit. It is *unitary* when it is a C^*-category and its tensor product and structural isomorphisms are compatible with the adjoint. A *braiding* is a natural family $c_{X,Y} : X \otimes Y \to Y \otimes X$ satisfying the hexagon identities. A *unitary modular tensor category* is a unitary ribbon fusion category with unitary nondegenerate braiding.

Exercise 12.6 (From Jones index to a modular category)**.** Assume that $\mathcal{A}$ is completely rational: it is split and strongly additive, with finite μ-index. Assume that charge transporters for localized finite-index DHR endomorphisms produce a unitary braiding and that the complete-rationality theorem makes their category a unitary modular tensor category with finitely many irreducible sectors ρ_i and

$$\mu(\mathcal{A}) = \sum_i d(\rho_i)^2.$$

Identify composition with tensor product, conjugate endomorphisms with duals, and charge transporters with the braiding. Explain where nondegeneracy enters modularity. Use Kawahigashi–Longo–Müger, *Communications in Mathematical Physics* **219** (2001). For the Ising net at $c = 1/2$, take the sectors $1, \psi, \sigma$ with dimensions $1, 1, \sqrt{2}$ and fusion rules

$$\psi^2 = 1, \qquad \psi\sigma = \sigma, \qquad \sigma^2 = 1 \oplus \psi.$$

Compute $\mu(\mathcal{A})$ and interpret ψ and σ as the chiral Majorana and spin sectors.

Exercise 12.7 (Fusion rules and braiding)**.** Prove that simple-object classes of a fusion category form a based ring under tensor product. Derive the pentagon and hexagon equations for associativity and braiding data. Compute quantum dimensions and braid-group representations for the Fibonacci fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$.

12.4 TQFT, Reshetikhin–Turaev theory, and Chern–Simons theory

Definition 12.9 (Bordisms and topological field theories)**.** The *bordism category* has closed oriented $(n-1)$-manifolds as objects, oriented n-dimensional bordisms as morphisms, and disjoint union as tensor product. An *n-dimensional topological field theory* is a symmetric monoidal functor from this category to vector spaces. An extended theory may assign categories to codimension-two manifolds. A TQFT is *invertible* when its state spaces are tensor-invertible (hence one-dimensional over a field) and every bordism acts invertibly; noninvertible TQFTs may instead carry nontrivial superselection and fusion sectors.

Definition 12.10 (The Reshetikhin–Turaev $SU(2)_k$ theory)**.** For $k \in \mathbb{N}$, let $\mathcal{C}_k = SU(2)_k$ be the unitary modular tensor category with simple objects $V_0, \ldots, V_k$, where V_j has spin $j/2$. The Reshetikhin–Turaev construction uses the colored ribbon-graph calculus of $\mathcal{C}_k$ and surgery to define a symmetric monoidal $(2+1)$-dimensional TQFT

$$Z_k^{\mathrm{RT}} = Z_{\mathcal{C}_k}$$

on bordisms equipped with the extra framing data needed to cancel the framing anomaly. It assigns invariants to closed framed three-manifolds and to framed links colored by the objects V_j. We call Z_k^{RT} the Reshetikhin–Turaev model of quantum $SU(2)$ Chern–Simons theory at level k; this terminology does not define it by a functional integral.

Definition 12.11 (Chern–Simons theory and Wilson loops)**.** Let G have an invariant inner product on $\mathfrak{g}$, and fix a trivialization of a principal G-bundle over a closed oriented three-manifold. For the resulting $\mathfrak{g}$-valued connection potential A, the *Chern–Simons functional* at level k is

$$S_{CS}(A) = \frac{k}{4\pi}\int_M \left(\langle A \wedge dA\rangle + \frac{1}{3}\langle A \wedge [A \wedge A]\rangle\right).$$

For a representation V and closed curve K, the *Wilson loop* is $W_V(K) = \mathrm{Tr}_V(\mathrm{Hol}_A(K))$.

Exercise 12.8 (Gauge invariance, level, and knot invariants)**.** For compact, connected, simply connected simple G, normalize the inner product so that the winding term of a gauge transformation lies in $2\pi\mathbb{Z}$. Compute the change of S_{CS} and derive $k \in \mathbb{Z}$ as the condition for $e^{iS_{CS}}$ to be gauge invariant. Prove that its critical points are flat connections.

Exercise 12.9 (Reshetikhin–Turaev links and Jones evaluations)**.** Use the Temperley–Lieb realization of $\mathcal{C}_k$ and its ribbon functor to show that the Z_k^{RT} invariant of a framed link in S^3, with a component colored by V_j, is the corresponding j-colored Jones evaluation at

$$q = e^{2\pi i/(k+2)}.$$

In particular, identify V_1 with the fundamental color giving the ordinary Jones polynomial. Compute the normalized invariants of the unknot, Hopf link, and trefoil, state the normalization of the colored unknot, and track the framing factor separately from the ambient-isotopy invariant.

Exercise 12.10 (Comparison with Chern–Simons quantization)**.** Do not use the formal Chern–Simons path integral as a definition. Assume the comparison theorem identifying the modular functor underlying Z_k^{RT} with the level-k $SU(2)$ Wess–Zumino–Witten conformal-block modular functor, the quantization associated with $SU(2)$ Chern–Simons theory; see Andersen–Ueno, *Construction of the Witten–Reshetikhin–Turaev TQFT from conformal field theory*. Explain why this gives a precise sense in which Z_k^{RT} realizes quantum Chern–Simons theory and matches its Wilson-loop labels and gluing rules. Also explain why the comparison theorem does not by itself construct a nonperturbative measure for the formal path integral

$$\int \mathcal{D}A\, e^{iS_{CS}(A)} \prod_r W_{V_{j_r}}(K_r).$$

Exercise 12.11 (Functorial gluing)**.** Show that the monoidal-functor axioms give multiplicativity under disjoint union, composition under gluing, duality under orientation reversal, and mapping-class-group representations on state spaces. In two dimensions, prove the equivalence with commutative Frobenius algebras.

In a gapped two-dimensional phase, the same categorical data describe anyon fusion, braiding, topology-dependent state spaces, and protected logical observables.

12.5 Anyons, topological order, and quantum Hall phases

Definition 12.12 (Anyons and topological degeneracy)**.** In a two-dimensional gapped phase, an *anyon type* is a simple object of its braided fusion category. It is *abelian* if its quantum dimension is 1 and *nonabelian* otherwise. The *topological degeneracy* on a closed surface is the dimension of the state space assigned by the associated topological field theory.

Exercise 12.12 (The toric code: error correction from topological order)**.** Give an $L \times L$ square cellulation of the torus, with $L \geq 2$, a qubit on every edge, and denote the resulting physical Hilbert space by $\mathcal{H}$. Let X and Z be self-adjoint unitaries satisfying $X^2 = Z^2 = 1$ and $XZ = -ZX$ on one qubit. Operators on distinct edges commute. For each vertex v and plaquette p, set

$$A_v = \prod_{e \ni v} X_e, \qquad B_p = \prod_{e \subset \partial p} Z_e, \qquad H = -\sum_v A_v - \sum_p B_p.$$

Prove that all A_v and B_p commute and that the ground space is their common $+1$ spectral subspace. Establish the two relations $\prod_v A_v = 1$ and $\prod_p B_p = 1$, prove that the remaining stabilizers are independent, find $E_0 = -2L^2$, and compute

$$\dim \ker(H - E_0) = 2^{2L^2 - (L^2 + L^2 - 2)} = 4.$$

Show that an open Z-string creates violated vertex terms only at its endpoints, while an open dual X-string creates violated plaquette terms only at its endpoints. Prove that contractible closed strings are products of stabilizers. Show that strings around the two noncontractible cycles give two pairs of logical operators, with a primal Z-loop and a dual X-loop anticommuting exactly when their intersection number is odd. Deduce that the code encodes two logical qubits and has distance L.

Let $\mathcal{S}$ be the stabilizer group generated by the A_v and B_p, let $\mathcal{S}' \subseteq B(\mathcal{H})$ be its commutant, and let P be the ground space projection. Prove that the logical observable algebra is

$$\mathcal{A}_{\log} = P\mathcal{S}'P = \operatorname{span}\{P\overline{X}_1^{i_1}\overline{Z}_1^{j_1}\overline{X}_2^{i_2}\overline{Z}_2^{j_2}P : i_r, j_r \in \{0,1\}\} \cong M_2(\mathbb{C}) \otimes M_2(\mathbb{C}),$$

where the barred operators are noncontractible logical loops. Equivalently, identify the logical Pauli group as the Pauli centralizer of $\mathcal{S}$ modulo $\mathcal{S}$. If an error channel has Kraus operators E_a for which every $E_a^* E_b$ is supported on fewer than L edges, prove that

$$PE_a^* E_b P \in \mathbb{C}P = \mathcal{A}_{\log}'$$

where, as in the definition of protected observable algebras, the prime is taken inside the code corner $PB(\mathcal{H})P$, and hence verify the exact OAQEC condition for $\mathcal{A}_{\log}$. Deduce in particular that the toric code corrects arbitrary errors on at most $\lfloor (L-1)/2 \rfloor$ edges.

Interpret the endpoint excitations as anyons e and m. Derive $e^2 = m^2 = 1$, set $\varepsilon = em$, and show that taking e once around m produces the phase -1. Explain how local error syndromes, global logical Wilson loops, and the fourfold TQFT degeneracy express the same topological structure; see Kitaev, *Fault-tolerant quantum computation by anyons*, and Dauphinais–Kribs–Vasmer, *Stabilizer Formalism for Operator Algebra Quantum Error Correction*, Quantum **8** (2024).

Exercise 12.13 (Abelian and nonabelian anyons)**.** Compute fusion spaces and braid actions for an abelian category and for the Fibonacci category. Prove that nonabelian fusion spaces can have dimension greater than one and that braiding gives linear braid-group representations on them. Explain why the corresponding physical gates are commonly regarded projectively after global phases are discarded. Compute the torus degeneracy from the simple anyon types in a modular category.

Exercise 12.14 (Quantum Hall phases and measured topological order)**.** Use flux insertion for a Laughlin state at filling $\nu = 1/m$ to derive quasiparticle charge e/m, exchange phase π/m, Hall conductance $\nu e^2/h$, and genus-g degeneracy m^g. Compare these consequences separately with conductance plateaus, fractional shot-noise charge measurements, and interferometric phase data, stating which topological predictions each observation tests and which it does not.

13 Topological Structures in Holography

Experimental observations.

- Ground states of one-dimensional critical systems have interval entanglement that grows logarithmically with interval length, whereas in gapped ground states it typically saturates once the interval is much longer than the correlation length.
- Black-hole thermodynamics assigns entropy proportional to horizon area rather than to the volume behind the horizon.

Entanglement entropy was introduced earlier as a property of a reduced state. Holography proposes that, in special quantum theories admitting a semiclassical gravitational dual, the same number can be computed by a bulk surface. Topology is indispensable even at leading order: the surface must not merely have the right boundary, but must be homologous to the chosen boundary region. This section isolates that topological content without assuming that every quantum field theory has a useful gravitational dual.

Throughout this section, use natural units $\hbar = c = k_B = 1$; in particular, entropy symbols denote the dimensionless entropy S/k_B relative to the thermodynamic convention used earlier.

13.1 Ryu–Takayanagi surfaces and entanglement wedges

Definition 13.1 (Relative homology and the Ryu–Takayanagi surface)**.** Let Σ be a static bulk time slice with conformal boundary B, and let $A \subset B$ be a spatial region. A codimension-one surface $\gamma \subset \Sigma$ satisfies the *RT homology constraint* for A if

$$\partial\gamma = \partial A \quad \text{and} \quad \partial r_A = A \cup \gamma$$

with the appropriate orientations for some bulk region $r_A \subset \Sigma$. Equivalently, after regulating the conformal boundary, the cycle $A - \gamma$ is null-homologous in the bulk.

In a static state with a semiclassical asymptotically anti-de Sitter dual, the *Ryu–Takayanagi prescription* at leading order is

$$S(A) = \frac{\mathrm{Area}(\gamma_A)}{4G_N},$$

where γ_A has least area among surfaces satisfying the anchoring and homology conditions. The cutoff used to regulate its area is the geometric counterpart of the field-theory ultraviolet cutoff.

Exercise 13.1 (An interval in AdS_3)**.** On the time-zero slice of Poincaré AdS_3, use

$$ds^2 = L^2 \frac{dz^2 + dx^2}{z^2}, \qquad z > 0,$$

and regulate the boundary at $z = \epsilon$. Show that the geodesic anchored at the endpoints of an interval of length ℓ is a semicircle orthogonal to $z = 0$, and compute

$$\mathrm{Length}(\gamma_A) = 2L \log \frac{\ell}{\epsilon} + O(\epsilon^2/\ell^2).$$

Using the Brown–Henneaux relation $c = 3L/(2G_3)$, derive

$$S(A) = \frac{c}{3} \log \frac{\ell}{\epsilon} + O(1).$$

Compare this with the replica calculation for a vacuum interval in a two-dimensional CFT. State clearly that the equality is a test of the holographic dictionary in its semiclassical regime, not a derivation of AdS/CFT for arbitrary CFTs; see Ryu–Takayanagi, *Physical Review Letters* **96** (2006) and Brown–Henneaux, *Communications in Mathematical Physics* **104** (1986).

Definition 13.2 (Entanglement wedges, HRT surfaces, and quantum corrections)**.** For a static RT surface, the *entanglement wedge* of A is the domain of dependence of the bulk region r_A bounded by $A \cup \gamma_A$. In a time-dependent spacetime, the *Hubeny–Rangamani–Takayanagi surface* is instead an extremal codimension-two spacetime surface anchored at ∂A and homologous to A, with least area among the appropriate extremal candidates.

Beyond leading classical order, a *quantum extremal surface* extremizes the generalized entropy

$$S_{\text{gen}}(\gamma) = \frac{\text{Area}(\gamma)}{4G_N} + S_{\text{bulk}}(r_A),$$

and the entropy is obtained from the least admissible extremum. Thus RT is the static classical limit, HRT is its covariant classical extension, and the quantum-extremal prescription includes bulk entanglement corrections.

Exercise 13.2 (Topology change for two intervals)**.** On a constant-time slice of Poincaré AdS_3, let $A = [x_1, x_2]$ and $B = [x_3, x_4]$ with $x_1 < x_2 < x_3 < x_4$. The two admissible pairings of boundary endpoints have regulated lengths

$$L_{\text{disc}} = 2L \log \frac{x_2 - x_1}{\epsilon} + 2L \log \frac{x_4 - x_3}{\epsilon},$$

$$L_{\text{conn}} = 2L \log \frac{x_4 - x_1}{\epsilon} + 2L \log \frac{x_3 - x_2}{\epsilon}.$$

Show that the RT surface changes topology when these two expressions cross, and derive

$$I(A : B) = \max\left\{0, \frac{c}{3} \log \frac{(x_2 - x_1)(x_4 - x_3)}{(x_4 - x_1)(x_3 - x_2)}\right\}.$$

Explain why vanishing leading-order mutual information corresponds to a disconnected classical entanglement wedge, while finite-G_N bulk correlations need not vanish exactly.

Exercise 13.3 (Entropy inequalities from cutting and regluing)**.** For disjoint A and B in a static bulk slice, draw minimal representatives for the regions needed in subadditivity and strong subadditivity. Cut the surfaces where they intersect and reglue the pieces into admissible, not necessarily minimal, surfaces for the opposite sides of

$$S(A) + S(B) \geq S(A \cup B),$$

and

$$S(AB) + S(BC) \geq S(B) + S(ABC).$$

Use minimality to prove the inequalities. Identify where the relative homology constraint is needed to ensure that the reglued surfaces are admissible. Explain why this argument is geometric and classical and why the bulk-entropy term must be included at quantum order.

13.2 Black-hole topology and entropy

Definition 13.3 (The BTZ horizon and thermofield-double topology)**.** The nonrotating BTZ black hole has metric

$$ds^2 = -\frac{r^2 - r_+^2}{L^2}\, dt^2 + \frac{L^2}{r^2 - r_+^2}\, dr^2 + r^2 d\phi^2, \qquad \phi \sim \phi + 2\pi,$$

temperature $T = r_+/(2\pi L^2)$, and horizon entropy

$$S_{\mathrm{BH}} = \frac{2\pi r_+}{4G_3}.$$

Its maximally extended geometry has two asymptotic boundaries and is dual, in the standard semiclassical dictionary, to a thermofield-double state of two CFT copies. The bifurcation circle is homologous to either complete boundary separately, but not to the union of both boundaries.

Exercise 13.4 (Homology, horizons, and black-hole entropy)**.** For the pure thermofield-double state on $H_L \otimes H_R$, prove for its reduced states that $S(\rho_L) = S(\rho_R)$ while the joint entropy vanishes. Apply the RT homology constraint to show that the surface for one complete boundary is the bifurcation circle, while the empty surface is admissible for the union of both boundaries. Recover $S(H_L) = 2\pi r_+/(4G_3)$ and interpret the result as entanglement between the two boundary theories. For a boundary circle of circumference $2\pi L$, use $\beta = 2\pi L^2/r_+$ and $c = 3L/(2G_3)$ to verify that the high-temperature CFT expression

$$S_{\mathrm{thermal}} = \frac{\pi c}{3}\frac{2\pi L}{\beta}$$

equals S_{BH}. In the Arithmetic part this thermal formula will be recovered from modular invariance and asymptotic state counting. Consult Hubeny–Rangamani–Takayanagi, *Journal of High Energy Physics* **07** (2007) for the covariant prescription and Faulkner–Lewkowycz–Maldacena, *Journal of High Energy Physics* **11** (2013) for the bulk quantum correction, and Engelhardt–Wall, *Journal of High Energy Physics* **01** (2015) for quantum extremal surfaces.

14 Gauge Duality and the Geometric Langlands Program

Experimental observations.

- In empty space Maxwell's equations treat electric and magnetic fields as two components of one field strength, and electromagnetic waves carry both. Electric sources break the continuous electric–magnetic rotation; a magnetic source would restore a more symmetric charge lattice.
- Electric charge is quantized, but no elementary magnetic monopole or supersymmetric partner has been observed. The supersymmetric duality used below is therefore a sharply constrained theoretical model, not an experimentally established extension of the Standard Model.

The gauge theory developed in Differential Geometry already supplies connections, curvature, Yang–Mills action, instantons, and characteristic classes. What it does not yet explain is how the same quantum gauge theory can admit two complementary descriptions, one organized by electric probes and the other by magnetic probes. Supersymmetry makes that proposed duality precise enough to survive a topological twist. On a product of two surfaces, the twisted four-dimensional theory reduces to a sigma model whose target is a moduli space of solutions to Hitchin's equations. Electric–magnetic duality then becomes mirror symmetry, and its exchange of Wilson and 't Hooft lines becomes the Hecke-eigenvalue condition of geometric Langlands. This is the physical chain developed in this section:

$$\mathcal{N}=4 \text{ Yang–Mills} \longrightarrow \text{topological twist} \longrightarrow \text{Kapustin–Witten equations} \\ \longrightarrow \mathcal{M}_H \longrightarrow \text{branes and Hecke eigensheaves.}$$

We use natural units $\hbar = c = 1$ in this section. Ordinary Yang–Mills facts refer back to Exercise 4.9; the genuinely new physical input is stated explicitly rather than hidden in exercises.

14.1 Supersymmetry and electric–magnetic duality

Definition 14.1 ($\mathcal{N}=4$ super Yang–Mills)**.** Let G be a compact Lie group and let $P \to M$ be a principal G-bundle over an oriented Riemannian four-manifold. The bosonic fields of $\mathcal{N}=4$ super Yang–Mills theory are a connection A and six adjoint-valued scalar fields $X_1, \dots, X_6$. With an invariant positive inner product on $\mathfrak{g}$, the bosonic action has the form

$$\frac{1}{g^2}\int_M \left(\frac{1}{2}|F_A|^2 + \frac{1}{2}\sum_I |d_A X_I|^2 + \frac{1}{2}\sum_{I<J} |[X_I, X_J]|^2\right) \mathrm{vol} - i\theta k(A),$$

supplemented by fermions and Yukawa couplings fixed by supersymmetry. The six scalars transform under the R-symmetry group $\mathrm{Spin}(6)_R \cong SU(4)_R$, and the theory has sixteen real supercharges. Its gauge coupling and theta angle combine into

$$\tau = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}.$$

The potential is minimized when the X_I commute, so scalar expectation values can break G to a maximal torus and make electric and magnetic charges visible simultaneously. A protected property special to this theory is that its beta function vanishes: τ does not run with energy. This scale independence is essential to the proposed exact duality below.

Supersymmetry is not being introduced merely to enlarge the particle list. It protects masses and operator products from uncontrolled quantum corrections, allowing a weakly coupled calculation to constrain a strongly coupled description. It also permits a scalar supercharge after twisting, which is the mechanism that will remove metric-dependent information while retaining topology.

Definition 14.2 (Langlands dual group)**.** Let $T \subset G$ be a maximal torus. Its character and cocharacter lattices are

$$X^*(T) = \mathrm{Hom}(T, U(1)), \qquad X_*(T) = \mathrm{Hom}(U(1), T).$$

The *Langlands dual group* ${}^L G$ is the compact form whose complexified root datum is dual to that of G: characters and cocharacters are exchanged, and roots are exchanged with coroots. Thus

$$X^*({}^L T) = X_*(T), \qquad X_*({}^L T) = X^*(T).$$

Ignoring global-form refinements, $U(1)$ is self-dual, ${}^L SU(n) = PSU(n)$, and the root systems B_n and C_n are exchanged. The complex reductive group used in geometric Langlands is ${}^L G_{\mathbb{C}}$. The definition is forced physically by Dirac quantization: magnetic charges of G are weights of ${}^L G$.

Definition 14.3 (Montonen–Olive duality as physical input)**.** The S-duality conjecture asserts an exact quantum equivalence between $\mathcal{N} = 4$ theories with dual gauge groups. For simply laced root systems its basic transformation is

$$(G, \tau) \longmapsto ({}^L G, -1/\tau).$$

It exchanges electrically charged states with magnetically charged states and Wilson line operators with 't Hooft line operators. For a non-simply-laced root system a root-length factor n_G enters, $\tau \mapsto -1/(n_G \tau)$. Global forms of the gauge group and discrete theta angles must also be tracked in a complete statement. We use the duality as a physical input, not as a proved theorem. Its origin is the proposal of Montonen and Olive, *Physics Letters B* **72** (1977).

Exercise 14.1 (Charge lattices and the dual group)**.** For a torus $T = U(1)^r$, prove that a representation restricts to characters $\lambda : T \to U(1)$ with $\lambda \in X^*(T)$. A magnetic monopole defines a transition map around its equator, hence a cocharacter $\mu : U(1) \to T$ with $\mu \in X_*(T)$. Show that composition gives an integer pairing

$$\langle \lambda, \mu \rangle \in \mathbb{Z},$$

which is the Dirac quantization condition. Verify directly that exchanging $X^*(T)$ and $X_*(T)$ exchanges electric and magnetic labels. For $G = SU(2)$, compare the weight and coweight lattices and explain why the global dual group is adjoint rather than simply connected.

14.2 The geometric Langlands twist

Definition 14.4 (Topological twisting and Q-cohomology)**.** In Euclidean signature the rotation group is locally $\mathrm{Spin}(4) = SU(2)_+ \times SU(2)_-$. A *topological twist* replaces it by a diagonal combination with a $\mathrm{Spin}(4)$ subgroup of $\mathrm{Spin}(6)_R$. Under the new rotation group, two combinations of the supersymmetry generators become scalars, say Q_ℓ and Q_r. Their projective linear combinations

$$Q_t = Q_\ell + t Q_r, \qquad t \in \mathbb{P}^1,$$

satisfy $Q_t^2 = 0$ on gauge-invariant quantities, up to a gauge transformation. Physical observables of the twisted theory are classes

$$H_{Q_t} = \ker Q_t / \operatorname{im} Q_t.$$

Because Q_t is a scalar, this cohomology can be defined on a curved four-manifold. Metric-dependent terms in the action become Q_t-exact, which is why the surviving correlation functions are topological.

After the twist, four of the six scalar fields transform as an adjoint-valued one-form $\phi \in \Omega^1(M, \operatorname{ad} P)$; the other two form a complex adjoint scalar. Thus the equations below use only geometric objects already available in the manuscript: a connection, its curvature, the covariant exterior derivative, and the Hodge decomposition of two-forms.

Definition 14.5 (Canonical parameter)**.** Although the twisted theory is presented using (τ, t), its topological sector depends on them through the single parameter

$$\Psi = \frac{\tau + \bar\tau}{2} + \frac{\tau - \bar\tau}{2}\,\frac{t - t^{-1}}{t + t^{-1}} = \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}\,\frac{t - t^{-1}}{t + t^{-1}}.$$

For the basic S transformation, the compatible twist parameter changes by $t \mapsto \pm t\tau/|\tau|$; the sign is immaterial because t and $-t$ define equivalent twists. This rule and $\tau \mapsto -1/(n_G \tau)$ give

$\Psi \mapsto -1/(n_G \Psi)$, where n_G is the ratio of the squared lengths of long and short roots and equals one in the simply laced case. Formally, the twisted action has the structure

$$S = \{Q_t, V\} + \frac{i\Psi}{4\pi} \int_M \mathrm{Tr}(F_A \wedge F_A).$$

Consequently, varying the coefficient of the first term does not change correlators of Q_t-closed observables, provided the measure and boundary conditions preserve Q_t.

Theorem 14.6 (Kapustin–Witten localization; used without proof)**.** *The bosonic localization equations of the geometric Langlands twist are*

$$\begin{aligned} (F_A - \phi \wedge \phi + t\, d_A\phi)^+ &= 0, \\ (F_A - \phi \wedge \phi - t^{-1} d_A\phi)^- &= 0, \\ d_A^* \phi &= 0. \end{aligned} \tag{KW}$$

The values $t = 0, \infty$ are interpreted by taking the corresponding limits. At $t = \pm i$, the first two equations say that the complex connection

$$\mathcal{A} = A \pm i\phi$$

is flat; the last equation is a real moment-map condition selecting a well-behaved representative modulo compact G-gauge transformations. The construction and conventions are those of Kapustin and Witten, Communications in Number Theory and Physics ***1** (2007).*

Exercise 14.2 (Why Q-exact deformations disappear)**.** Let $S_s = S_0 + s\{Q, V\}$ and let $\mathcal{O}$ be Q-closed. Assuming a Q-invariant measure and no boundary contribution in field space, differentiate $\langle \mathcal{O} \rangle_s$ with respect to s and prove that the result vanishes. Explain why the same calculation fails for an observable that is not Q-closed. Next substitute $t = i$ into (KW) and verify directly that its first two lines combine into

$$F_{A+i\phi} = F_A - \phi \wedge \phi + i\, d_A\phi = 0.$$

Finally, compute Ψ at $t = \pm 1$ and at $t = \pm i$, and verify algebraically that S-duality acts by $\Psi \mapsto -1/(n_G \Psi)$.

14.3 Dimensional reduction and Hitchin moduli space

Let the four-manifold be $M = \Sigma \times C$, where C is a closed Riemann surface. When C is made small compared with Σ, finite-energy fields vary slowly along Σ and at each point of Σ solve a two-dimensional gauge equation on C. The remaining field is therefore a map from Σ into a moduli space. This is the physical reason that a four-dimensional gauge theory produces a two-dimensional sigma model.

Definition 14.7 (Hitchin equations and Higgs bundles)**.** For a connection A on a principal G-bundle over C and $\phi \in \Omega^1(C, \mathrm{ad}\, P)$, the *Hitchin equations* are

$$F_A - \phi \wedge \phi = 0, \qquad d_A\phi = 0, \qquad d_A^*\phi = 0. \tag{H}$$

Writing the $(1,0)$ part of ϕ as φ, these become, up to a conventional normalization of φ,

$$\bar\partial_A \varphi = 0, \qquad F_A + [\varphi, \varphi^\dagger] = 0.$$

Thus (E, φ) is a $G_{\mathbb{C}}$-*Higgs bundle*: E is a holomorphic principal $G_{\mathbb{C}}$-bundle and

$$\varphi \in H^0(C, K_C \otimes \operatorname{ad} E).$$

The quotient of solutions to (H) by compact gauge transformations is the *Hitchin moduli space* $\mathcal{M}_H(G, C)$. Equivalently, after the appropriate stability condition is imposed, it is the moduli space of polystable $G_{\mathbb{C}}$-Higgs bundles.

Exercise 14.3 (Hitchin equations from four-dimensional instantons)**.** On $\mathbb{R}^4$ take a connection invariant under translations in x^3, x^4 and write $X = A_3$, $Y = A_4$. Decompose the anti-self-duality equations into

$$F_{12} + [X, Y] = 0, \qquad D_1 X - D_2 Y = 0, \qquad D_1 Y + D_2 X = 0,$$

up to the sign determined by the chosen orientation. With $z = x^1 + ix^2$ and a suitable choice of $\varphi = (X + iY)\, dz$, show that the last two equations give $\bar{\partial}_A \varphi = 0$ and the first gives the real Hitchin equation. This calculation is not the full supersymmetric compactification, but it exhibits the same dimensional-reduction mechanism using the instantons already studied in Exercise 4.9.

Theorem 14.8 (Hyperkähler descriptions; used without proof)**.** *On its smooth locus, $\mathcal{M}_H(G, C)$ carries a metric and complex structures I, J, K obeying the quaternion relations. In complex structure I it is the moduli space of Higgs bundles. In complex structure J it is the moduli space of reductive flat $G_{\mathbb{C}}$-connections, represented by $\mathcal{A} = A + i\phi$. If $p_1, \ldots, p_r$ generate the invariant polynomials on $\mathfrak{g}_{\mathbb{C}}$ and have degrees $d_1, \ldots, d_r$, the* Hitchin map *is*

$$h : \mathcal{M}_H(G, C) \longrightarrow \mathcal{B}_G = \bigoplus_{j=1}^{r} H^0(C, K_C^{d_j}), \qquad (E, \varphi) \longmapsto (p_1(\varphi), \ldots, p_r(\varphi)).$$

Its generic fibers are abelian varieties. The generic fibers for G and ${}^L G$ are dual, which is the geometric seed of mirror symmetry. These facts originate in Hitchin, Proceedings of the London Mathematical Society ***55*** *(1987).*

There is a useful symplectic explanation of the three equations. The affine space of pairs (A, ϕ) has three symplectic forms, and the gauge group acts with three moment maps. Their simultaneous vanishing is exactly (H), so

$$\mathcal{M}_H(G, C) = \{(A, \phi) : (\mathrm{H})\}/\mathcal{G}$$

is an infinite-dimensional hyperkähler quotient. This extends the ordinary symplectic reduction developed earlier: one now imposes three compatible moment-map equations rather than one.

Exercise 14.4 (The abelian Hitchin system)**.** Take $G = U(1)$ and let C have genus g_C. Show that all commutator terms vanish and that a solution consists of a flat unitary line bundle together with a holomorphic one-form. Define

$$\operatorname{Jac}(C) = H^1(C; \mathbb{R})/H^1(C; \mathbb{Z})$$

as the torus of degree-zero unitary flat line bundles, using Hodge theory to identify the displayed quotient with its usual complex structure. Deduce

$$\mathcal{M}_H(U(1), C) \cong T^* \operatorname{Jac}(C)$$

as a complex manifold in complex structure I, and compute its real dimension as $4g_C$. Identify the Hitchin map and its fibers. Explain why the duality of these fibers reduces here to the self-duality of a torus.

14.4 From gauge theory to branes

Definition 14.9 (Sigma-model reduction)**.** A two-dimensional sigma model on Σ with target X has fields $\Phi : \Sigma \to X$. In the adiabatic limit of the twisted gauge theory on $\Sigma \times C$, its low-energy configurations are slowly varying solutions of (H), hence maps

$$\Phi : \Sigma \longrightarrow \mathcal{M}_H(G, C).$$

The twist determines which topological sigma model survives. At $\Psi = \infty$ it is the B-model in complex structure J; at $\Psi = 0$ it is the A-model with symplectic form ω_K.

The B-model remembers complex geometry: its basic branes are described, schematically, by holomorphic subspaces carrying complexes of coherent sheaves. The A-model remembers symplectic geometry: a basic brane may be supported on a Lagrangian submanifold with a flat bundle, while more general coisotropic branes carry curvature compatible with the symplectic form. These are boundary conditions for the sigma model, not additional particles in four-dimensional spacetime.

Definition 14.10 (Mirror-symmetry form of S-duality)**.** Compactification converts the four-dimensional S-duality input into the equivalence

$$B_J(\mathcal{M}_H(G, C)) \simeq A_K(\mathcal{M}_H({}^L G, C)), \tag{M}$$

where the subscripts record the complex or symplectic structure. Reversing G and ${}^L G$ gives the direction used below: a point brane associated to a flat ${}^L G_{\mathbb{C}}$-connection is transformed into an A-brane on $\mathcal{M}_H(G, C)$. Equation (M) is the physical mirror-symmetry input; it is not proved by the preceding finite-dimensional theory.

Exercise 14.5 (Elementary brane geometry)**.** Let X be a smooth manifold. Prove that the zero section of T^*X is Lagrangian for the canonical symplectic form, and that the graph of a one-form α is Lagrangian exactly when $d\alpha = 0$. If X is a complex manifold, show that a point is a complex submanifold and hence is a possible support for a B-brane. Apply both observations to the local cotangent-bundle model of $\mathcal{M}_H(G, C)$ to see why mirror symmetry can exchange branes with supports of very different dimensions.

14.5 Line operators, Hecke modifications, and eigensheaves

Definition 14.11 (Wilson and 't Hooft lines)**.** A Wilson line measures the holonomy of a connection in a representation R:

$$W_R(\gamma) = \mathrm{Tr}_R \, \mathrm{Hol}_\gamma(\mathcal{A}).$$

At $\Psi = \infty$, the topological line uses the flat complex connection $\mathcal{A} = A + i\phi$ and is labeled by a representation of G. A *'t Hooft line* is defined instead by prescribing a magnetic singularity transverse to the line. Its charge is a cocharacter of G, equivalently a weight of ${}^L G$. The S transformation exchanges these two kinds of probes. When a line lies along a curve in Σ at a point $x \in C$, it acts on branes in the reduced sigma model.

Definition 14.12 (Hecke modification)**.** Let $x \in C$. A *Hecke modification* of a holomorphic principal $G_{\mathbb{C}}$-bundle E at x is another bundle E' together with an isomorphism

$$E|_{C \setminus \{x\}} \cong E'|_{C \setminus \{x\}}.$$

Its relative position at x is labeled by a dominant coweight of G. Under dimensional reduction, the singular boundary condition of an 't Hooft line changes E in precisely this way. Thus magnetic line operators become *Hecke functors* H_V, labeled by representations V of ${}^L G$.

Exercise 14.6 (Local Hecke modifications for GL_n)**.** Let z be a coordinate centered at x and let $\mathcal{O} = \mathbb{C}[[z]]$. For integers $m_1 \geq \cdots \geq m_n$, compare the lattices

$$\mathcal{O}^n, \qquad z^{m_1}\mathcal{O}e_1 \oplus \cdots \oplus z^{m_n}\mathcal{O}e_n$$

inside $\mathbb{C}((z))^n$. Prove that they define vector bundles isomorphic away from x, and compute the change of degree as $\sum_i m_i$, with the sign fixed by your convention for $\mathcal{O}(x)$. Identify $(m_1, \ldots, m_n)$ as a coweight. For $n = 1$, show that every modification is tensoring by $\mathcal{O}(mx)$.

Definition 14.13 (D-modules and Hecke eigensheaves)**.** Let $\mathrm{Bun}_G(C)$ denote the moduli stack of holomorphic $G_{\mathbb{C}}$-bundles on C; a stack is needed because bundles can have automorphisms. A D-module on it is, schematically, a sheaf carrying a compatible action of differential operators. Let $\mathcal{E}$ be a flat ${}^LG_{\mathbb{C}}$-bundle on C. A *Hecke eigensheaf* with eigenvalue $\mathcal{E}$ is a D-module $\mathcal{F}_{\mathcal{E}}$ equipped, for every representation V of ${}^LG$, with compatible isomorphisms

$$\mathsf{H}_V(\mathcal{F}_{\mathcal{E}}) \cong V_{\mathcal{E}} \boxtimes \mathcal{F}_{\mathcal{E}}, \tag{HE}$$

where $V_{\mathcal{E}}$ is the flat vector bundle obtained from $\mathcal{E}$ through V. Compatibility with tensor products of representations is part of the definition.

The word "eigen" in (HE) is literal but categorified. A commuting family of operators ordinarily sends an eigenvector to scalar multiples of itself. Here the operators are Hecke functors, the eigenvector is a sheaf, and the eigenvalues form the fibers of a flat vector bundle as the insertion point x moves around C. Flatness follows physically because moving a topological line on C cannot change its local action, though it can produce global monodromy.

Definition 14.14 (The Kapustin–Witten dictionary)**.** Begin on the ${}^LG$ side at $\Psi = \infty$. A flat ${}^LG_{\mathbb{C}}$-connection $\mathcal{E}$ is a point of $\mathcal{M}_H({}^LG, C)$ in complex structure J, so it supports a B-brane. Wilson lines act on this point brane through the vector spaces $V_{\mathcal{E},x}$. Mirror symmetry carries it to an A-brane on $\mathcal{M}_H(G, C)$, while S-duality carries Wilson lines to 't Hooft lines. The image is therefore a simultaneous eigenbrane for the Hecke operators.

To turn that eigenbrane into a D-module, one studies its open strings with the *canonical coisotropic brane* $\mathcal{B}_{cc}$. The endomorphism algebra of $\mathcal{B}_{cc}$ quantizes functions along the cotangent directions and yields a sheaf of differential operators on $\mathrm{Bun}_G(C)$, with a canonical twisting by a determinant or half-canonical line. Open strings from $\mathcal{B}_{cc}$ to the eigenbrane then form the twisted D-module $\mathcal{F}_{\mathcal{E}}$ satisfying (HE). This is the physical construction relating electric–magnetic duality to the geometric Langlands correspondence. Derived, stack-theoretic, and singular-locus refinements are suppressed here.

Exercise 14.7 (From an eigenbrane to the Hecke condition)**.** Suppose topological line operators $L_V(x)$ obey the representation-ring law

$$L_V(x)L_W(x) \cong L_{V\otimes W}(x)$$

and a brane $\mathcal{B}$ obeys $L_V(x)\mathcal{B} \cong \mathcal{B} \otimes E_{V,x}$. Show that the spaces $E_{V,x}$ must satisfy $E_{V\otimes W,x} \cong E_{V,x} \otimes E_{W,x}$. Explain why topological motion of x makes E_V a local system on C. After Wilson–'t Hooft duality, rewrite this statement as (HE). Identify exactly which step uses the unproved physical duality input.

This endpoint also clarifies the relation to the Arithmetic part. The geometric Langlands program is a categorical correspondence for bundles and local systems on a curve; the arithmetic Langlands program relates automorphic representations to Galois data over number fields and function fields. They share dual groups, Hecke operators, and harmonic-analysis structures, but the four-dimensional gauge theory above directly motivates the *geometric* correspondence. Passing from this picture to arithmetic Langlands requires additional algebro-geometric and number-theoretic machinery; it is not a consequence of supersymmetry alone.

Part IV

Arithmetic

15 Partitions and Generating Functions

Experimental observations.

- When independent bosonic modes have equally spaced excitation energies, many occupation patterns have exactly the same total energy; the degeneracy grows rapidly with the excitation number.
- Finite-size spectra at criticality organize into towers whose repeated level degeneracies can be read as coefficients of power series.

Arithmetic enters physics first as exact counting. A generating function does more than list degeneracies: its product form records independent modes, its sum form records constrained many-body states, and its analytic behavior controls the entropy of highly excited states. The tension between these descriptions is the source of the q-series identities used later in conformal field theory and integrable models.

15.1 Partition generating functions and finite boxes

Definition 15.1 (Partitions and q-Pochhammer symbols)**.** An *integer partition* of $n \geq 0$ is a finite nonincreasing sequence $\lambda = (\lambda_1, \dots, \lambda_r)$ of positive integers with $|\lambda| = \sum_i \lambda_i = n$. Let $p(n)$ be the number of partitions, with $p(0) = 1$. For a formal variable q, set

$$(a;q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \qquad (a;q)_\infty = \prod_{j=0}^{\infty} (1 - aq^j).$$

Analytically, the infinite product converges for $|q| < 1$. Euler's partition generating function is

$$P(q) = \sum_{n \geq 0} p(n) q^n = \frac{1}{(q;q)_\infty}.$$

Exercise 15.1 (Bosonic and fermionic occupation numbers)**.** Consider oscillator modes with one-particle energies $\hbar\omega, 2\hbar\omega, 3\hbar\omega, \dots$. Show that a bosonic Fock state of excitation energy $n\hbar\omega$ is a partition of n, and derive

$$\prod_{m \geq 1} \frac{1}{1 - q^m} = \sum_{n \geq 0} p(n) q^n.$$

For fermionic occupation numbers in $\{0, 1\}$, show that $\prod_{m \geq 1}(1 + q^m)$ counts partitions into distinct parts. Construct a bijection, by separating powers of two in the multiplicities, between partitions into distinct parts and partitions into odd parts, and conclude

$$\prod_{m\geq 1}(1+q^m)=\prod_{m\geq 1}\frac{1}{1-q^{2m-1}}.$$

Interpret the two products as different quasiparticle descriptions with the same exact degeneracies.

Definition 15.2 (Gaussian binomial coefficients)**.** For $0\leq r\leq n$, the *Gaussian binomial coefficient* is

$$\begin{bmatrix} n \\ r \end{bmatrix}_q=\frac{(q;q)_n}{(q;q)_r(q;q)_{n-r}},$$

and it is zero outside this range. Despite its quotient presentation, it is a polynomial in q with nonnegative integer coefficients.

Exercise 15.2 (Partitions in a box)**.** Prove the recurrence

$$\begin{bmatrix} n \\ r \end{bmatrix}_q=\begin{bmatrix} n-1 \\ r \end{bmatrix}_q+q^{n-r}\begin{bmatrix} n-1 \\ r-1 \end{bmatrix}_q.$$

Use either this recurrence or Ferrers-diagram conjugation to show that $\left[\begin{smallmatrix} a+b \\ a \end{smallmatrix}\right]_q$ is the generating polynomial for partitions whose diagram fits in an $a\times b$ rectangle. Deduce $\left[\begin{smallmatrix} a+b \\ a \end{smallmatrix}\right]_q=\left[\begin{smallmatrix} a+b \\ b \end{smallmatrix}\right]_q$ and prove the finite q-binomial theorem

$$\prod_{j=0}^{n-1}(1+zq^j)=\sum_{r=0}^{n}q^{r(r-1)/2}\begin{bmatrix} n \\ r \end{bmatrix}_q z^r.$$

Explain why finite-volume state counts are polynomials even when their thermodynamic limits are infinite products.

15.2 Rogers–Ramanujan identities and exclusion rules

Definition 15.3 (The Rogers–Ramanujan functions)**.** For $|q|<1$, define

$$G(q)=\sum_{r\geq 0}\frac{q^{r^2}}{(q;q)_r},\qquad H(q)=\sum_{r\geq 0}\frac{q^{r(r+1)}}{(q;q)_r}.$$

The Rogers–Ramanujan identities are

$$G(q)=\frac{1}{(q;q^5)_\infty(q^4;q^5)_\infty},\qquad H(q)=\frac{1}{(q^2;q^5)_\infty(q^3;q^5)_\infty}.$$

They equate fermionic-looking sums, in which r behaves like a particle number subject to exclusion, with bosonic-looking products over allowed mode numbers.

Exercise 15.3 (Difference conditions and Rogers–Ramanujan counting)**.** Fix the number r of parts. Subtract the staircase $(2r-1,2r-3,\ldots,1)$ to prove that $q^{r^2}/(q;q)_r$ generates partitions with exactly r parts whose successive parts differ by at least two. Use the staircase $(2r,2r-2,\ldots,2)$ for the same difference condition with smallest part at least two. Interpret the product sides as partitions into parts congruent to 1 or 4 modulo 5, and to 2 or 3 modulo 5, respectively. Taking the analytic Rogers–Ramanujan identities as supplied, deduce the two partition theorems. Later, identify the same functions as exact CFT characters and as functions occurring in an integrable lattice model; see Andrews, *Proceedings of the National Academy of Sciences* **78** (1981).

16 Theta Functions and Modular Forms

Experimental observations.

- Diffraction by a crystal produces sharp peaks on a reciprocal lattice, so real-space lattice geometry is encoded by arithmetic data in momentum space.
- A particle on a compact circle admits both a momentum-mode expansion and a winding-path expansion; Poisson summation makes the two thermal partition functions equal.

The same dual description appears throughout arithmetic physics. Lattice points may be counted directly, or reorganized through the dual lattice. The modular transformation $\tau \mapsto -1/\tau$ is the complex-analytic form of this exchange, while $\tau \mapsto \tau + 1$ records integrality and parity.

16.1 Lattice theta series and Poisson duality

Definition 16.1 (Integral lattices and theta series)**.** A *Euclidean lattice* $L \subset \mathbb{R}^d$ is the integer span of a basis of $\mathbb{R}^d$. Its dual is

$$L^* = \{y \in \mathbb{R}^d : \langle x, y\rangle \in \mathbb{Z} \text{ for every } x \in L\}.$$

It is *integral* if $L \subseteq L^*$, *unimodular* if $L = L^*$, and *even* if $\|x\|^2 \in 2\mathbb{Z}$ for every $x \in L$. For τ in the upper half-plane and $q = e^{2\pi i\tau}$, its theta series is

$$\Theta_L(\tau) = \sum_{x\in L} q^{\|x\|^2/2}.$$

Its coefficient of q^n counts lattice vectors of squared length $2n$.

Exercise 16.1 (Poisson duality for lattice theta series)**.** Apply Exercise 5.5 to a Gaussian and prove

$$\Theta_L(-1/\tau) = \frac{(-i\tau)^{d/2}}{\mathrm{vol}(L)}\Theta_{L^*}(\tau),$$

using the branch positive on the positive imaginary axis. Show that evenness gives $\Theta_L(\tau+1) = \Theta_L(\tau)$. Interpret the first identity as momentum–winding duality for a compact free boson and as the reciprocal-lattice relation behind diffraction. Explain why a general lattice theta series transforms with its dual rather than being a scalar modular form by itself.

16.2 Modular forms, theta functions, and eta products

Definition 16.2 (The modular group and modular forms)**.** The modular group $SL_2(\mathbb{Z})$ acts on the upper half-plane by

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix},$$

and is generated by $S : \tau \mapsto -1/\tau$ and $T : \tau \mapsto \tau + 1$. A *modular form of integral weight k for $SL_2(\mathbb{Z})$* is a holomorphic function f on the upper half-plane satisfying

$$f(\gamma\tau) = (c\tau + d)^k f(\tau)$$

and holomorphic at the cusp, meaning that its Fourier expansion $f(\tau) = \sum_{n\geq 0} a_n q^n$ has no negative powers. It is a *cusp form* if $a_0 = 0$.

Definition 16.3 (Jacobi theta functions and the Dedekind eta function)**.** Set

$$\vartheta_2(\tau) = \sum_{n\in\mathbb{Z}} q^{(n+1/2)^2/2},$$
$$\vartheta_3(\tau) = \sum_{n\in\mathbb{Z}} q^{n^2/2}, \qquad\qquad \eta(\tau) = q^{1/24}(q;q)_\infty.$$
$$\vartheta_4(\tau) = \sum_{n\in\mathbb{Z}} (-1)^n q^{n^2/2},$$

These functions have weight 1/2 transformation laws with phases or permutations, rather than being integral-weight scalar modular forms for the full modular group. In particular,

$$\eta(\tau+1) = e^{\pi i/12}\eta(\tau), \qquad \eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau).$$

Exercise 16.2 (Theta transformations and the Jacobi triple product)**.** Use Poisson summation on shifted and phase-twisted Gaussians to derive

$$\vartheta_2(-1/\tau) = (-i\tau)^{1/2}\vartheta_4(\tau), \quad \vartheta_3(-1/\tau) = (-i\tau)^{1/2}\vartheta_3(\tau), \quad \vartheta_4(-1/\tau) = (-i\tau)^{1/2}\vartheta_2(\tau).$$

Prove the T-transformation laws directly from the series. Establish the Jacobi triple product, first as an identity of formal Laurent series,

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2/2} = \prod_{m\geq 1}(1-q^m)(1+zq^{m-1/2})(1+z^{-1}q^{m-1/2}),$$

and use specializations to recover product forms for the theta functions. Explain how the two sides encode charge sectors and independent oscillator modes, respectively.

16.3 Eisenstein series and arithmetic applications

Definition 16.4 (Eisenstein series and the discriminant)**.** Let $\sigma_r(n) = \sum_{d|n} d^r$. The normalized Eisenstein series of weights four and six are

$$E_4(\tau) = 1 + 240\sum_{n\geq 1}\sigma_3(n)q^n, \qquad E_6(\tau) = 1 - 504\sum_{n\geq 1}\sigma_5(n)q^n.$$

The modular discriminant is the weight-twelve cusp form

$$\Delta(\tau) = \eta(\tau)^{24} = q\prod_{n\geq 1}(1-q^n)^{24} = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728}.$$

Exercise 16.3 (The ring of modular forms and the E_8 lattice)**.** Construct Eisenstein series by summing $(m\tau+n)^{-k}$ over nonzero lattice points for even $k > 2$, prove their modular transformation law, and obtain the displayed Fourier expansions for E_4 and E_6. Taking the valence formula as supplied, prove

$$M_*(SL_2(\mathbb{Z})) \cong \mathbb{C}[E_4, E_6]$$

as a graded ring. For a positive-definite even unimodular lattice, first use consistency of the S and T laws to prove that its dimension d is divisible by eight, and then show that its theta series is a modular form of weight $d/2$. For the E_8 lattice, use $\dim M_4 = 1$ to deduce $\Theta_{E_8} = E_4$, and conclude that E_8 has 240 vectors of squared length two.

Exercise 16.4 (Partition asymptotics from modular inversion)**.** Put $q = e^{-t}$ and use the modular transformation of η to prove

$$P(e^{-t}) \sim \sqrt{\frac{t}{2\pi}} \exp\left(\frac{\pi^2}{6t}\right) \qquad (t \downarrow 0).$$

Apply a saddle-point argument, or an appropriate Tauberian theorem with its hypotheses stated, to derive the Hardy–Ramanujan formula

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Interpret $\log p(n)$ as the microcanonical entropy of the bosonic modes in the first exercise of this part. Distinguish this asymptotic result from an exact coefficient formula; see Hardy–Ramanujan, *Proceedings of the London Mathematical Society* **17** (1918).

17 CFT Characters and Integrable Models

Experimental observations.

- At a one-dimensional quantum critical point, finite-size energy levels fall into conformal towers with reproducible degeneracies rather than an unstructured continuum.
- Exactly solved lattice models exhibit identities between constrained particle configurations and infinite products that are invisible in a generic numerical diagonalization.

The coefficients of a CFT character count states, while its modular transformation law relates low and high energies. This makes characters a meeting point of representation theory, statistical mechanics, and arithmetic. They are generally components of a vector-valued modular function, not scalar modular forms in the preceding definition.

17.1 Virasoro representations and character formulas

Definition 17.1 (The Virasoro algebra and its characters)**.** The *Virasoro algebra* has generators L_n for $n \in \mathbb{Z}$ and a central element acting by c, with

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}(m^3 - m)\delta_{m+n,0}.$$

A highest-weight vector $|h\rangle$ satisfies $L_0|h\rangle = h|h\rangle$ and $L_n|h\rangle = 0$ for $n > 0$. For a positive-energy module H_a with finite-dimensional L_0 eigenspaces, its *character* is

$$\chi_a(\tau) = \mathrm{Tr}_{H_a}\, q^{L_0 - c/24}.$$

The shift by $-c/24$ is the Casimir energy produced by mapping the plane to the cylinder.

Exercise 17.1 (Descendants, partitions, and null states)**.** Use the Poincaré–Birkhoff–Witt theorem to order Virasoro descendants as

$$L_{-\lambda_1} \cdots L_{-\lambda_r}|h\rangle, \qquad \lambda_1 \geq \cdots \geq \lambda_r \geq 1.$$

Before quotienting by null vectors, deduce the Verma-module character

$$\chi^{\mathrm{Verma}}_{c,h}(\tau) = \frac{q^{h-c/24}}{(q;q)_\infty}.$$

For the vacuum, impose $L_{-1}|0\rangle = 0$ and find the resulting generic vacuum descendant product. Explain why additional null vectors subtract states and turn unrestricted partition numbers into the more structured q-series of minimal models.

17.2 Modular invariance and exact state counting

Definition 17.2 (Rational characters and modular invariants)**.** In a rational chiral CFT, assume there are finitely many irreducible positive-energy sectors H_a and that their characters span a representation of the modular group,

$$\chi_a(-1/\tau) = \sum_b S_{ab}\chi_b(\tau), \qquad \chi_a(\tau+1) = \sum_b T_{ab}\chi_b(\tau).$$

A full torus partition function has the form

$$Z(\tau,\bar\tau) = \sum_{a,b} M_{ab}\chi_a(\tau)\overline{\chi_b(\tau)}$$

with nonnegative integer multiplicities. It is modular invariant when M commutes with S and T. Under the standard rationality hypotheses, these matrices agree with the modular data governing the associated superselection category developed in Quantum Topology.

Exercise 17.2 (Exact state counting in the Ising CFT)**.** For the Ising theory with $c = 1/2$ and primary weights $0, 1/2, 1/16$, take

$$\begin{aligned}
\chi_0 &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} + \sqrt{\frac{\vartheta_4}{\eta}}\right),\\
\chi_{1/2} &= \frac{1}{2}\left(\sqrt{\frac{\vartheta_3}{\eta}} - \sqrt{\frac{\vartheta_4}{\eta}}\right),\\
\chi_{1/16} &= \frac{1}{\sqrt{2}}\sqrt{\frac{\vartheta_2}{\eta}}.
\end{aligned}$$

Expand each as $q^{h-c/24}$ times a power series and interpret its first coefficients as exact descendant degeneracies. Derive

$$S = \frac{1}{2}\begin{pmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & -\sqrt{2} \\ \sqrt{2} & -\sqrt{2} & 0 \end{pmatrix}, \qquad T_{aa} = e^{2\pi i(h_a - c/24)}.$$

Verify the modular invariance of $|\chi_0|^2 + |\chi_{1/2}|^2 + |\chi_{1/16}|^2$ and recover the Ising fusion rules from the Verlinde formula. Relate these towers to the critical Ising spectroscopy exercise in Quantum Topology.

Exercise 17.3 (The Cardy density of states)**.** Assume a modular-invariant unitary CFT with discrete spectrum, unique vacuum, and left and right central charges $c, \bar{c}$. Set $\tau = i\beta/(2\pi)$ and use S-invariance to show that the high-temperature partition function is controlled by the vacuum at inverse temperature $4\pi^2/\beta$. A saddle-point inverse Laplace transform then gives, for large left and right excitation numbers,

$$\log \rho(N, \bar{N}) \sim 2\pi\sqrt{\frac{cN}{6}} + 2\pi\sqrt{\frac{\bar{c}\bar{N}}{6}}.$$

State how c is replaced by $c_{\text{eff}} = c - 24h_{\min}$ in a nonunitary theory. For a nonrotating state with $c = \bar{c}$ and $N = \bar{N}$, recover the thermal entropy used in the BTZ exercise and hence reproduce the horizon entropy from boundary-state counting; see Cardy, *Nuclear Physics B* **270** (1986).

Exercise 17.4 (Rogers–Ramanujan characters of the Yang–Lee model)**.** The nonunitary minimal model $M(2,5)$ has $c = -22/5$ and primary weights 0 and $-1/5$. Show, by comparing leading powers and null-state degeneracies, that its two irreducible characters are

$$\chi_0(\tau) = q^{11/60}H(q), \qquad \chi_{-1/5}(\tau) = q^{-1/60}G(q).$$

Use both the sum and product sides of the Rogers–Ramanujan identities to give exclusion-rule and allowed-mode descriptions of the same conformal towers. Compute $c_{\text{eff}} = 2/5$ and explain why high-energy growth is controlled by the lowest conformal weight rather than by the vacuum in this nonunitary example. Do not interpret the model as a unitary quantum Hamiltonian despite its exact arithmetic structure.

17.3 Integrable lattice models

Definition 17.3 (Transfer matrices and integrability)**.** For a two-dimensional lattice model on a cylinder, a *row transfer matrix* $T_N(u)$ propagates one width-N row to the next and depends on a spectral parameter u. The model is *integrable* when it admits a nontrivial commuting family

$$[T_N(u), T_N(v)] = 0 \qquad \text{for every } u, v,$$

usually obtained from a Yang–Baxter or star–triangle relation. Commuting transfer matrices permit simultaneous spectral analysis, but integrability alone does not guarantee that every finite-size degeneracy has a simple closed formula.

Exercise 17.5 (Hard hexagons and exact finite-volume counting)**.** In the hard-hexagon model, occupied vertices of a triangular lattice may not be adjacent. On an $N \times M$ torus define

$$Z_{N,M}(z) = \sum_{I \text{ independent}} z^{|I|}.$$

Index a row transfer matrix by cyclic binary rows and set its matrix element to zero when two consecutive rows violate nearest-neighbor exclusion, and otherwise to $z^{(|a|+|b|)/2}$. Prove $Z_{N,M}(z) = \operatorname{Tr}(T_N(z)^M)$ and compute the polynomial for several small sizes. Show that the coefficients are exact state counts, whereas the largest eigenvalue controls the thermodynamic free energy.

Take Baxter's commuting deformation and exact thermodynamic solution as input. Verify that its critical activity is

$$z_c = \frac{11 + 5\sqrt{5}}{2} = \left(\frac{1+\sqrt{5}}{2}\right)^5,$$

and identify where Rogers–Ramanujan functions enter the solution. Explain why their appearance is a special consequence of the integrable structure, not a feature of arbitrary lattice gases; see Baxter, *Journal of Physics A* **13** (1980).

18 Root Systems, DAHA, and Macdonald Operators

Experimental observations.

- A linear-optics simulator sent the same photon polarization through two wave-plate circuits representing the two sides of a spectral Yang–Baxter equation. The measured output fidelity reached one, within error, precisely along the predicted relation among the three spectral parameters; away from it the two circuits produced different outputs.

Thought experiment. Place n identical particles on a line and temporarily label them so that their trajectories can be followed. The coincidence sets $x_i = x_j$ divide the labeled configuration space into $n!$ chambers, one for each ordering of the particles. Crossing a wall exchanges two labels. If many-body scattering factorizes into two-body exchanges, two different sequences with the same net exchange must give the same operator; this is the braid relation. In models where a two-body exchange has two eigenamplitudes, its normalized operator may also satisfy a quadratic Hecke relation. Closing the line into a ring repeats the collision walls periodically and introduces the affine translations below.

Thus the route to double affine Hecke algebras is not "introduce a large algebra and hope it is useful." Collision walls first produce a reflection group. In suitable factorized-scattering models its group algebra deforms to a Hecke algebra. Putting the particles on a periodic space makes the walls affine, while quantizing both periodic position and momentum produces two lattice directions. A DAHA packages those two directions and generates commuting difference Hamiltonians.

18.1 Root systems, scattering, and Hecke algebras

Definition 18.1 (Root systems and affine Weyl groups)**.** Let V be a finite-dimensional Euclidean space. A finite set $R \subset V \setminus \{0\}$ is a *reduced crystallographic root system* if it spans V, is preserved by the reflections

$$s_\alpha(v) = v - \langle v, \alpha^\vee \rangle \alpha, \qquad \alpha^\vee = \frac{2\alpha}{\langle \alpha, \alpha \rangle},$$

has $R \cap \mathbb{R}\alpha = \{\alpha, -\alpha\}$, and satisfies $\langle \beta, \alpha^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in R$. The group $W = \langle s_\alpha : \alpha \in R \rangle$ is its *Weyl group*, and the *coroot lattice* is $Q^\vee = \mathrm{span}_{\mathbb{Z}}\{\alpha^\vee : \alpha \in R\}$.

The hyperplanes

$$H_{\alpha,m} = \{v \in V : \langle v, \alpha \rangle = m\}, \qquad \alpha \in R, \quad m \in \mathbb{Z},$$

are the *affine reflection hyperplanes*. Reflections in them generate the *affine Weyl group* $W_{\mathrm{aff}} = Q^\vee \rtimes W$. Its connected chambers are called *alcoves*.

Exercise 18.1 (Type A as particle exchange)**.** In $V = \{u \in \mathbb{R}^n : \sum_i u_i = 0\}$, take

$$R = A_{n-1} = \{e_i - e_j : i \neq j\}.$$

Show that $s_{e_i - e_j}$ exchanges the ith and jth coordinates and hence $W \cong S_n$. Prove that the simple reflections $s_i = s_{e_i - e_{i+1}}$ satisfy

$$s_i^2 = 1, \qquad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}, \qquad s_i s_j = s_j s_i \quad (|i - j| > 1).$$

Identify the finite chambers with possible orderings of labeled particles. On the universal cover of a ring configuration, interpret $u_i - u_j = m$ as periodic copies of collision walls and explain why translations join the permutation group in the affine Weyl group.

Definition 18.2 (Hecke deformation of reflections)**.** Let $(W, \{s_i\})$ be a Coxeter system, and let m_{ij} be the order of $s_i s_j$. Over $\mathbb{C}(t^{1/2})$, its *Hecke algebra* $\mathcal{H}_t(W)$ is generated by T_i with the same braid relations as the s_i and with

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

At $t = 1$ this becomes $T_i^2 = 1$, so the algebra specializes to the group algebra of W. For generic t, a reflection has been replaced by an operator with two allowed eigenvalues. This algebraic fact matches a two-channel exchange matrix only in models whose scattering actually obeys the displayed quadratic relation; it is not a claim about generic particle collisions.

Exercise 18.2 (From factorized exchange to a Hecke algebra)**.** Let R_i be operators describing adjacent exchanges. Assume distant exchanges commute, three-particle consistency gives $R_i R_{i+1} R_i = R_{i+1} R_i R_{i+1}$, and every R_i has the same two nonzero eigenvalues a and b. Rescale the R_i so that the eigenvalues become $t^{1/2}$ and $-t^{-1/2}$, when this is possible. Prove that the resulting operators define a representation of $\mathcal{H}_t(S_n)$. Explain exactly which assumption would fail for a two-body scattering matrix with three distinct eigenchannels.

The Hecke algebra describes the exchange operators, but it does not yet say why those operators coexist with a coproduct, tensor-product representations, and commuting transfer matrices. The algebraic dual of the compact quantum-group picture from Operator Space Theory supplies that missing symmetry.

18.2 Quantum groups and the Yang–Baxter equation

Definition 18.3 (Quantized enveloping algebras and R-matrices)**.** For generic q, the Hopf algebra $U_q(\mathfrak{sl}_2)$ is generated by E, F, K, K^{-1} with

$$KEK^{-1} = q^2 E, \qquad KFK^{-1} = q^{-2} F, \qquad [E, F] = \frac{K - K^{-1}}{q - q^{-1}},$$

and coproduct

$$\Delta(K) = K \otimes K, \quad \Delta(E) = E \otimes 1 + K \otimes E, \quad \Delta(F) = F \otimes K^{-1} + 1 \otimes F.$$

It is paired algebraically with the dense coordinate Hopf algebra of $SU_q(2)$ introduced in Operator Space Theory: one side acts on states, while the other is made from their matrix coefficients.

A quasitriangular completion of a Hopf algebra has an invertible *universal R-matrix* $\mathcal{R}$ satisfying

$$\mathcal{R}\Delta(x) = \Delta^{\mathrm{op}}(x)\mathcal{R}, \quad (\Delta \otimes \mathrm{id})(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{23}, \quad (\mathrm{id} \otimes \Delta)(\mathcal{R}) = \mathcal{R}_{13}\mathcal{R}_{12}.$$

On finite-dimensional modules, the braid operator $\check{R} = P(\pi \otimes \pi)(\mathcal{R})$, with P exchanging tensor factors, satisfies

$$\check{R}_{12}\check{R}_{23}\check{R}_{12} = \check{R}_{23}\check{R}_{12}\check{R}_{23}.$$

When $\check{R}$ has exactly two eigenvalues, a normalization may turn its minimal polynomial into the Hecke quadratic relation. This implication requires the two-eigenvalue hypothesis; a general quantum-group R-matrix need not factor through a Hecke algebra.

Exercise 18.3 (Coproduct consistency and the Yang–Baxter equation)**.** Check that the displayed coproduct preserves every defining relation of $U_q(\mathfrak{sl}_2)$ and is coassociative. Derive the quantum Yang–Baxter equation

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$$

from the two coproduct identities, and then derive the braid relation for $\check{R}$. Interpret the two sides as the two factorizations of a three-body scattering process. Prove that a two-eigenvalue $\check{R}$ can be rescaled to a Hecke generator exactly under the compatibility condition found in the preceding exercise.

Exercise 18.4 (A direct optical Yang–Baxter test)**.** Consider the unitary polarization transformations

$$A(\theta) = \begin{pmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \qquad B(\theta) = \begin{pmatrix} \cos\theta & -i\sin\theta \\ -i\sin\theta & \cos\theta \end{pmatrix}.$$

Multiply the matrices and prove that, away from singular boundary cases,

$$A(\theta_1)B(\theta_2)A(\theta_3) = B(\theta_3)A(\theta_2)B(\theta_1)$$

exactly when

$$\tan\theta_2 = \frac{\sin(\theta_1+\theta_3)}{\cos(\theta_1-\theta_3)}.$$

For $\theta_1 = 56^\circ$ and $\theta_3 = 23^\circ$, obtain $\theta_2 \approx 49.49^\circ$. For an arbitrary input qubit $|\psi\rangle$, show that the output-state overlap has absolute value one when the relation holds and is generally smaller away from it. Compare these predictions with the wave-plate experiment of Zheng et al., *Journal of the Optical Society of America B* **30** (2013). State clearly that this experiment tests a two-dimensional Temperley–Lieb reduction of the Yang–Baxter equation, not the full representation theory of a quantum group or DAHA.

18.3 Quantum spherical functions and qKZ transport

Definition 18.4 (Quantum spherical functions)**.** Let $H = U_q(\mathfrak{g})$ and let $B \subseteq H$ be a right coideal subalgebra, so $\Delta(B) \subseteq B \otimes H$. It plays the role of the stabilizer algebra of a quantum homogeneous space. If an H-module V contains a vector v satisfying

$$bv = \varepsilon(b)v \qquad (b \in B),$$

then the matrix coefficient

$$\varphi_V(x) = \langle v, \pi_V(x)v\rangle$$

is a *quantum zonal spherical function.* It belongs to the coordinate Hopf dual of H and is invariant under the left and right quantum stabilizer actions.

Radial parts of central or invariant elements act on these coefficients by commuting q-difference operators. For important quantum symmetric pairs, their Weyl-invariant joint eigenfunctions are Macdonald polynomials on special parameter subfamilies. This is the quantum-group deformation of the Harish–Chandra spherical transform developed earlier.

Exercise 18.5 (From spherical analysis to Macdonald polynomials)**.** First let $q \to 1$ and replace B by the enveloping algebra of a compact subgroup $K \subseteq G$. Recover the K-fixed vector and zonal spherical coefficient used in Harish–Chandra theory. Explain why the radial parts of invariant operators commute on G/K.

Taking quantum symmetric-pair harmonic analysis as supplied, identify in Noumi's conventions the zonal spherical functions of the quantum analogues of $GL(n)/SO(n)$ and $GL(2n)/Sp(2n)$ with type-A Macdonald polynomials on the parameter lines $t = q^{1/2}$ and $t = q^2$, respectively; see Noumi, *Advances in Mathematics* **123** (1996). Compare their radial eigenvalue with the eigenvalue of $D_{q,t}$ below. Taking the general quantum symmetric-pair result as input, explain how the same mechanism extends across root systems; see Letzter, *Advances in Mathematics* **189** (2004).

On the separate nonarchimedean branch, take as supplied that spherical functions of split reductive groups over a p-adic field are Hall–Littlewood-type functions whose parameters are fixed by root multiplicities and the residue-field cardinality. Explain why this is a genuinely arithmetic specialization, whereas a quantum symmetric space usually imposes a relation between q and t. Conclude that neither construction alone supplies two independent generic parameters.

An R-matrix with spectral parameter also defines a difference connection: its exchange equations transport a vector-valued function between adjacent ordering chambers, and the Yang–Baxter equation is precisely the flatness condition for three successive exchanges. With periodic transport these become quantum Knizhnik–Zamolodchikov (qKZ) equations. For affine Hecke modules, the Matsuo–Cherednik correspondence converts suitable qKZ solutions into joint eigenfunctions of Cherednik–Macdonald difference operators. In spin representations the same equations become the qKZ equations of XXZ-type integrable chains; see Stokman, *Communications in Mathematical Physics* **338** (2015).

Exercise 18.6 (Flatness, qKZ transport, and affine Hecke monodromy)**.** Let a vector-valued function satisfy the adjacent exchange rule

$$\Psi(\dots, z_i, z_{i+1}, \dots) = \check{R}_i(z_i/z_{i+1})\Psi(\dots, z_{i+1}, z_i, \dots).$$

Show that consistency of the two routes reversing three adjacent variables is exactly the spectral Yang–Baxter equation. Add a periodic rule that returns z_1 at the end as qz_1, and explain why translations enlarge the finite braid action to an affine one. If the R-matrices satisfy a Hecke quadratic, prove that the exchange monodromy factors through an affine Hecke algebra. Taking the Matsuo–Cherednik correspondence as supplied, track how Weyl-invariant qKZ solutions become scalar joint eigenfunctions of the commuting difference operators. This is the precise bridge from quantum-group scattering data to the Macdonald spectral problem.

Quantum groups and p-adic groups therefore explain why Macdonald-type functions arise as spherical functions, but only on constrained parameter families. DAHA is not itself a compact quantum group. Its role is to organize the full two-parameter difference harmonic analysis, including two affine directions and the Fourier-like exchange between them.

18.4 DAHA and Macdonald spectral theory

Definition 18.5 (The GL_n form of type-A DAHA)**.** Let

$$\mathcal{P}_n = \mathbb{C}(q, t^{1/2})[x_1^{\pm 1}, \dots, x_n^{\pm 1}],$$

let X_j act by multiplication by x_j, and let s_i exchange x_i and x_{i+1}. Define the Demazure–Lusztig operators

$$T_i = t^{1/2}s_i + (t^{1/2} - t^{-1/2})\frac{s_i - 1}{x_i/x_{i+1} - 1}, \qquad 1 \le i < n,$$

and the affine cyclic shift

$$(\pi f)(x_1,\dots,x_n) = f(x_2,\dots,x_n,qx_1).$$

The quotients in the formula for T_i are Laurent polynomials because $s_i f - f$ vanishes when $x_i = x_{i+1}$. For generic q,t, the operator algebra generated by $X_j^{\pm 1}$, T_i, and $\pi^{\pm 1}$ is the faithful polynomial realization of the GL_n form of the type-A_{n-1} *double affine Hecke algebra*, or DAHA. Passing to the semisimple A_{n-1} version removes the common center-of-mass coordinate.

The first affine lattice is visible in the commuting multiplication operators X_j. Words in the T_i and π produce a second commuting family Y_j of q-difference-reflection operators. The Hecke generators mediate between the two. This X–Y symmetry is the meaning of "double affine"; it does not mean merely that the algebra has two parameters. Cherednik used this structure to prove Macdonald's inner-product and constant-term conjectures; see Cherednik, *Annals of Mathematics* **141** (1995).

Exercise 18.7 (Checking the polynomial representation)**.** Show directly that each T_i preserves $\mathcal{P}_n$ and satisfies

$$(T_i - t^{1/2})(T_i + t^{-1/2}) = 0.$$

Verify the braid relation first on Laurent monomials for $n = 3$, and then prove it in general. Check that

$$\pi X_j \pi^{-1} = X_{j+1} \quad (j < n), \qquad \pi X_n \pi^{-1} = qX_1.$$

Interpret this last wraparound factor as the affine translation. Finally, set $t = 1$ and identify the remaining reflection and shift operators.

Definition 18.6 (Macdonald operators and polynomials)**.** Let T_{q,x_i} shift one variable,

$$(T_{q,x_i} f)(x_1,\dots,x_n) = f(x_1,\dots,qx_i,\dots,x_n).$$

The first type-A *Macdonald difference operator* is

$$D_{q,t} = \sum_{i=1}^{n} \left(\prod_{j\neq i} \frac{tx_i - x_j}{x_i - x_j} \right) T_{q,x_i}.$$

Although its summands are rational, their sum preserves symmetric Laurent polynomials. In DAHA it comes from a Weyl-invariant symmetric function of the commuting Y-operators; higher symmetric functions give a full commuting family.

For generic q,t, the *Macdonald polynomial* $P_\lambda(x;q,t)$ is the unique symmetric polynomial triangular in the monomial-symmetric basis and satisfying

$$D_{q,t} P_\lambda = \left(\sum_{i=1}^{n} q^{\lambda_i} t^{n-i} \right) P_\lambda, \qquad \lambda_1 \geq \cdots \geq \lambda_n \geq 0.$$

Thus partitions label exact joint eigenstates of commuting arithmetic difference operators. These polynomials simultaneously deform Schur, Hall–Littlewood, and Jack polynomials; see Macdonald, *Séminaire Lotharingien de Combinatoire* B20a (1988).

Exercise 18.8 (The two-particle Macdonald spectrum)**.** For $n = 2$, prove that the apparent pole of $D_{q,t} f$ cancels whenever $f(x_1,x_2)$ is symmetric. Check

$$D_{q,t} 1 = (1 + t), \qquad D_{q,t}(x_1 + x_2) = (1 + qt)(x_1 + x_2),$$

and find the coefficient $c(q,t)$ for which $x_1^2 + x_2^2 + c(q,t)x_1x_2$ is an eigenfunction with leading partition $(2,0)$. Investigate the specializations $q = t$, $q = 0$, and $q \to 1$ with $t = q^g$. Which familiar symmetric-polynomial families appear?

The physical payoff is now visible. Multiplication by x_i is a periodic position observable, while T_{q,x_i} is a finite translation in its conjugate logarithmic coordinate. The Macdonald operators are therefore finite-difference many-body Hamiltonians. After a change of measure they are the trigonometric Ruijsenaars operators, the relativistic deformation of Calogero–Sutherland dynamics. Their mutual commutativity is inherited from the commuting Y-family rather than guessed from the displayed formula; see Ruijsenaars, *Communications in Mathematical Physics* **110** (1987).

Exercise 18.9 (The Calogero–Sutherland limit)**.** Write $x_i = e^{z_i}$, set $q = e^{\epsilon}$ and $t = e^{g\epsilon}$, and expand $D_{q,t}$ through order ϵ^2. After subtracting its scalar value on 1 and removing the center-of-mass drift, show that the nontrivial differential operator is proportional to

$$\mathcal{L}_g = \sum_i \partial_{z_i}^2 + g \sum_{i<j} \coth\left(\frac{z_i - z_j}{2}\right) (\partial_{z_i} - \partial_{z_j}).$$

On the circle, put $z_i = i\theta_i$ and conjugate by

$$\Psi_0(\theta) = \prod_{i<j} \left| 2 \sin \frac{\theta_i - \theta_j}{2} \right|^g .$$

Recover, up to its ground-state energy, the Sutherland Hamiltonian

$$H_g = -\sum_i \partial_{\theta_i}^2 + \frac{g(g-1)}{2} \sum_{i<j} \frac{1}{\sin^2((\theta_i - \theta_j)/2)}.$$

Show that $P_\lambda(x; q, q^g) \to P_\lambda^{(1/g)}(x)$ as $q \to 1$, with the stated Jack-parameter convention. This limit explains physically why root systems, partitions, q-difference operators, and an exactly solvable many-body spectrum occur in the same theory.

18.5 From Gustafson integrals to elliptic Yang–Baxter identities

The Macdonald operators are the spectral side of root-system harmonic analysis. Their orthogonality measures and normalization constants form a parallel integral side. In rank one this begins with the Askey–Wilson q-beta integral; Gustafson's multivariable extensions expose the root system in its integrand. Elliptic deformation then replaces each q-factor by a theta factor. At that level the resulting identities also become local consistency relations for exactly solvable lattice models.

For a list of parameters write $(a_1, \dots, a_r; q)_\infty = \prod_s (a_s; q)_\infty$, and use the shorthand $f(z^{\pm 1}) = f(z)f(z^{-1})$ and $f(z_i^{\pm 1} z_j^{\pm 1}) = \prod_{\epsilon,\delta=\pm 1} f(z_i^{\epsilon} z_j^{\delta})$.

Theorem 18.7 (Askey–Wilson and Gustafson; used without proof)**.** *Let $|q|, |t|, |a_r| < 1$. The rank-one Askey–Wilson integral is*

$$\frac{1}{2\pi i} \oint_{|z|=1} \frac{(z^{\pm 2}; q)_\infty}{\prod_{r=1}^4 (a_r z^{\pm 1}; q)_\infty} \frac{dz}{z} = 2 \frac{(a_1 a_2 a_3 a_4; q)_\infty}{(q; q)_\infty \prod_{1 \le r < s \le 4} (a_r a_s; q)_\infty}.$$

Its BC_n extension is obtained from the Weyl-invariant density

$$\Delta_{BC_n}(z) = \prod_{i=1}^n \frac{(z_i^{\pm 2}; q)_\infty}{\prod_{r=1}^4 (a_r z_i^{\pm 1}; q)_\infty} \times \prod_{1 \le i < j \le n} \frac{(z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}{(t z_i^{\pm 1} z_j^{\pm 1}; q)_\infty}.$$

With unit circles separating the two sequences of poles, Gustafson's evaluation is

$$\begin{aligned}\frac{1}{2^n n!}\oint_{\mathbb{T}^n}\Delta_{BC_n}(z)\prod_{i=1}^{n}\frac{dz_i}{2\pi i z_i}\\ =\prod_{j=1}^{n}\frac{(t,t^{n+j-2}a_1a_2a_3a_4;q)_\infty}{(t^j,q;q)_\infty\prod_{1\le r<s\le 4}(t^{j-1}a_ra_s;q)_\infty}.\end{aligned}$$

The factor $2^n n!$ is the order of the BC_n Weyl group. Related A_n, $SU(n)$, and $Sp(n)$ integrals and multilateral sums give root-system extensions of classical beta integrals and bilateral hypergeometric identities; see Gustafson, SIAM Journal on Mathematical Analysis ***25*** *(1994).*

The variables z_i lie on a maximal torus. Permutations and inversions $z_i \mapsto z_i^{-1}$ form its BC_n Weyl symmetry, while the factors $z_i^{\pm 1}z_j^{\pm 1}$ run over the pair roots $\pm e_i \pm e_j$. Consequently the integral can be read as the exact normalization of a q-deformed interacting eigenvalue ensemble. It is also the constant term giving the norm of the zero-partition polynomial in the Macdonald–Koornwinder orthogonality measure. Thus the appearance of the same root systems on the operator and integral sides is structural rather than a coincidence of formulas.

Exercise 18.10 (Root systems in the Gustafson integral)**.** Prove directly that Δ_{BC_n} is invariant under permutations and independent inversions of the z_i. Identify its factors with the roots $\pm e_i$, $\pm 2e_i$, and $\pm e_i \pm e_j$. Set $n = 1$ in Gustafson's formula and recover the displayed Askey–Wilson integral, including the Weyl-group factor. Set $t = q^g$, use

$$\frac{(q^x;q)_\infty}{(q^{x+g};q)_\infty}\sim(1-q)^g\frac{\Gamma(x+g)}{\Gamma(x)}\qquad(q\to 1^-),$$

and explain how the pair factors tend to the power-law repulsion of a Selberg-type density. Interpret the product evaluation as an exactly known partition-function normalization, without claiming that a generic interacting ensemble is integrable.

For $|p| < 1$, define the multiplicative theta function and elliptic shifted factorial by

$$\theta(z;p)=(z;p)_\infty(p/z;p)_\infty,\qquad (a;q,p)_m=\prod_{j=0}^{m-1}\theta(aq^j;p).$$

Since $\theta(z;0) = 1 - z$, the limit $p \to 0$ recovers the q-shifted factorial. The elliptic gamma function

$$\Gamma(z;p,q)=\prod_{j,k\ge 0}\frac{1-p^{j+1}q^{k+1}/z}{1-p^jq^kz}$$

satisfies $\Gamma(qz;p,q) = \theta(z;p)\Gamma(z;p,q)$ and the analogous equation with p and q exchanged. It is therefore the two-period deformation of the q-gamma factors appearing in beta integrals.

Theorem 18.8 (Elliptic beta integral and star–triangle relation; used without proof)**.** *If $|p|, |q|, |t_r| < 1$ and $\prod_{r=1}^{6} t_r = pq$, then*

$$\frac{(p;p)_\infty(q;q)_\infty}{4\pi i}\oint_{\mathbb{T}}\frac{\prod_{r=1}^{6}\Gamma(t_rz^{\pm 1};p,q)}{\Gamma(z^{\pm 2};p,q)}\frac{dz}{z}=\prod_{1\le r<s\le 6}\Gamma(t_rt_s;p,q).$$

With a suitable grouping of the six parameters into three rapidity pairs, this evaluation is the star–triangle relation: integration over the spin at a star gives the product of weights on the dual triangle.

The same identity supplies a Coxeter relation for elliptic integral transforms and hence an integral-operator solution of the Yang–Baxter equation. These statements, including the required contours and positivity regimes, are taken as input; see Spiridonov, Elliptic beta integrals and solvable models of statistical mechanics *and Derkachov–Spiridonov,* Yang–Baxter equation, parameter permutations, and the elliptic beta integral*.*

Multivariable elliptic beta integrals on root systems similarly degenerate to Gustafson-type q-beta integrals as $p \to 0$, sometimes after parameters are scaled. In rank one the displayed elliptic integral first gives the Nassrallah–Rahman integral; a further specialization gives Askey–Wilson. This qualification matters: the arrows below describe families and their degenerations, not a claim that every displayed formula limits directly to the next displayed formula without parameter choices.

There is a complementary finite-sum branch. In additive notation set

$$[x]_{h,\tau} = \frac{\vartheta_{11}(hx,\tau)}{\vartheta_{11}(h,\tau)}.$$

Products of these elliptic numbers replace ordinary rising factorials in terminating balanced very-well-poised series. Frenkel and Turaev call the resulting functions *modular hypergeometric functions*: "modular" refers to the simultaneous transformation of the elliptic parameters (h,τ), not to an ordinary scalar modular form in τ alone.

Theorem 18.9 (Frenkel–Turaev; used without proof)**.** *Elliptic $6j$-symbols with spectral parameter can be expressed by terminating balanced very-well-poised modular hypergeometric series. Their tetrahedral symmetries give elliptic Bailey transformations and Jackson-type summations, while their recoupling consistency gives an elliptic spectral Yang–Baxter equation. Under degeneration they yield trigonometric $6j$-symbols and basic hypergeometric identities. These spectral-parameter symbols deform recoupling data of solvable face models; they are not being identified with the fixed F-symbols of an arbitrary modular tensor category. See Frenkel–Turaev,* Elliptic solutions of the Yang–Baxter equation and modular hypergeometric functions*.*

At the trigonometric level, the corresponding Bailey transformations are the same kind of iteration mechanism that produces Rogers–Ramanujan-type q-series. The identities used earlier for exclusion-rule counting and the identities enforcing integrable recoupling therefore belong to one deformation framework, even though a particular Rogers–Ramanujan identity need not itself be a Yang–Baxter equation.

Exercise 18.11 (The rational–trigonometric–elliptic hierarchy)**.** Use the Jacobi triple product to prove

$$[x]_{h,\tau} \longrightarrow \frac{\sin(\pi hx)}{\sin(\pi h)} \quad (\operatorname{Im}\tau \to \infty), \qquad \frac{\sin(\pi hx)}{\sin(\pi h)} \longrightarrow x \quad (h \to 0).$$

Prove $\theta(z;0) = 1 - z$ and $(a;q,p)_m \to (a;q)_m$ as $p \to 0$. Hence organize the three levels as

$$\begin{gathered}\text{elliptic hypergeometric identities and elliptic Yang–Baxter data}\\ \downarrow\ p \to 0\\ \text{basic hypergeometric identities, Gustafson integrals, and trigonometric } R\text{-matrices}\\ \downarrow\ q \to 1\\ \text{ordinary hypergeometric identities, Selberg integrals, and rational data.}\end{gathered}$$

Explain why the integral/star–triangle branch and the $6j$-symbol/recoupling branch are complementary realizations of Yang–Baxter consistency rather than two names for the same construction. Finally relate the trigonometric middle level to the Macdonald–Ruijsenaars operators above and the elliptic level to their elliptic deformation.

19 Cyclotomic and Aperiodic Structure

Experimental observations.

- Electron diffraction from rapidly cooled aluminum–manganese alloys shows sharp Bragg peaks with icosahedral symmetry, even though no three-dimensional periodic lattice can have that rotational symmetry.
- Spectra of quasiperiodically modulated chains develop recursively organized gaps whose labels are constrained by irrational frequencies.

Arithmetic order lies between periodicity and randomness. Roots of unity produce the algebraic phases and quantum dimensions of finite topological data, while irrational algebraic numbers generate substitutions and cut-and-project structures with long-range order but no translation period. The resulting diffraction and spectral gaps are experimentally visible.

19.1 Algebraic numbers, roots of unity, and cyclotomic fields

Definition 19.1 (Algebraic numbers and cyclotomic fields)**.** A complex number α is *algebraic* over $\mathbb{Q}$ if it is a root of a nonzero polynomial in $\mathbb{Q}[x]$, and an *algebraic integer* if it is a root of a monic polynomial in $\mathbb{Z}[x]$. Its *minimal polynomial* is the monic irreducible polynomial in $\mathbb{Q}[x]$ that it satisfies. A *Galois conjugate* is another root of that polynomial.

For $n \geq 1$, let $\zeta_n = e^{2\pi i/n}$. The *nth cyclotomic field* is $\mathbb{Q}(\zeta_n)$, and the *nth cyclotomic polynomial* $\Phi_n(x)$ is the minimal polynomial of ζ_n. It satisfies

$$x^n - 1 = \prod_{d|n} \Phi_d(x), \qquad [\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n).$$

Exercise 19.1 (Roots of unity in topological data)**.** Prove the cyclotomic factorization recursively and show that $\mathrm{Gal}(\mathbb{Q}(\zeta_n)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$ by $\zeta_n \mapsto \zeta_n^a$. For the $SU(2)_k$ theory from Quantum Topology, put $\xi = e^{\pi i/(k+2)}$ and prove that the quantum dimensions

$$d_j = \frac{\sin((j+1)\pi/(k+2))}{\sin(\pi/(k+2))}, \qquad 0 \leq j \leq k,$$

belong to the real subfield of $\mathbb{Q}(\xi)$ and are algebraic integers. At $k = 3$, identify the nontrivial dimension $2\cos(\pi/5) = (1+\sqrt{5})/2$ and recover its polynomial $x^2 - x - 1$. Show directly that the topological twists in $SU(2)_k$ are roots of unity up to the common framing phase. Explain what this proves for this family without asserting that arbitrary algebraic numbers are physical topological data.

19.2 Diophantine structure and substitutions

Definition 19.2 (Continued fractions and Pisot numbers)**.** For irrational α, its simple continued fraction is

$$\alpha = [a_0; a_1, a_2, \ldots] = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots}},$$

with convergents $p_n/q_n = [a_0; \ldots, a_n]$. A real algebraic integer $\beta > 1$ is a *Pisot number* if every other Galois conjugate has absolute value less than one.

Exercise 19.2 (The golden ratio as a Diophantine scale)**.** Let $\phi = (1+\sqrt{5})/2$. Prove

$$\phi = [1; 1, 1, 1, \ldots], \qquad \frac{F_{n+1}}{F_n} \longrightarrow \phi, \qquad F_{n+1}F_{n-1} - F_n^2 = (-1)^n.$$

Derive the convergent estimate $|\phi - F_{n+1}/F_n| < 1/F_n^2$ and the corresponding near recurrences of phases $e^{2\pi i m/\phi}$. Show that ϕ is a Pisot number because its other conjugate is $-1/\phi$. Explain why good rational approximants produce large approximate periods without producing an exact translation period.

Definition 19.3 (Substitutions and the Fibonacci word)**.** A *substitution* on a finite alphabet replaces each letter by a finite word and extends by concatenation. Its incidence matrix records how many copies of each letter occur in every substituted letter. The Fibonacci substitution is

$$\sigma(a) = ab, \qquad \sigma(b) = a, \qquad M_\sigma = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Its one-sided fixed word is the limit of $a, ab, aba, abaab, \ldots$.

Exercise 19.3 (Frequencies and aperiodicity of the Fibonacci chain)**.** Diagonalize M_σ, prove that its Perron–Frobenius eigenvalue is ϕ, and derive letter frequencies

$$\mathrm{freq}(a) = \frac{1}{\phi}, \qquad \mathrm{freq}(b) = \frac{1}{\phi^2}.$$

Prove that every finite subword recurs with bounded gaps. Show that the infinite word cannot be periodic because a periodic word has rational letter frequencies. If tiles of lengths ϕ and 1 replace a and b, show that substitution followed by rescaling by ϕ reproduces the tiling. Separate exact inflation symmetry from ordinary translation symmetry.

19.3 Cut-and-project order and diffraction

Definition 19.4 (Cut-and-project sets)**.** Let $\mathcal{L}$ be a lattice in $\mathbb{R}^d \times \mathbb{R}^m$, with projections $\pi_\parallel$ and $\pi_\perp$ onto physical and internal space. Assume $\pi_\parallel$ is injective on $\mathcal{L}$ and $\pi_\perp(\mathcal{L})$ is dense. For a compact window $W \subset \mathbb{R}^m$ with nonempty interior, define the *model set*

$$\Lambda(W) = \{\pi_\parallel(x) : x \in \mathcal{L},\ \pi_\perp(x) \in W\}.$$

It is *regular* when the boundary of W has measure zero. The higher-dimensional lattice supplies long-range order, while the irrational orientation of physical space generally destroys translation periodicity.

Exercise 19.4 (The Fibonacci chain by cut and project)**.** Let $\phi' = -1/\phi$ and embed $\mathbb{Z}^2$ by

$$(m,n) \longmapsto (m + n\phi,\ m + n\phi') \in \mathbb{R}_\parallel \times \mathbb{R}_\perp.$$

Prove the injectivity and density conditions in the definition. Choose a generic half-open interval window of length $1 + 1/\phi$, with boundary disjoint from the projected lattice, and order the resulting physical points. Determine the two possible nearest-neighbor spacings and show, after an overall rescaling and a possible exchange of letters, that their sequence is a Fibonacci tiling. Track how translating the window changes the local pattern without changing its set of finite patches. Interpret the internal coordinate as an arithmetic phase invisible in physical space.

Definition 19.5 (Autocorrelation and diffraction)**.** For a uniformly discrete point set $\Lambda \subset \mathbb{R}^d$, its Dirac comb is $\omega = \sum_{x\in\Lambda} \delta_x$. When the vague limit exists, its *autocorrelation* is

$$\gamma = \lim_{R\to\infty} \frac{\omega|_{B_R} * \widetilde{\omega|_{B_R}}}{\mathrm{vol}(B_R)}, \qquad \widetilde{\mu}(f) = \overline{\mu(\widetilde{f})}, \quad \widetilde{f}(x) = \overline{f(-x)}.$$

The positive measure $\widehat{\gamma}$ is its *diffraction measure*; its atoms are Bragg peaks. A regular model set has pure-point diffraction supported on a projected dual-lattice module, although that module need not be a periodic reciprocal lattice.

Exercise 19.5 (Sharp diffraction without periodicity)**.** For a finite sample Λ_R, show that the kinematic scattering intensity is

$$I_R(k) = \left| \sum_{x\in\Lambda_R} e^{-ik\cdot x} \right|^2$$

and relate its volume-normalized limit to $\widehat{\gamma}$. Recover the ordinary reciprocal lattice for a periodic crystal. Taking the pure-point diffraction theorem for regular model sets as supplied, derive the projected dual module and window Fourier-transform intensities for a cut-and-project set. Explain how a dense Fourier module can carry locally finite intensity and sharp peaks without implying real-space periodicity.

Compare this mechanism with the sharp noncrystallographic diffraction reported in Shechtman et al., *Physical Review Letters* **53** (1984). State what diffraction establishes—long-range coherent order and its symmetry—and what it does not establish without a structural model.

19.4 Quasicrystal spectra and gap labelling

Definition 19.6 (The Fibonacci Hamiltonian and integrated density of states)**.** Let $\alpha = 1/\phi$ and

$$v_\theta(n) = \mathbf{1}_{[1-\alpha,1)}(\theta + n\alpha \bmod 1).$$

The *Fibonacci Hamiltonian* on $\ell^2(\mathbb{Z})$ is

$$(H_\theta\psi)_n = \psi_{n+1} + \psi_{n-1} + \lambda v_\theta(n)\psi_n.$$

For the uniquely ergodic rotation of θ, its *integrated density of states* is

$$N(E) = \int_0^1 \langle \delta_0, \mathbf{1}_{(-\infty,E]}(H_\theta)\delta_0\rangle \, d\theta.$$

It is the limiting number of states below E per lattice site.

Let Ω be the closure of the shift orbit of a Fibonacci sequence in $\{0,1\}^{\mathbb{Z}}$, and let $S : \Omega \to \Omega$ be the shift. The associated observable algebra is the crossed product $\mathcal{A}_\alpha = C(\Omega) \rtimes_S \mathbb{Z}$, generated by $C(\Omega)$ and a unitary U satisfying $UfU^* = f \circ S^{-1}$. The unique invariant measure μ gives the canonical trace

$$\tau\left(\sum_n f_n U^n\right) = \int_\Omega f_0 \, d\mu.$$

The Hamiltonians above are the covariant representations of one element of this algebra.

Exercise 19.6 (Diophantine gap labels)**.** Prove that $N(E)$ is nondecreasing and is constant on every open spectral gap. Approximate α by Fibonacci convergents and interpret the finite periodic approximants as large-unit-cell crystals. Taking the one-dimensional gap-labeling theorem as supplied, show that every gap value satisfies

$$N(E) \in (\mathbb{Z} + \alpha\mathbb{Z}) \cap [0, 1].$$

Explain why the theorem restricts possible labels but, by itself, does not assert that every allowed gap is open. Relate the label group to the range of τ_* on $K_0(\mathcal{A}_\alpha)$, thereby connecting Diophantine frequency to the operator K-theory developed in Index Theory; see Bellissard–Bovier–Ghez, *Reviews in Mathematical Physics* **4** (1992).